A guide to the Global Macroeconomic Analysis Projections and Scenarios:

conditional forecasts, global scenarios, and policy counterfactuals in the browser[1]

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Source code: https://github.com/matyasfarkas/BVARScenarioTool

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Abstract

This document describes the Global Macroeconomic Analysis Projections and Scenarios, a browser-based tool for quarterly macroeconomic projection and scenario analysis covering about fifty economies. Each economy is a Bayesian vector autoregression estimated with the hierarchical prior of Giannone et al. (2015) under a COVID-as-missing data-augmentation treatment; thirty-three of them are coupled into a global system with an endogenous United States, solved as a linear fixed point at display time. The user conditions any subset of variables on hard or soft paths — a mouse drag, an imported outside forecast with a chosen degree of trust — and a Durbin–Koopman smoother (Durbin and Koopman, 2002) propagates the constraints to every other variable and, on request, across borders. Public debt and the external position are derived from the estimated drivers by their accounting identities; long-run anchors shift the companion drift exactly and instantly; and monetary or fiscal counterfactuals are evaluated with impulse responses from the Macroeconomic Model Data Base (Wieland et al., 2012). We state the methods with their exact implemented conventions, show how each is operated from the interface, and document the data, the out-of-sample evaluation, and the replication path. All computation runs client-side; there is no server.

Keywords: BVARs, Conditional Forecasting, GVAR, Missing Values, Scenario Analysis, Kalman Filter, Policy Counterfactuals, DSGE Models

JEL classification: C32, C53, E37, E52, E58, F47

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Contents

1 Introduction

The tool described in this document is a single web page. It loads the estimated models of about fifty economies into the browser and lets the user read, bend, and export their quarterly forecasts. There is nothing to install and no server-side computation: every number on the screen is produced client-side, so the tool can be hosted as a static website and used offline once loaded. The source code, the estimation pipeline, and the data files are in one repository,

https://github.com/matyasfarkas/BVARScenarioTool

with the deployed application served from its `app/` static export. The tool comes as is; it is a research platform, not an official forecast of any institution.

The economic machinery is organized in three layers, applied in order:

(i) **Baseline.** Each economy is an n-variable Bayesian vector autoregression (BVAR) estimated on quarterly data with the hierarchical Minnesota prior of Giannone et al. (2015), treating the pandemic quarters as missing data (Section 3). Thirty-three economies are additionally coupled into a global VAR (GVAR) with a fully endogenous United States (Section 5). The baseline forecast is computed once, offline, and shipped with the model.

(ii) **Scenario.** The user conditions any subset of variables on desired paths. Conditioning is a partial-observation problem in the companion-form state space, solved by a Kalman filter and Durbin–Koopman smoother (Bańbura et al., 2015; Durbin and Koopman, 2002); a pinned value can be *hard* (exact) or *soft* (an outside forecast held with a chosen degree of trust), and it propagates to the other variables — and, on request, to the other economies — through the estimated covariance and trade structure (Sections 4 and 5).

(iii) **Counterfactual.** A monetary or fiscal policy counterfactual is overlaid on the scenario using impulse responses from a DSGE model of the user's choice, drawn from the Macroeconomic Model Data Base (MMB; Wieland et al., 2012), and re-conditioned through the BVAR so the remaining variables stay consistent with the estimated co-movements (Section 9).

On top of these, two accounting blocks derive what should not be estimated freely: public debt from the budget identity (Section 6) and the net external position from the balance-of-payments identity (Section 7); and a long-run anchor mechanism shifts the far end of any forecast onto a user-chosen value, exactly and instantly (Section 8).

Three display rules govern everything on screen, and we state them here because the rest of the document relies on them. *Data are data*: observed history is drawn solid; any model-filled value — a not-yet-published quarter estimated from everything already observed — is drawn in amber and labelled, never disguised as an observation. *The model is frozen within the year*: quarterly data updates advance the forecast origin but never silently change the estimated coefficients; re-estimation happens once a year on a complete final year of data (Section 11). *Judgment is visible*:

whenever the user moves a forecast away from the model — a scenario pin, a long-run target, a counterfactual — the chart says so with a badge.

This document is both the user guide and the methodological documentation. It is organized so that each section states a method with its exact implemented conventions — the equations below were checked against the code, and file names are cited where the correspondence matters — and then shows how that method is operated from the interface, in numbered Examples that name the on-screen controls exactly. Section 2 gets a new user from a blank page to a first scenario. Sections 3–4 cover the estimation and the conditional-forecasting engine, Section 5 the cross-country system, Sections 6–7 the accounting blocks, Section 8 the long-run anchors, and Section 9 the policy counterfactuals. Section 11 documents the data and the vintage policy, Section 12 the out-of-sample evaluation, Section 13 the implementation and the replication path, and Section 14 concludes with limitations and known issues. Notation is collected in Appendix A; the per-economy variable tables — generated directly from the deployed data files — are Appendix B; a glossary is Appendix C.

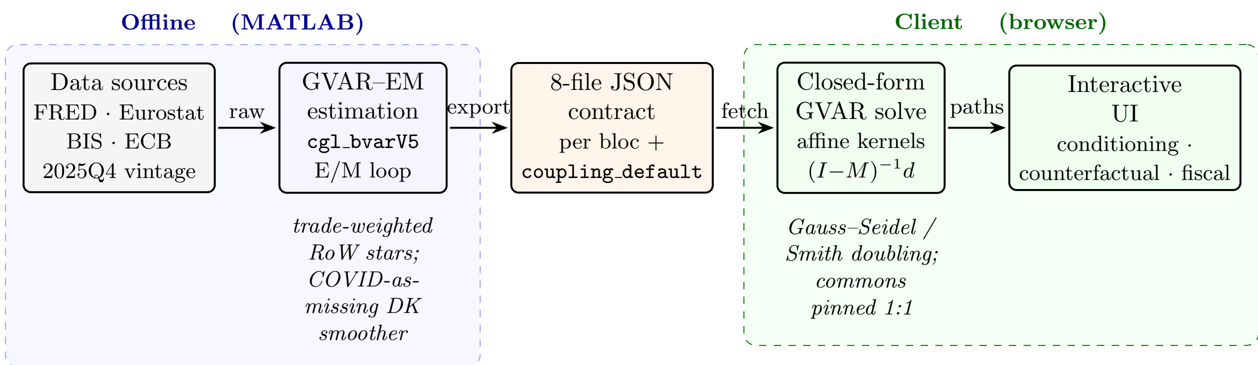

Figure 1: End-to-end system architecture of the EM-Bayesian Global VAR platform. Harmonized macro-financial data (FRED, Eurostat, BIS, ECB; common 2025Q4 *estimation* vintage, with the observed panel subsequently refreshed to 2026Q1 with no re-estimation, §11) feed the *offline* MATLAB GVAR–EM estimator: per-bloc Bayesian VARs (`cgl_bvarV5`) are re-fit in an EM loop that, at each sweep, rebuilds the trade-weighted rest-of-world "star" variables (ROWDEM/ROWINF/ROWFIN) from partners' Durbin–Koopman-completed panels and re-estimates every bloc with the 2020Q2–Q4 *macro* columns masked (the foreign-rate star ROWFIN and all domestic financials kept observed; COVID-as-missing), iterating to $\max|\Delta \text{star}| < 0.03$ (cf. §5.2). Each converged bloc is serialized to the eight-file JSON contract (`spec`, `companion_mean`, `companion_draws`, `baseline`, `baseline_levels`, `historical`, `historical_levels`, `transformations`; plus `fiscal_meta` and the precomputed `coupling_default.json.gz` coupling artifact). In the *client* (browser), the cross-country system is solved in closed form: the scenario delta obeys the affine fixed point $x = Mx + d$, evaluated by Gauss–Seidel sweeps or directly via $(I - M)^{-1}d$ (Smith doubling), with global commons (US WTI oil, US 10Y, US BAA, US equity) pinned 1:1 onto every RoW bloc. The interactive UI exposes conditioning, counterfactuals, and the stock-flow fiscal block (§6).

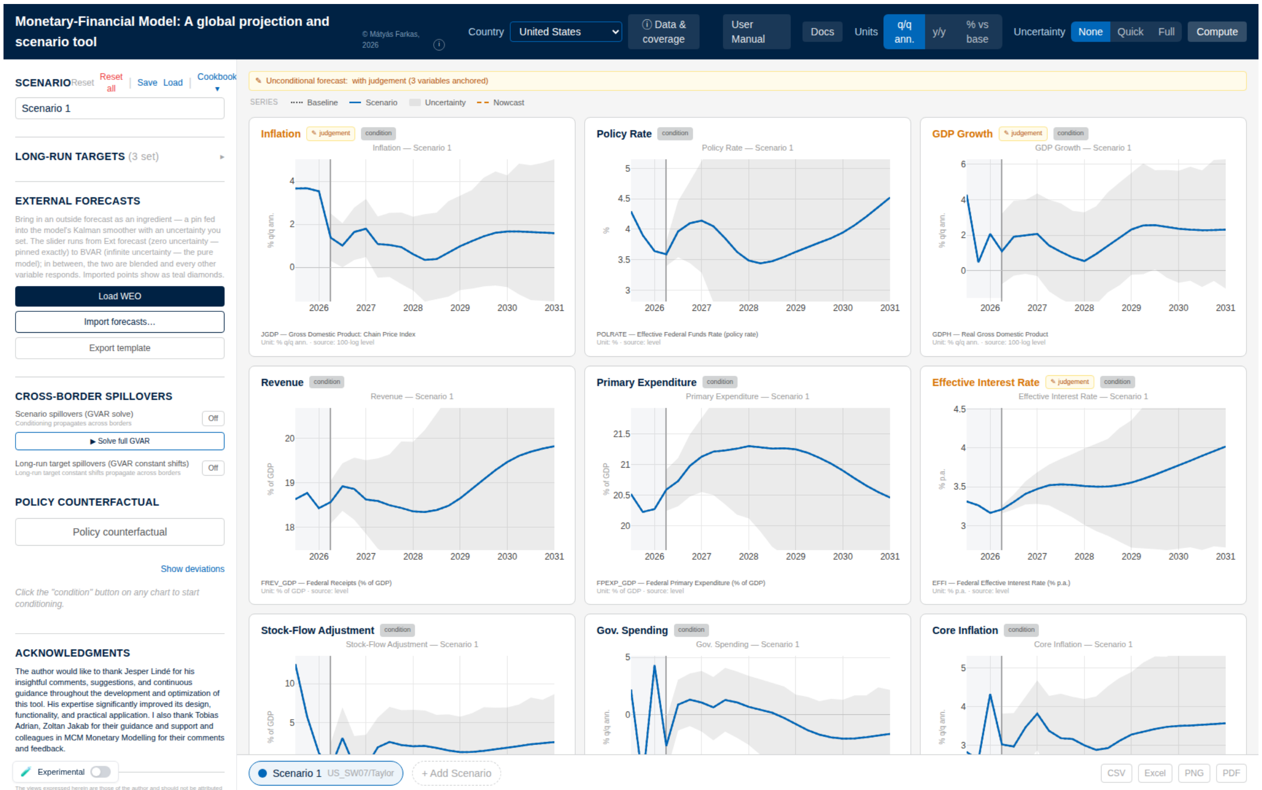

Figure 2: The screen at load (United States). The top bar holds the country selector, the *Data & coverage* provenance panel, the display-unit toggles (q/q annualized, y/y, % vs base), the uncertainty setting (None/Quick/Full), and Compute. The left rail holds the scenario manager, long-run targets, external forecasts (§4.5), cross-border spillovers (§5.5), and the policy counterfactual (§9). Each chart shows history, the dotted baseline, the solid scenario path, and the predictive band; judgement chips mark anchored variables (§8) and every chart carries a condition chip (§4.4).

2 Getting started

2.1 The screen

Open the tool's address in a modern browser and give the first visit a few seconds; the model files load into the page. The screen has four parts:

- the *top bar*: the **Country** dropdown, the display **Units** toggle, the **Uncertainty** switch (**None/Quick/Full**), the **Compute** button, the **Data & coverage** dialog, and links to this manual, the online documentation (**Docs**), and the release-impact **Monitor**;
- the *left control panel*: the **Scenario** block (with **Reset**, **Reset all**, **Save**, **Load**, and the **Cookbook** of ready-made scenarios), **Long-run targets**, **External forecasts**, **Cross-border spillovers**, the **Policy Counterfactual** block, and — once a variable is conditioned — the quarter-by-quarter conditioning list;
- the *chart grid*: one chart per variable; this is where the user clicks and drags;

- the *scenario bar* along the bottom: one tab per scenario (up to four, **+ Add Scenario**) and the **CSV**, **Excel**, **PNG**, and **PDF** export buttons.

The **Country** dropdown has two groups. Economies under **GVAR** belong to the coupled cross-country system of Section 5; a scenario built there can reach every other member. Economies under **Other countries** — including the individual euro-area member states, which receive shocks through the euro-area aggregate (Section 5.6) — are forecast on their own. Switching country never deletes work: each economy keeps its scenario settings until reset.

2.2 Reading one chart

Every chart follows the same grammar, and reading it correctly is most of the tool:

- *History* sits on a lightly shaded background; the solid line there is observed data. If a series' latest quarters have not been published, its history ends in an *amber dashed* tail with open circles — model-filled values (a *nowcast*, Section 11.2), never disguised as data. Hovering one reads *Model-implied nowcast (imputed)*.

- The *dotted grey line* is the **Baseline**: the model's own forecast, untouched by the user. The pale grey fan around it is the model's predictive uncertainty, shipped with the estimation (Section 3.5); it is there before anything is computed.

- The *solid coloured line* is the active scenario. Until the user conditions something it lies exactly on the baseline.

- The *header* carries a **condition** button (click to make the variable steerable; it turns into **CONDITIONED**), a **derived** badge on charts computed by an accounting identity, and an amber **judgement** badge whenever a long-run target has moved the variable off the model's estimate.

- The *footer* gives the variable's definition and unit.

The **Units** toggle changes the lens, not the numbers: **q/q ann.** is the quarter-on-quarter annualized rate (the model's native display unit, equation (2) below); **y/y** the change on the same quarter a year earlier; **% vs base** the *difference* between scenario and baseline, centred at zero — a read-only impact view in which a second toggle, **Level** versus **Rate (IRF)**, selects between the cumulative level effect and the quarter-by-quarter response (Section 9.3). Charts that are already rates (the policy rate, unemployment) ignore the toggle, and a handful of index series (oil, exchange-rate, equity, and house-price indices) are always drawn as levels, because a trade-weighted index quoted q/q-annualized is not an informative number.

Example 1 (A first scenario in five minutes)**.** *Open the tool; the* ***Country*** *dropdown reads* ***United States****, with* ***Units*** *on* ***q/q ann.*** *and* ***Uncertainty*** *on* ***None****. (1) On the* ***Policy Rate*** *chart click* ***condition****; the button turns blue and reads* ***CONDITIONED****. (2) Press the mouse on the* *forecast region about four quarters out and drag half a percentage point above the dotted baseline; release. A dot marks the pinned quarter, and the solid scenario line passes through it — only that quarter is imposed, and the model fills in the most plausible route through the point. (3) Scroll the grid: every chart has updated, not just the rate — growth cools, inflation eases with a lag — because the smoother re-draws the whole forecast consistently with the pin (Section 4). Other countries have not moved; cross-border transmission is off until requested (Section 5.5). (4) Set **Uncertainty** to **Quick** and press **Compute**: after a few seconds a band in the scenario's colour wraps the solid line — the middle two-thirds of plausible outcomes — and collapses onto the line at the pinned quarter, because an imposed value carries no uncertainty (Section 4.6). (5) Clean up: in the left panel click **Reset** (clears the active scenario for this country only) and set **Uncertainty** back to **None**. A right-click on an unconditioned chart is the shortcut for steps (1)–(2): it conditions the variable and plants the pin at the cursor in one gesture.*

Remark 1 (When to press Compute). The central scenario path updates live with every pin; **Compute** is never needed to see it. The button exists for the two calculations that are too heavy to run on every mouse move: the scenario uncertainty bands (Section 4.6) and the complete cross-country solve (Section 5.5). It is greyed out until either **Quick/Full** uncertainty is selected or a long-run target is set. Bands are a snapshot — change a pin afterwards and the band no longer matches until **Compute** is pressed again.

3 The country models

3.1 Reduced form and data spaces

Each economy ("bloc") c is an n-variable VAR with p lags in reduced form,

$$y_t = c + \sum_{\ell=1}^{p} B_\ell \, y_{t-\ell} + u_t, \qquad u_t \sim \mathcal{N}(0, \Sigma), \tag{1}$$

where $y_t \in \mathbb{R}^n$ collects the bloc's observables, $c \in \mathbb{R}^n$ is the intercept, $B_\ell \in \mathbb{R}^{n\times n}$ are the autoregressive matrices, and $\Sigma \in \mathbb{R}^{n\times n}$ is the innovation covariance. Throughout, $p = 3$ and the forecast horizon is $H = 20$ quarters. The deployed dimensions run from $n = 16$ (Costa Rica) to $n = 40$ (the United States); Appendix B tabulates every bloc. The estimated coefficients are stored as a $(1+np)\times n$ matrix $\hat\beta$ whose *first* row is the intercept and whose remaining np rows stack the lag coefficients $[B_1; \ldots; B_p]$ — a convention the reader needs only twice, for the companion construction (7) and the model signature of Section 5.3.

Variables enter the VAR in one of two spaces, recorded per variable in the bloc's specification (`spec.json`) and never changed at run time:

- `log`: the stored series is $\ell_t = 100 \ln Y_t$. Displayed growth is quarter-on-quarter annualized,

$$g_t \;=\; 4\,(\ell_t - \ell_{t-1}) \;=\; 400\,(\ln Y_t - \ln Y_{t-1}), \tag{2}$$

and year-on-year growth is $\ell_t - \ell_{t-4}$. The factor of 4 on the stored $100 \ln$ space — never 400, which would re-apply the 100 — is the single most load-bearing convention in the tool; every pin, band, and cross-country link below respects it.

- `lin`: the series enters in levels (interest rates in percent, unemployment, ratios to GDP). No growth conversion is applied.

Display units are chosen per variable for readability: activity and price series are shown as growth and inflation via (2); rates and ratios as levels in percent; and index series (the trade-weighted dollar, equity, house prices, oil, commodity indices) as the index level $\exp(\ell_t/100)$, with conditioning still performed in the stored $100 \ln$ space. Unemployment-rate bands are floored at zero; policy-rate bands are not, since negative nominal rates occur.

3.2 The hierarchical prior

Estimation follows the hierarchical approach of Giannone et al. (2015), henceforth GLP: the Minnesota prior (Litterman, 1986; Doan et al., 1984) with its tightness treated as a hyperparameter and selected by maximizing the marginal likelihood. The prior centers each equation on a univariate own-lag model, with a sum-of-coefficients component (Doan et al., 1984) shrinking toward "a unit increase in every level raises its own forecast one for one." Two hyperparameters are sampled — the overall tightness λ and the sum-of-coefficients weight μ — by a random-walk Metropolis step on the marginal likelihood of the *completed* data (Section 3.4), initialized at the `csminwel` optimum; the remaining prior settings are held at their GLP defaults. The estimator is the bundled MATLAB implementation (`BVAR/MFBVAR/bvarGLPmf.m`, called through the COVID-aware wrapper `cgl_bvarV5.m`); no external MATLAB packages are required.

Own-lag prior mean: random walk or white noise. The prior mean on variable i's own first lag is δ_i, with $\delta_i = 1$ (random walk, `RW`) the default: for (near-)integrated series — log activity, log prices, nominal yields — "the best forecast of the level is the current level." For a genuinely *stationary* series, a mean-reverting ratio that fluctuates around a constant, the random-walk center is the wrong null: it shrinks toward a unit root and lets the projection drift at long horizons. Such variables carry `prior:"WN"` in the specification and are estimated with $\delta_i = 0$ (white noise): "the best forecast is the unconditional mean." The mechanism is one line in the prior construction — the own-lag prior-mean diagonal is zeroed at the `WN` indices (`diagb(pos)=0` in `bvarGLPmf.m`, with `pos` built from the per-variable tags) — and it moves only the prior *center*, not a hard constraint: a persistent series whose data favour a near-unit root still obtains a large posterior own-lag coefficient at the estimated tightness. In the deployed panel the `WN` tag is carried by the three fiscal GDP-ratios (Section 6); every other variable is `RW` (Appendix B).

3.3 Seasonal dummies

A subset of the panel is reported *not seasonally adjusted*, so a regular quarter-of-year cycle sits in the level: NSA consumer prices (HICP/CPI for several economies and euro members), unemployment rates, house-price indices, India's government series, and the UK and Korean stock-flow adjustments. Left untreated, a pure autoregression can reproduce such a cycle only through a near-unit root close to the seasonal frequency, which then *explodes* the projection into a spurious sawtooth out of sample. The deterministic remedy is a set of quarterly dummies. For a bloc flagged seasonal, let $q(t) \in \{1, 2, 3, 4\}$ be the calendar quarter of observation t and collect the three non-base indicators

$$D_t = \big[\, \mathbf{1}\{q(t) = 2\},\ \mathbf{1}\{q(t) = 3\},\ \mathbf{1}\{q(t) = 4\} \,\big], \tag{3}$$

with the first quarter as the base. Each equation's regressor vector becomes $x_t = [\,1,\ D_t,\ y'_{t-1}, \dots, y'_{t-p}\,]$, so the estimated coefficient block acquires three deterministic rows between the constant and the lag matrices,

$$\widehat{\beta} = [\,\underbrace{c}_{1};\ \underbrace{\gamma_{Q2};\ \gamma_{Q3};\ \gamma_{Q4}}_{n_{\rm sr}=3};\ \underbrace{B_1; \dots; B_p}_{n\ \text{lag blocks}}\,], \qquad k = 1 + n_{\rm sr} + n\,p, \tag{4}$$

where n is the number of endogenous variables, $p = 3$ the lag order, and $n_{\rm sr}$ the number of "seasonal rows" (3 when the bloc is flagged, 0 otherwise — the legacy layout, left byte-identical). The dummy coefficients $\gamma_{Q2}, \gamma_{Q3}, \gamma_{Q4}$ carry a *diffuse* prior — the intercept variance V_c, centred at zero — and are zeroed in the sum-of-coefficients dummy observations, so the shrinkage machinery treats them like the constant and never pulls them toward a unit root (`BVAR/MFBVAR/Support/BVAR.m`; the COVID-aware wrapper `cgl_bvarV5.m` passes them through the `qoy` argument and carries the seasonal term in the simulation-smoother artificial path, so the Durbin–Koopman completion of Section 3.4 stays exact).

Which blocs are flagged is data-driven: a panel scan tags a bloc seasonal when at least one series shows calendar-$R^2 \geq 0.40$ with a display amplitude ≥ 2, and every coupon-lumpy bloc of Remark 6 is added because its `SFA` inherits the cash-versus-accrual timing. In the deployed system this is sixteen coupled blocs (CL, CN, CO, CR, EA, ID, IL, IN, IS, JP, KR, MX, NO, TR, UK, ZA) and six euro members (ES, IT, GR, PT, SI, BE); all remaining blocs keep $n_{\rm sr} = 0$.

In the projection the three coefficients generate a *bounded*, repeating quarter-of-year cycle rather than an explosive root: the tool precomputes four quarter-specific transition constants and picks the one matching each forecast step's quarter-of-year phase (`c2ByQ`/`c2AtStep` in `app/src/lib/compute/kalman.ts`). The phase itself is always recomputed from the tool's own date grid — the quarter following the last history point, wrapped 1...4 — never read from the shipped estimation-grid value, which sits one quarter behind after the production data shift. Every $n_{\rm sr} = 0$ path is unchanged to the byte.

3.4 COVID as missing data

The 2020 pandemic quarters are extreme outliers that would otherwise contaminate both the autoregressive coefficients and Σ. Rather than dummying or rescaling them, the estimator treats the pandemic window as *missing* and imputes it inside the estimation, by a data-augmentation Gibbs sampler in the sense of Tanner and Wong (1987): each sweep alternates

(i) a Durbin–Koopman simulation-smoother draw (Durbin and Koopman, 2002) of the missing cells given the current coefficients, treating them as `NaN` observations in the companion-form state space of Section 4.1 (Algorithm 1);

(ii) a Metropolis–Hastings update of (λ, μ) on the completed-data marginal likelihood; and

(iii) a draw of (β, Σ) from the completed data.

The surrounding quarters stay in the sample and inform the imputation, so the pandemic history is preserved while the outlier does not drive the dynamics.

Two implementation choices matter. First, the production window is the *fixed* three quarters 2020Q2–Q4 for every bloc. (A legacy single-bloc variant self-calibrates the window to the GDP trough searched within 2019Q4–2021Q1 — restricting the search matters, since an unrestricted trough lands on the wrong crisis for some economies, e.g. Korea's 2008Q4 — but the deployed GVAR estimator uses the fixed window.) Second, the masking is *macro-only*: over the window the domestic macro columns and the two macro star variables of Section 5.1 are set missing, while the financial columns (rates, yields, equity, house prices, spreads; the `isFinancial` tag in Appendix B) and the financial star stay *observed*. The smoother therefore imputes the pandemic macro holes conditional on financial series that carry genuine signal through 2020. The same device handles non-pandemic artefacts: the UK effective-rate spike of 2008–09, a mechanical consequence of inflation uplift on index-linked gilts being recorded as accrued interest, is NaN'd and imputed identically (Section 6).

Algorithm 1 Durbin–Koopman simulation smoother with partial observations (one draw inside the data-augmentation Gibbs sweep; `cgl_bvarV5.m`).

Input: system matrices T, Z, Q of Section 4.1; partial observations $\{y_t^{(o)}\}$ (COVID-masked and ragged cells absent); current draw of (β, Σ).
Output: one draw $\tilde{\alpha}_{1:T}$ from $p(\alpha_{1:T} \mid y^{(o)}, \beta, \Sigma)$, hence imputed values for the missing cells.

1: ▷ (1) Forward simulation of an unconditional path.
2: Draw α_0^+ from the initial law; **for** $t = 1, \ldots, T$: draw shocks and set $\alpha_t^+ = T\alpha_{t-1}^+ + \eta_t$, $y_t^+ = Z\alpha_t^+$.
3: Form synthetic partial observations $y_t^{+(o)}$ on the *same* missing pattern as the data.
4: ▷ (2) Kalman filter (Section 4.2) on the artificial series $y_t^{(o)} - y_t^{+(o)}$, processing only the observed rows at each date.
5: ▷ (3) Backward smoothing recursion (equation (12) of Section 4.2) $\Rightarrow$ smoothed discrepancy $\hat{\alpha}_t$.
6: ▷ (4) Mean correction. $\tilde{\alpha}_t \leftarrow \alpha_t^+ + \hat{\alpha}_t$; read the imputed cells off $\tilde{y}_t = Z\tilde{\alpha}_t$.

3.5 What ships: posterior output and predictive bands

For the deterministic engine the posterior is summarized by the means $(\hat{\beta}, \hat{\Sigma})$. For uncertainty the chain is thinned to $D = 500$ retained draws $\{(\hat{\beta}^{(d)}, \hat{\Sigma}^{(d)})\}$, shipped per bloc in `companion_draws.json` and loaded lazily — only when the user first presses **Compute** (the US file is the largest at roughly 34 MB).

The shipped baseline carries *predictive* bands, not parameter-only bands: the Gibbs sampler emits a forecast density that integrates over parameter uncertainty, future innovations, and the missing-data imputation simultaneously, and the exported quantiles are

$$\big(q_{05},\, q_{16},\, q_{50},\, q_{84},\, q_{95}\big)_{T+h} \;=\; \mathrm{Quantile}\big\{\, \tilde{y}^{(d)}_{T+h} \,\big\}_{d=1}^{D}, \tag{5}$$

drawn on the charts as the grey fan (the 68% band (q_{16}, q_{84}) shaded, the 90% band (q_{05}, q_{95}) fainter). The *central* path remains the deterministic forecast from the posterior means, so an untouched screen shows the stored baseline exactly. Scenario bands — computed in the browser, conditional on the user's pins — are constructed differently and are described with the conditioning engine (Section 4.6).

Every bloc is serialized to one eight-file JSON contract — `spec`, `companion_mean`, `companion_draws`, `baseline`, `baseline_levels`, `historical`, `historical_levels`, `transformations` — fiscal blocs adding `fiscal_meta` and external blocs `external_meta`, with every float compacted to six significant figures. The deployed application reads only this contract and never re-estimates (Section 13).

4 Conditional forecasting

Conditioning a forecast on a chosen path is, in state-space form, the same problem as filtering with missing data: a pinned cell is an *observation* of the future, a free cell is a missing one (Bańbura et al., 2015; Waggoner and Zha, 1999). This section states the state space, the filter and smoother exactly as implemented (`app/src/lib/compute/kalman.ts`), the display construction, and the two kinds of pin — hard and soft — the interface exposes.

4.1 Companion-form state space

Stack the current and lagged observables into the state $\alpha_t = (y_t', y_{t-1}', \ldots, y_{t-p+1}')' \in \mathbb{R}^{np}$. The VAR (1) is then the first-order system

$$\alpha_t = T\,\alpha_{t-1} + c_\alpha + \eta_t, \qquad y_t = Z\,\alpha_t + \varepsilon_t, \tag{6}$$

with

$$T = \begin{pmatrix} B_1 & B_2 & \cdots & B_{p-1} & B_p \\ & I_{n(p-1)} & & & 0 \end{pmatrix}, \qquad c_\alpha = \begin{pmatrix} c \\ 0 \end{pmatrix}, \qquad Z = \big(I_n \mid 0_{n\times n(p-1)}\big), \tag{7}$$

$\eta_t \sim \mathcal{N}(0, Q)$ where Q carries Σ in its top-left $n \times n$ block and zeros elsewhere, and $\varepsilon_t \sim \mathcal{N}(0, R)$ with

$$R = \sigma_\varepsilon^2 I_n, \qquad \sigma_\varepsilon^2 = 10^{-12}. \tag{8}$$

The top block of T is filled from the stored coefficients as $T_{ij} = \hat{\beta}_{j+1,\,i}$ (the transpose of the lag rows, skipping the intercept row), and c_α from the intercept row. The measurement variance (8) is the *hard-pin* noise: small enough that a pinned value is honoured to machine precision, nonzero for numerical stability. The state is initialized at the last p observed rows, $\alpha_0 = (y'_T, \ldots, y'_{T-p+1})'$, with $P_0 = 10^{-7} I_{np}$.

4.2 Filter and smoother with partial observations

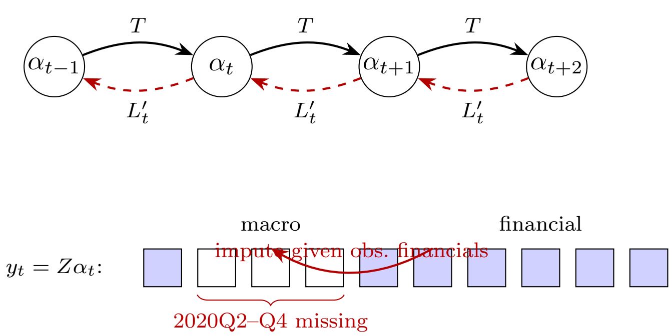

Figure 3: Companion-form state space with the Durbin–Koopman filter/smoother under the COVID-as-missing treatment. States α_t (§4.1) evolve forward through the transition matrix T (solid, Kalman filter, Eq. (9)); the backward smoother propagates $r_{t-1} = (Z_t^{(o)})' F_t^{-1} v_t + L'_t r_t$ (dashed, Eq. (12)). In the observation row $y_t = Z\alpha_t$ (§4.1), over the fixed window 2020Q2–Q4 (three quarters) the *macro* cells (real activity, prices, and the macro stars ROWDEM/ROWINF) are set missing (hollow) while the *financial* cells (rates, yields, equity, house prices, and the foreign-rate star ROWFIN) remain observed (filled). Because the filter processes only the observed rows $Z_t^{(o)}$ of each partially-observed vector, the macro holes are imputed conditional on the observed financials.

Collect the user's pins in a matrix $Y^* \in \mathbb{R}^{H \times n}$ with $Y^*_{t,j}$ the pinned value of variable j at forecast quarter t and `NaN` where free. At each t the filter processes only the observed rows: let Z_t stack the rows of Z for the pinned cells (plus any linear-combination rows, below) and R_t the corresponding diagonal of measurement variances. For $t = 1, \ldots, H$:

$$a_{t|t-1} = T\, a_{t-1|t-1} + c_\alpha, \qquad P_{t|t-1} = T\, P_{t-1|t-1}\, T' + Q, \tag{9}$$

$$v_t = y_t^* - Z_t\, a_{t|t-1}, \qquad F_t = Z_t\, P_{t|t-1}\, Z'_t + R_t, \tag{10}$$

$$K_t = P_{t|t-1}\, Z'_t\, F_t^{-1}, \qquad a_{t|t} = a_{t|t-1} + K_t\, v_t, \qquad P_{t|t} = P_{t|t-1} - K_t\, Z_t\, P_{t|t-1}, \tag{11}$$

with $a_{t|t} = a_{t|t-1}$, $P_{t|t} = P_{t|t-1}$ at quarters with no pins. F_t^{-1} is computed by LU decomposition with partial pivoting (singularity threshold 10^{-15}), and the covariance updates are symmetrized.

The Durbin–Koopman fixed-interval smoother then runs backward with $r_H = 0$:

$$L_t = T\big(I - P_{t|t-1} Z_t' F_t^{-1} Z_t\big), \qquad r_{t-1} = Z_t' F_t^{-1} v_t + L_t' \, r_t, \tag{12}$$

($r_{t-1} = T' r_t$ at pin-free quarters), and the smoothed state is $\hat{\alpha}_{t|H} = a_{t|t-1} + P_{t|t-1} \, r_{t-1}$. The smoothed *variance* is delivered by the companion N-recursion,

$$N_{t-1} = Z_t' F_t^{-1} Z_t + L_t' \, N_t \, L_t, \qquad V_t = P_{t|t-1} - P_{t|t-1} \, N_{t-1} \, P_{t|t-1}, \tag{13}$$

with $N_H = 0$; V_t is the state covariance conditional on *all* pins, so uncertainty about a quarter *preceding* a future pin is correctly reduced by it. The conditional forecast is the observation block of $\hat{\alpha}_{t|H}$; the display shading of the impact view (Section 4.6) reads V_t.

Linear-combination pins. A constraint on a linear combination $h' y_t = v$ — used by the fiscal block to pin the primary balance, $h_{\text{rev}} = +1$, $h_{\text{pexp}} = -1$ (Section 6) — enters the same machinery as one extra observation row $(h'Z, \, v)$ with measurement variance $\sum_k h_k^2 \, \sigma_\varepsilon^2$. No separate variable is created.

4.3 The display: exact deltas on the stored baseline

To guarantee that an untouched screen reproduces the shipped baseline exactly, the tool displays

$$\hat{y}_{T+h}^{\text{display}} = \bar{y}_{T+h}^{\text{baseline}} + \big(\hat{y}_{T+h}^{\text{cond}} - \hat{y}_{T+h}^{\text{uncond}}\big), \tag{14}$$

the stored baseline plus the smoother *delta* between a conditional and an unconditional pass (the latter cached and reused). Zero pins give a zero delta and hence the stored baseline to the last digit. For `log` variables the delta is computed in the $100 \ln$ level space and scaled to displayed growth by the factor 4 of (2); a dragged growth pin is correspondingly cumulated into the level pin, one quarter's worth (deviation/4) per active quarter.

4.4 Operating hard pins

A pin is set on the chart (drag on a **CONDITIONED** chart; right-click to condition and plant in one gesture; right-click again to toggle a quarter's pin off) or in the conditioning list, which shows one row per forecast quarter, labelled **NC**, **t+1**, . . . , **t+19**. **NC** — the nowcast — is the current, not-yet-published quarter (Section 11.2). Each row has an on/off toggle and a value slider; **Show deviations/Show levels** switches between entering distances from baseline and absolute values. Pins are what-you-set-is-what-you-see: a pinned quarter sits exactly at its number, always; the quarters left off are filled by the model. Multiple variables stack, and all pins are imposed *jointly* — the smoother finds the single most plausible forecast through every pin at once, so mutually strained pins show their cost in the unconditioned charts. Up to four scenarios can be held side by

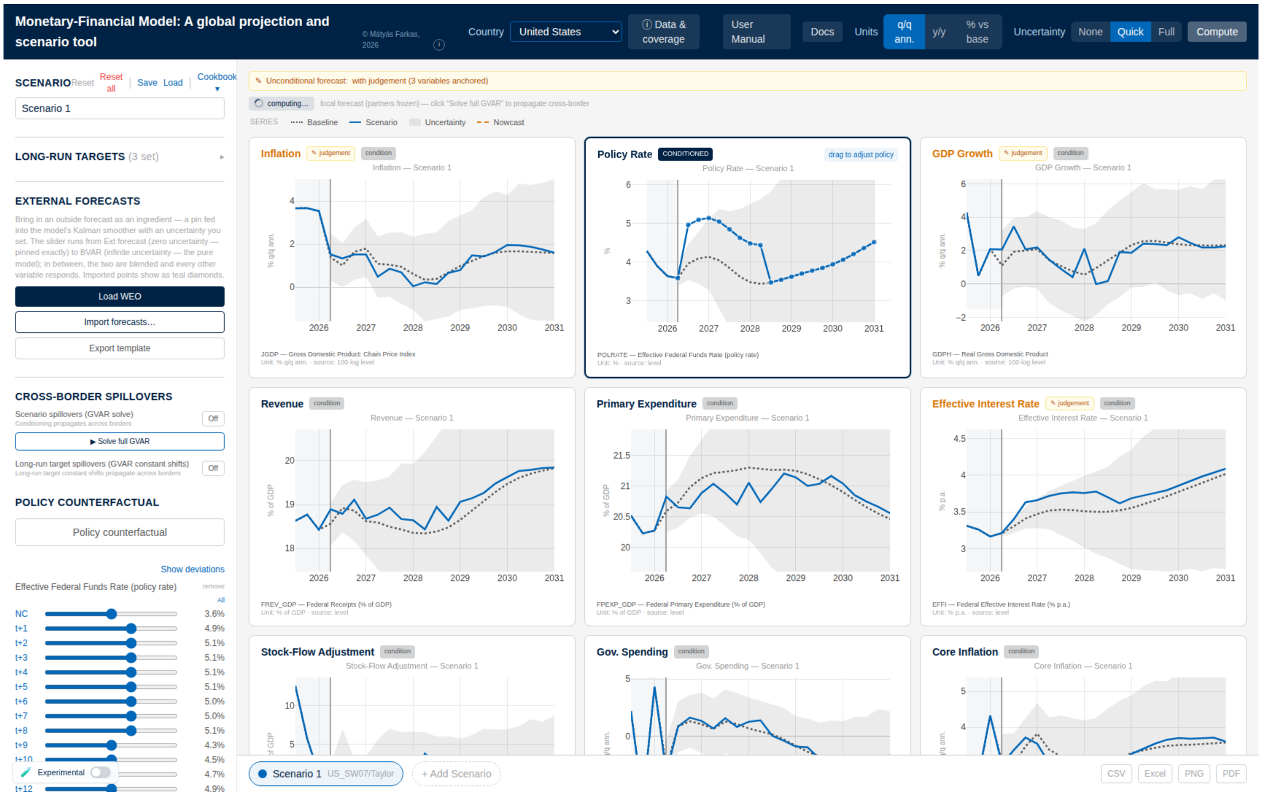

Figure 4: Hard conditioning in action. The policy-rate chart is put in CONDITIONED mode and the path is raised by +100 bp for eight quarters — by dragging the dots on the chart or the per-quarter sliders that appear in the left rail (here $t{+}1, \ldots, t{+}8$). After Compute, every panel redraws the smoothed scenario (solid) against the frozen baseline (dotted): output growth dips, inflation eases, the effective interest rate on debt climbs, and the fiscal ratios respond — all through the estimated cross-correlations of §4.2; nothing is re-estimated.

side (the bottom bar); **Reset** clears the active scenario for the current country, **Reset all** everything everywhere, and **Save/Load** round-trip one country's conditioning to an Excel file.

Example 2 (A ready-made scenario, and what identification buys). *The **Cookbook** (left panel, **Scenario** header) loads documented, source-calibrated scenarios; each recipe offers **Load naive** — pin only the headline variable — and **Load + restriction** — add the short-run pins that shape the shock into its economically meaningful form. The credit-crunch recipe (calibrated to the Federal Reserve's 2020 severely-adverse scenario) illustrates on the United States what the extra pins buy.* Naive*: pin only the corporate bond yield +400 bp. The model reads a corporate-yield spike as generic rates tightening — the whole curve rises (at the 2026Q1 vintage, the policy rate peaks around* $7\frac{1}{2}\%$*, the 10-year near* $8\frac{1}{2}\%$*) and inflation moves up on impact: comovement alone cannot tell a credit-supply dislocation from a broad rate shift.* Restricted*: add the documented crunch signature — safe 10-year yield* down *(flight to safety), output down, inflation down, equity down, the policy rate cut. The same +400 bp spread now produces a deep recession (GDP trough near −6% q/q ann.), disinflation, and easing — the policy rate and the safe yield move in the* opposite *direction from* *the naive read. Loading a recipe* replaces *the active scenario's pins; run the two variants on two scenario tabs to compare.*

4.5 Soft pins: outside forecasts with a trust dial

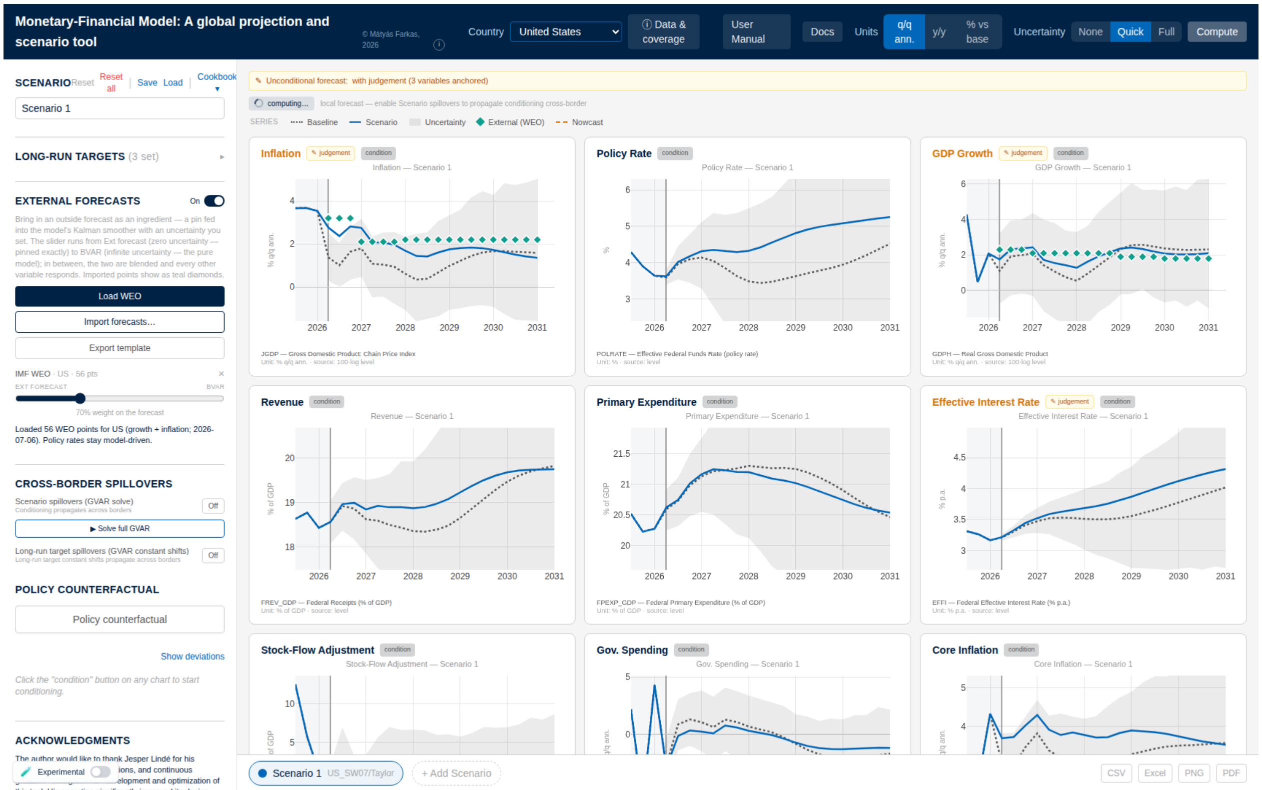

Figure 5: An outside forecast as an ingredient (Load WEO). The teal diamonds are the imported WEO growth and inflation points; the trust dial (left rail, here at 70% weight on the forecast) maps to the measurement-error variance of Eq. (15), so the scenario blends the outside path with the model instead of pinning it; policy rates stay model-driven.

An imported outside forecast — IMF WEO, a consensus survey, desk numbers — enters as a *soft* pin: an observation with a measurement variance set by the user's trust in the source, so the model reconciles the outside number with everything else rather than being overridden by it. For a cell of variable j at horizon h held with trust $\tau \in (0, 1)$, the measurement variance in (10) is

$$R^{\text{ext}}_{h,j} = \widehat{\text{Var}}_{h,j} \, \frac{1-\tau}{\tau}, \qquad \widehat{\text{Var}}_{h,j} = \Big(\max\big(\tfrac{q_{84}-q_{16}}{2},\, 10^{-6}\big)\Big)^2, \tag{15}$$

where $q_{84} - q_{16}$ is the model's own shipped predictive band of (5) at that cell — so the dial needs no units: $\tau = \frac{1}{2}$ makes the outside number exactly as informative as the model's own predictive density, $\tau \geq 0.999$ collapses to the hard pin (8), and $\tau \to 0$ is a no-op (τ is clamped to $[0.001, 0.999]$; the default is 0.7). Display-unit values are converted to level-space pins by the conventions of Section 4.3 (growth targets cumulate; any active long-run drift of Section 8 is netted out first, so the *displayed* line lands on the imported number). A cell the user has pinned by hand always wins over an imported one (`externalForecast.ts`).

In the interface (**External forecasts**, left panel): **Load WEO** imports the IMF World Economic Outlook real-GDP-growth and inflation forecast for the viewed economy, each annual figure applied softly to its four quarters; **Export template** writes an Excel file pre-filled with the model baseline, to be overwritten and re-imported; **Import forecasts...** accepts that file or a small JSON. Each source appears with its trust slider; imported points draw as teal diamonds with an error bar that grows as trust falls; the source's × (or zero trust) removes it. **Export session/Import session** bundle all countries' conditioning, targets, and imports into one JSON for hand-over. On a coupled economy an imported forecast also propagates cross-border, with one performance caveat documented in Section 5.7.

4.6 Scenario uncertainty bands

Pressing **Compute** with **Quick** or **Full** selected draws $N_b = 50$ or 200 predictive paths and bands the scenario. The default construction is deliberately simple and fast:

(i) for each band draw, sample a posterior draw $(\hat{\beta}^{(d)}, \hat{\Sigma}^{(d)})$ from the $D = 500$ shipped draws (with replacement) and iterate the companion forward from the last observed state *with* innovation shocks $\eta_h \sim \mathcal{N}(0, \hat{\Sigma}^{(d)})$ — a full predictive draw carrying both parameter and shock uncertainty;

(ii) convert to display units and shift the whole path by the (draw-invariant) mean scenario delta of (14);

(iii) report the empirical 16th and 84th percentiles per cell.

The band is therefore the model's own predictive range *re-centred on the scenario path* — directly comparable to the baseline fan — and at the actively pinned quarters of a conditioned variable it is collapsed onto the line, since an imposed value carries no uncertainty. On the **% vs base** impact view the shading instead uses the smoothed conditional standard deviation from (13): it pinches at hard pins, shrinks partially at soft pins in proportion to (15), and tapers *into* a future pin — the honest conditional-variance picture.

Remark 2 (Why the default band is not the per-draw conditional cloud)*.* Re-solving the conditional forecast draw by draw produces a genuinely conditional cloud, but with only a point or two pinned it collapses far too tightly under the estimated prior — a band the user reads as false confidence. The default recentring preserves the full predictive width; the per-draw conditional construction is available for coupled economies as the **Experimental** switch (bottom-left), which re-solves the entire cross-country system per draw (Section 5.8). Toggling it clears any band on screen, because bands built the two ways must not be read against each other.

Remark 3 (Long-run targets translate bands rigidly). A long-run anchor (Section 8) shifts the central path and every band quantile by the same deterministic amount, instantly and without recomputation — band *widths* are invariant by construction. Only pins change conditional uncertainty; targets move where the forecast is headed, not how well it is known.

5 The global model

Thirty-three of the deployed economies are coupled into a global VAR (GVAR) in the tradition of Pesaran et al. (2004) and Dees et al. (2007): the United States, the euro area as a single aggregate bloc, and thirty-one further economies, spanning roughly ninety percent of world GDP (the roster is in Appendix B). Two design choices organize everything in this section. First, *coupling is imposed at solve time, not at estimation time*: each bloc is an independently estimated BVAR, and cross-country spillovers enter only through the scenario *deviation*, which propagates as a linear fixed point. Second, the United States is *fully endogenous*: it carries its own trade-weighted foreign variables and iterates inside the fixed point like every other bloc, so shocks abroad feed back into the US — the channel a classical frozen dominant unit cannot represent.

The euro area enters as one aggregate by necessity: a fully disaggregated euro GVAR is non-stationary (the coupling operator's spectral radius exceeds one, $\rho \approx 1.089$, in the disaggregated experiment), while aggregation restores contraction. The seventeen euro-area member states are therefore *standalone* models outside the coupled system, connected to it through a separate euro-area doorway (Section 5.6).

5.1 Star variables and trade weights

Every coupled bloc carries three trade-weighted *star* variables summarizing its foreign environment, the weak-exogeneity device of the GVAR literature: foreign demand (`ROWDEM`), foreign inflation (`ROWINF`), and foreign financial conditions (`ROWFIN`). At estimation time the stars are built as levels from the partners' *completed* panels (Section 5.2): with w_{cj} the trade weight of partner j in receiver c and $\ell_{j,t}$ the partner's stored $100 \ln$ level,

$$\mathrm{ROWDEM}_{c,t} = 4 \sum_j \tilde{w}_{cj} \left(\ell^{\mathrm{gdp}}_{j,t} - \ell^{\mathrm{gdp}}_{j,t-1}\right), \qquad \mathrm{ROWFIN}_{c,t} = \sum_j \tilde{w}_{cj}\, r_{j,t}, \tag{16}$$

(`ROWINF` as `ROWDEM` on price levels) — annualized growth for the two macro stars, the factor 4 of (2), and a level average of partner short rates $r_{j,t}$ for the financial star. The weights $\tilde{w}_{cj}$ are renormalized *per component and per quarter* over the partners actually present and eligible, so absent or excluded mass is reabsorbed rather than dropped: rate-less blocs (Switzerland, Norway) leave their partners' `ROWFIN`; Argentina's hyperinflation episode is excluded from partners' `ROWINF` and `ROWFIN` while its real demand stays in `ROWDEM`. (A deflate-by-subtraction rule exists for blocs reporting nominal GDP; since China was re-sourced to a real chain-linked volume index in the June-2026 re-estimate, the set of such blocs is currently empty.)

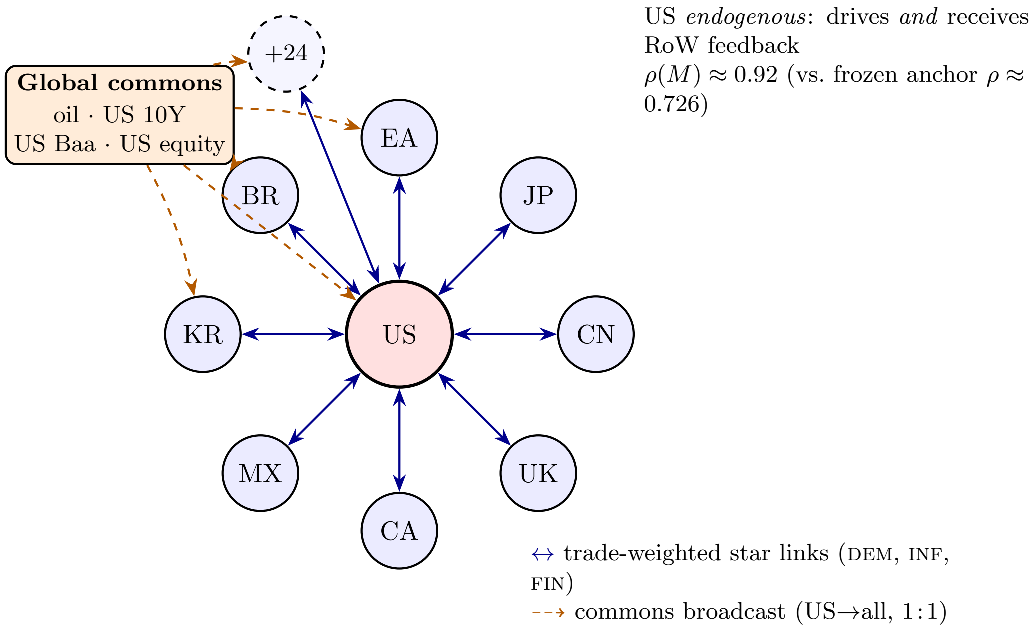

Figure 6: GVAR bloc topology. A central, dominant United States bloc is coupled to a ring of 32 Rest-of-World partner blocs (eight labelled—EA, JP, UK, CA, CN, MX, KR, BR—plus 24 further blocs) through *bidirectional*, trade-weighted "RoW star" links carrying foreign demand (DEM), inflation (INF), and the short rate (FIN). A small global-commons node (oil, the long US rate, US corporate credit, US equity) broadcasts *one-directionally* and identically $(1:1)$ to every bloc. The US is now a *fully endogenous* coupled bloc rather than the classical frozen anchor: it feeds the rest of the world *and* receives feedback, raising the coupling operator's spectral radius from $\rho(M) \approx 0.726$ (frozen-US, RoW-only) to $\rho(M) \approx 0.92$ at the full 33-bloc endogenous-US configuration (the value stored in the shipped `coupling_default.json.gz`; it moves slightly with each refit), both contractive so the fixed point $x = Mx + d$ is well defined (Sections 5.1 and 5.3).

The bilateral weights come from the IMF's International Merchandise Trade Statistics (exports plus imports, 2021–2023 average, euro area as a single counterpart) and ship as one row per coupled bloc — the US row included — each summing to one with a zero diagonal (`weights_24bloc.json`; the file name is historical, the file lists all 33 blocs). The topology is *not* hardcoded: a bloc is a coupled member if and only if its specification carries the three `row`-tagged stars *and* it has a weights row, and the US is treated as endogenous if and only if its own model satisfies the same test. Absent either, the engine falls back to the legacy frozen-anchor solve unchanged.

Time-varying weights (a gated capability). The single fixed-window row above can be replaced, *at estimation time*, by an annual bilateral panel: when the artifact `weights_timevarying.json` is present (and `GVAR_TV_WEIGHTS`$\neq$0), `buildstars` reads each quarter's stars off that quarter's calendar-year row, clamped to the file's span, with a partner absent from a given year assigned weight zero and reabsorbed by the same per-component, per-quarter renormalization as above (`export_bvar_gvar24_loop.m`). The weights then reach the star *history* on which the blocs are estimated, rather than freezing the 2021–2023 average across the whole sample. The staged panel is annual IMTS bilateral trade for 1995–2024 over 39 blocs. The empirical picture it carries is not the textbook monotone story: the US-to-China trade share rises to a 2017 peak (0.176) and then *falls* after the 2018 tariff episode (to 0.119 by 2024), while the US-to-euro-area share does *not* decline but drifts up to 0.178. This remains a gated capability: absent the artifact (or with `GVAR_TV_WEIGHTS`=0) the static row is used everywhere, byte-identical to the shipped models, and the solve-time coupling of Section 5.3 still reads the single current-year row — promoting the time-varying panel to the live solver is a separate step.

5.2 Estimation: an EM fixed point in the completed stars

Before the formal statement, here is the idea in plain terms. To estimate any one country's model you need its *foreign environment* — what its trading partners' demand, inflation, and interest rates are doing. But each partner's path is itself the output of that partner's model, which needs *its* foreign environment, and so on around the world: everything depends on everything, so no country can be estimated on its own and first. The way out is to treat the unknown foreign environment as *missing data* and fill it in by iterating. Start from a rough guess of the foreign environment for every country; estimate each country's model against its guess (filling in the pandemic holes in the same step); then read off from the freshly-estimated countries what the foreign environment *actually* turned out to be, and re-estimate against that. Repeat. Each pass the guesses become more mutually consistent, and the loop stops when the environment each country *assumed* matches the one its partners *produce* — when the guesses have come true. That self-consistent set of country models is the estimate.

A concrete image helps: fifty country desks share one room, each forecasting its own economy but needing its neighbours' forecasts as inputs. Everyone pencils in a rough guess for the neighbours and produces a first forecast; the forecasts are passed back around the room and each desk redoes its own with the updated inputs; the numbers keep circulating until nobody's change any more. This is exactly the expectation–maximization logic of Dempster et al. (1977) with the "rest of the world" — and the pandemic gaps — as the missing data: an **E-step** that fills in the blanks given the current models, and an **M-step** that re-fits the models given the filled-in blanks, iterated to a fixed point. Two things are worth keeping straight. Nothing is accepted or rejected *across* countries: coherence is the loop's stopping rule, not an acceptance test, and the loop settles on one consistent set of paths rather than a distribution of draws. The genuine Bayesian sampling — the simulation smoother that draws the missing cells and the Metropolis step that updates the shrinkage hyperparameters — lives *inside* each country's single fit (Section 3.4), one level below the loop.

The stars cannot be built directly from observed data: the panels are unbalanced (blocs enter the sample at different dates), the pandemic macro cells are deliberately missing (Section 3.4), and several financial series start late. A star built only on complete quarters would truncate the sample to the common balanced span — exactly the information loss the data-augmentation design avoids. The estimator therefore iterates completion and re-estimation to a fixed point in the stars (in the spirit of Dempster et al., 1977):

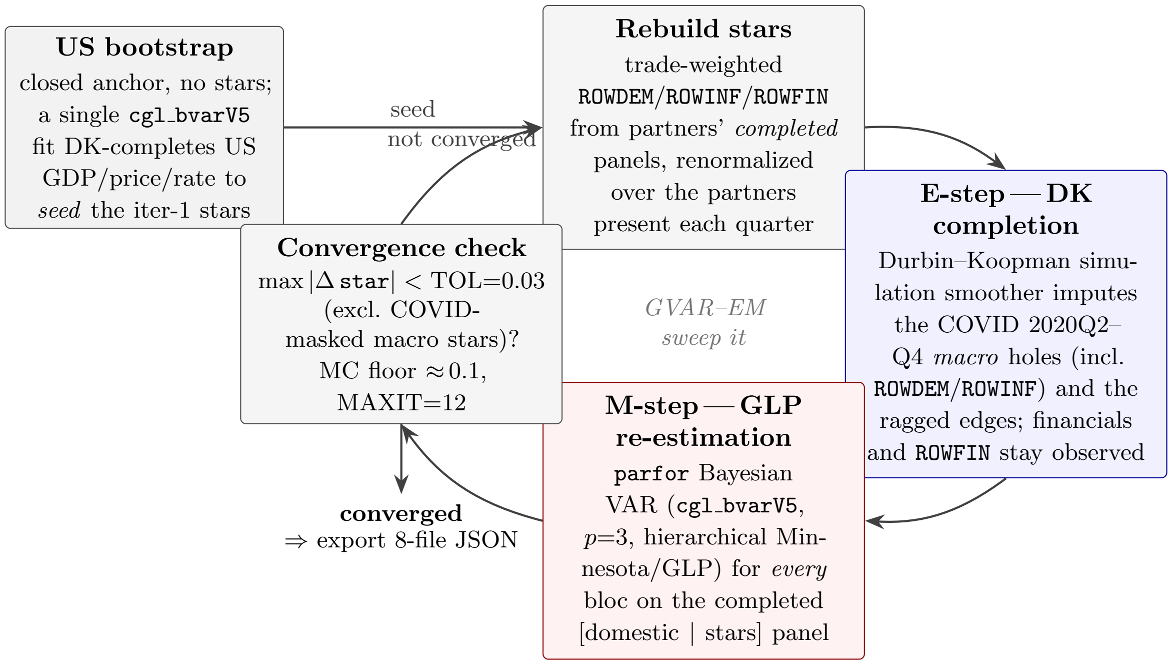

Figure 7: The GVAR-EM estimation loop under pervasive missingness (Giannone et al., 2019; Durbin and Koopman, 2002). A closed US bootstrap (no stars) is estimated once to seed the first trade-weighted foreign variables. Each sweep then (i) rebuilds the `ROWDEM/ROWINF/ROWFIN` stars from partners' completed panels, renormalized over the partners present each quarter, and (ii) re-estimates every bloc by hierarchical-prior Bayesian VAR (Gibbs, run in `parfor`). Within each bloc's single `cgl_bvarV5` pass the **E-step**—a Durbin–Koopman simulation-smoother completion that imputes the COVID macro holes (the fixed 2020Q2–Q4 window, with financial series and `ROWFIN` held observed) and the ragged, unbalanced-panel edges—and the **M-step**—the posterior re-estimation of the VAR coefficients on the completed [domestic | stars] panel—are carried out jointly by the same sampler. The sweep repeats until the foreign components stop moving, max $|\Delta\, \texttt{star}| <$ TOL $= 0.03$ (evaluated after the second sweep and excluding the COVID-masked macro stars, whose cross-iteration churn is Monte-Carlo noise with floor ≈ 0.1), capped at MAXIT $= 12$ iterations.

- **E-step (completion).** For each bloc, one call to the data-augmentation sampler of Section 3.4 on the panel $[\,\text{domestic} \mid \text{stars}\,]$ imputes every missing cell (COVID macro mask included) by the Durbin–Koopman simulation smoother; the posterior-mean completed panel is retained.

- **M-step (re-estimation).** The same call re-estimates the bloc's coefficients under the GLP prior on the completed data; the posterior means $(\hat{\beta}_c, \hat{\Sigma}_c)$ and the thinned draws are what ships.

- **Outer loop.** All blocs' stars are rebuilt via (16) from the *frozen* start-of-sweep completions, the blocs are re-estimated in parallel (a Jacobi sweep, order-independent), and the loop repeats until the stars stop moving: $\max_c \|x_c^{*(m)} - x_c^{*(m-1)}\|_\infty < \textsc{tol} = 0.03$ (COVID-masked macro-star cells excluded from the metric — their cross-iteration churn is imputation Monte-Carlo noise, with an observed floor near 0.13, so convergence in practice is a plateau at the noise floor), capped at MAXIT = 12 sweeps. A one-shot, no-stars US completion seeds the first sweep.

Algorithm 2 states the loop (`pipeline/export_bvar_gvar24_loop.m`). Setting MAXIT = 1 recovers the single-pass mode used for the standalone blocs, whose stars are read off data once. Relative to the classical fixed-weight GVAR — stars built once from observed data on a balanced panel, one OLS pass per country — every departure here serves the unbalanced panel: model-based completion flowing *into* the stars, per-quarter renormalized weights, Bayesian shrinkage with the COVID-as-missing treatment, and the star fixed point itself. The out-of-sample evaluation of Section 12 races exactly this apparatus against its classical ancestor.

Algorithm 2 GVAR-EM estimation loop (`export_bvar_gvar24_loop.m`).

Input: per-bloc observed panels with `NaN` holes; trade weights; eligibility rules; TOL = 0.03, MAXIT = 12; fixed COVID window 2020Q2–Q4.

Output: per-bloc posterior means and draws; converged stars; the eight-file export.

$\triangleright$ Bootstrap. Complete the US panel with no stars (COVID macro cells masked); its posterior-mean completion seeds the first-sweep stars. Under US_ENDOGENOUS the US then joins the swept set with its own `ROW*_US` stars.
for $m = 1, \ldots, \text{MAXIT}$:
    rebuild every receiver's stars (16) from the frozen start-of-sweep completed panels;
    parfor each bloc c: mask the COVID macro cells (domestic macro columns + `ROWDEM`, `ROWINF`; financials and `ROWFIN` stay observed); one `cgl_bvarV5` call $\Rightarrow$ completed panel + $(\hat{\beta}_c, \hat{\Sigma}_c)$ + draws;
    commit the new completions; mc $\leftarrow \max_c \|\Delta x_c^*\|_\infty$ (COVID macro-star cells excluded);
    if $m > 1$ **and** mc $<$ TOL: **break**.
Export each bloc to the eight-file JSON contract.

A two-stage E-step. The imputation inside each M-step (Section 3.4) can be run in either of two ways, and the loop uses both in sequence. During the outer sweeps the Durbin–Koopman completion is carried out with a *fixed companion*: the smoother's state-space matrices are built once from the posterior-mode fit and held across the Gibbs draws (`fixedcomp`= 1 in `estimate_bloc`, `export_bvar_gvar24_loop.m`). With the sampler's random stream reset per call, consecutive sweeps then share their random numbers, so a bloc's star moves only for *systematic* reasons and the Jacobi fixed point contracts to TOL — a run descends, e.g., 14.7 → 1.07 → 0.20 → 0.11 to a plateau at the Monte-Carlo floor. The textbook alternative rebuilds the companion from the current (β, Σ) draw at every Gibbs sweep — exact Tanner–Wong data augmentation (`cgl_bvarV5.m`; `CGL_FIXED_COMPANION`=0) — but re-randomizing the completions each sweep breaks the common-random-number coupling, leaving a star-churn floor that never meets TOL. The coupled estimator therefore uses the fixed-companion sweeps throughout, and *ships the fixed-companion posterior* from the converged outer loop. An optional exact-data-augmentation *polish* pass at the frozen stars is available (`GVAR_POLISH_EXACT`=1) but is *off by default for the coupled system*: exact augmentation marginally raises companion persistence and lifts the coupled operator's spectral radius above one (§5.3), so it would leave pinned-scenario spillovers non-contractive. Standalone single-bloc estimators (the euro members) run exact data augmentation directly, having no outer loop — and no coupled operator — to keep contractive.

5.3 Solve-time coupling: a linear fixed point

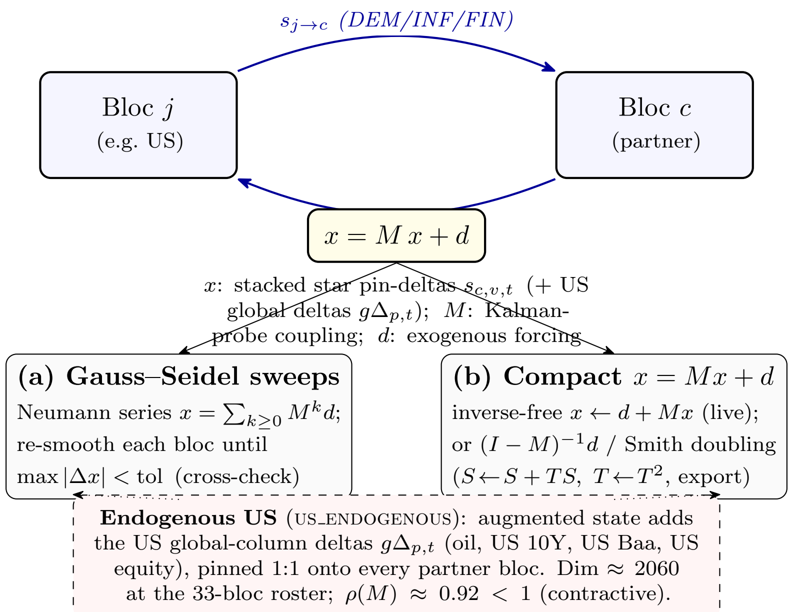

Figure 8: Solve-time cross-country coupling as the affine fixed point $x = M\,x + d$. Two coupled blocs exchange trade-weighted star pin-deltas (DEM/INF/FIN), forming the feedback cycle; the deviation state x stacks these pins (plus the US global-column deltas under an endogenous US). The fixed point admits two equivalent solutions—(a) Gauss–Seidel/Neumann sweeps through the Durbin–Koopman smoother, used as the cross-check, and (b) the compact M-state recursion, solved inverse-free by the Neumann fixed-point $x \leftarrow d + Mx$ on the production path (the explicit $(I-M)^{-1}d$ / Smith doubling are reserved for the small frozen-anchor case and the export-time precompute, since the in-browser $O(\dim^3)$ inverse is prohibitive at dim $\approx$ 2060). With the US endogenous the augmented state has dimension $\approx$ 2060 and spectral radius $\rho(M) \approx 0.92 < 1$, so the iteration contracts and the two routes agree to $\sim 10^{-9}$.

At solve time the coefficients are frozen, and the key observation is that the conditional-forecast operator of Section 4 is *affine in the pinned values for a fixed pin pattern*: the filter and smoother are linear in the data. Each bloc's level forecast can therefore be written

$$x_c \;=\; a_c \;+\; \sum_{v} \sum_{s=0}^{H-1} K_c[v][s]\; \Delta^*_{c,v,s}, \tag{17}$$

where a_c is the bloc's forecast with its *own* pins active and its stars and global columns held at baseline, $\Delta^*_{c,v,s}$ is the deviation pinned on star (or global) column v at horizon s, and the kernel $K_c[v][s] \in \mathbb{R}^{H\times n}$ is the smoothed response to a unit bump of that single cell. The kernels are computed from one filter–smoother pass per bloc by value-independent delta recursions (`affineBlocKernels`, numerically identical to explicit probing to below 10^{-9} and an order of magnitude faster), and they depend only on the bloc's coefficients and its own-pin *pattern* — never on pin values or on partners — so they are cached and reused across every value change of an interactive drag.

The solver never rebuilds star *levels* from partner levels. It pins each receiver's star column at its own baseline plus the trade-weighted partner *deviation*:

$$\Delta^*_{c,\mathrm{DEM},t} = \sum_j \tilde{w}_{cj} \cdot 4\left[\left(x^{\mathrm{scen}}_{j,t} - x^{\mathrm{scen}}_{j,t-1}\right) - \left(x^{\mathrm{base}}_{j,t} - x^{\mathrm{base}}_{j,t-1}\right)\right], \qquad \Delta^*_{c,\mathrm{FIN},t} = \sum_j \tilde{w}_{cj}\left(r^{\mathrm{scen}}_{j,t} - r^{\mathrm{base}}_{j,t}\right), \tag{18}$$

read off the partners' gdp/price/rate columns (at $t = 0$ the shared last in-sample level cancels in the difference). Stacking all star deviations — and, with the US endogenous, the deviations of the four US-determined global columns (Section 5.4) — into a vector x, the deviation system is the linear fixed point

$$x = M\,x + d, \tag{19}$$

where d collects the exogenous forcing (each bloc's own pins evaluated at zero partner deviation, plus any long-run drift, Section 8.1) and M carries three coupling families: star $\leftarrow$ partner star (through the kernels and weights, equations (17)–(18), the US on both sides), the US-global definition rows (a global column's deviation as the US's own response to its star pins), and star $\leftarrow$ global (a partner's star response to its own global pin). At the deployed roster the state dimension is $33\times3\times20+4\times20 = 2060$, and the shipped operator has spectral radius

$$\rho(M) = 0.945 \; < \; 1 \tag{20}$$

(the value stored in the shipped coupling artifact, 0.945494, eigendecomposition-exact — before 2026-07-17 the field read 0.935044 from an under-converged power iteration on a complex dominant pair; it moves with each re-estimate), so the map is contractive, $(I - M)$ is nonsingular, and the fixed point is unique.

The fixed point is solved three equivalent ways, all implemented in `gvarSolve.ts`:

(a) **Gauss–Seidel sweeps**: re-condition each bloc in turn on the freshest partner deviations until the maximum level change falls below 10^{-4} (at most 12 sweeps) — the term-by-term Neumann sum $\sum_k M^k d$, and the reference the other routes are validated against;

(b) **closed form** $x = (I - M)^{-1} d$, using a precomputed inverse (below), followed by one conditional-forecast pass per bloc to reconstruct all free variables;

(c) **inverse-free Neumann iteration** $x \leftarrow d + Mx$ (tolerance 10^{-10} relative, capped at 2000 iterations) when the precomputed inverse is unusable — the in-browser $O(\dim^3)$ inverse at $\dim = 2060$ would take the better part of an hour and is never built live.

The routes agree to 10^{-6} or better in the validation battery. Divergence is *detected, not prevented*: if a future roster or degenerate pin set pushed $\rho(M) \geq 1$, the iterations' convergence flag trips, a console warning fires, and the result is not cached — but a path is still returned (Section 14.2).

The coupling artifact and the model signature. M depends on the models and the pin *pattern*, not the pin values, so for the default (empty) pattern both M and $(I - M)^{-1}$ are precomputed offline — the inverse by a BLAS solve in MATLAB (`app/scripts/invert_coupling.m`; a NumPy fallback ships alongside) — rounded to six significant figures, gzipped (`coupling_default.json.gz`, about 30 MB compressed), and decompressed in the browser at load. Hydrating it reduces the first complete solve from a cold build of many minutes to a few seconds; it is mandatory infrastructure, not an optimization. A stale artifact is refused by a *model signature*: per bloc, the dimensions (n_c, p_c, H_c) and three six-digit fingerprints of the posterior mean — the first equation's intercept $\hat{\beta}_{0,0}$, the far corner of the last lag row $\hat{\beta}_{n_c p_c,\, n_c-1}$ (zero-indexed; recall the intercept is row zero), and the leading residual variance $\hat{\Sigma}_{0,0}$. Any re-estimation flips the signature, and the solver falls back to route (c). A pin on any coupled bloc changes the pattern and likewise bypasses the cached inverse; only the changed bloc's kernels are re-probed.

5.4 Global commons

Four US-determined series are carried by every bloc: the WTI oil price (`GLB_OIL`), the US 10-year yield (`GLB_USLR`), the US Baa corporate yield (`GLB_USBAA`), and US equity (`GLB_USEQ`). In the US model they are ordinary endogenous variables; every other bloc carries the same identifiers (matched by name, never by position) as weakly exogenous columns that the solver pins *by deviation*:

$$\widehat{\mathrm{GLB}}_{c,t} = \mathrm{G\bar{L}B}_{c,t} + \big(\mathrm{GLB}^{\mathrm{scen}}_{\mathrm{US},t} - \mathrm{G\bar{L}B}_{\mathrm{US},t}\big), \tag{21}$$

each bloc's own baseline plus the US deviation. Pinning the US *level* instead would inject the blocs' baseline gaps (tens of 100 ln units) as spurious shocks; the deviation convention gives a zero-scenario no-op exactly and an identical common shock under conditioning. With the US endogenous the global deviations are unknowns of (19) — this is precisely the second and third coupling family — so a partner shock that moves the US moves the commons, which feed back to every other bloc, without double counting. On screen the commons are conditionable on the US and read-only elsewhere.

5.5 Operating the global solve

By default a scenario moves only the economy on screen; partners stay frozen, and a status line says so. Cross-border transmission is explicit and two-tier:

- **Scenario spillovers (GVAR solve)** (left panel, **Cross-border spillovers**) switches on live propagation. While editing, the tool shows the *direct preview* — one cross-country hop through

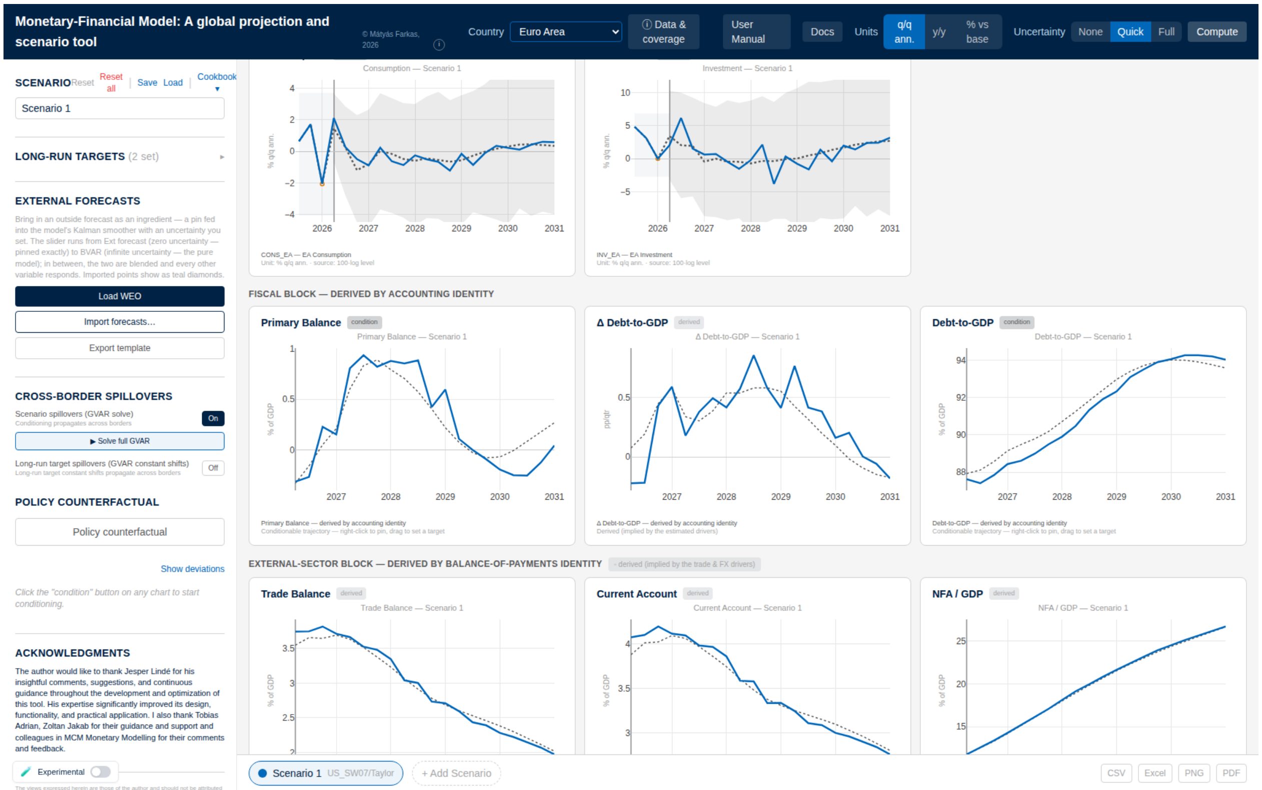

Figure 9: The euro-area view after Solve full GVAR on a US tightening scenario (spillovers On). The conditioning entered in the United States has propagated cross-border: euro-area consumption and investment deviate from baseline (dotted), and the derived satellite blocks — the fiscal block of §6 and the external-sector block of §7 — re-solve on the coupled paths. Switching countries after a full solve always shows the same converged system state.

the cached kernels, no fixed-point feedback, sub-second — flagged by an amber **GVAR DIRECT preview** badge above the grid.

- **Solve full GVAR** (or **Compute**) runs the complete fixed point (19) — all feedback loops, including RoW→US — in a few seconds, and turns the badge blue (**GVAR full solve**), with the sweep count and solve time in the read-out. The full solve reflects the pins at the moment it runs; edit afterwards and press it again.

For small shocks preview and full solve are close (the second-round echoes are smaller than the first hop); for large shocks, or when the *receiving* country is the object of interest, run the full solve before quoting numbers.

Example 3 (A global shock reaching a partner)**.** *Load the **Cookbook** energy-supply recipe on the United States (**Load + restriction**): an oil-price spike with its documented supply-shock signature. Switch the **Country** dropdown to the euro area and set **Units** to **% vs base**: the euro-area GDP line is flat at zero — the shock has not been propagated. Return to the US, press **Solve full GVAR**, and switch back: euro-area GDP now shows the imported hit, at the 2026Q1 vintage building to* *roughly −0.8 percent of the baseline level by the end of the horizon. The difference between the flat line and the solved one is exactly the content of the coupled system: a country-only read cannot see a foreign shock at all.*

5.6 Euro-area members: one doorway

The euro members do not set their own interest rate, and the tool respects this: the area-wide instruments — the ECB policy rate, the short rate, the area corporate yield — are *endogenous* in the EA-aggregate bloc and enter each member's model as *weakly exogenous* columns, alongside the global commons and the member's foreign stars. Shocks reach a member through one doorway, by *deviation transmission*: after a full solve, the EA-aggregate deviation is mapped identifier-by-identifier onto the member's corresponding columns (policy rate onto policy rate, area yield onto area yield, area activity and prices onto the member's rest-of-euro-area aggregates), each pinned at the member's own baseline plus the area deviation, and the member's own estimated dynamics translate the area shock into its GDP, inflation, and yields. A member never updates silently: an amber note above its grid says when a US or EA scenario has not yet been passed through, and **Solve full GVAR** runs the pass.

The reverse direction exists for two national instruments. Pinning a member's ECB-rate column mirrors one-for-one into the area policy rate; and a member sovereign-spread scenario feeds the area long yield by the member's GDP weight,

$$\Delta\mathrm{YIELD10}_{\mathrm{EA},t} = \Delta\mathrm{Bund}_t + w_m\,\Delta\mathrm{spread}_{m,t}, \tag{22}$$

the two channels additive, with w_m the member's share in euro-area GDP (real Eurostat current-price weights, shipped in `ea_gdp_weights.json`; a missing weight suppresses the channel rather than inventing one). An Italian spread scenario thus moves the area yield about forty times more than an Estonian one.

5.7 Outside forecasts across borders

A hard external forecast (trust at 100%) is indistinguishable from a hand pin and rides the cached closed-form path. A *soft* one changes the effective filter gains through (15), which the value-independent kernels of (17) cannot represent; the solver therefore routes any request with a finite trust variance to the Gauss–Seidel path, threading the per-cell variances into every sweep's conditional forecast. Exact but slower — a few seconds across the panel — so an uncertain import propagates cross-border only on **Solve full GVAR** or **Compute**, never live while typing; the viewed economy still updates instantly.

5.8 Per-draw coupled bands

With the **Experimental** switch on, **Compute** re-solves the whole coupled system once per posterior draw (N_b of the D shipped draws, selected by deterministic stratification; parameter uncertainty only, no innovation shocks; only the viewed bloc's model and pins are rebuilt per draw, partners held at their means) and reports the empirical 16th/84th percentiles of the resulting conditional cloud. This band pinches naturally at every pin — and can collapse too tightly when only a point or two is pinned, which is why it is opt-in (Remark 2). Each draw runs the full Gauss–Seidel budget (tolerance 10^{-4}, 12 sweeps), so **Full** on a coupled bloc is the slowest computation in the tool.

5.9 Identification: what is assumed where

The separable estimate-then-couple design rests on stated assumptions. *Weak exogeneity of the stars*: each bloc is estimated conditional on its trade-weighted foreign aggregates, valid when no single partner dominates the weights — the granularity condition of Chudik and Pesaran (2016); the per-component renormalization is designed to preserve it. The EM loop relaxes the one-shot version of this assumption by iterating the conditioning information to mutual consistency, and the endogenous US removes the frozen-dominant-unit asymmetry altogether; what keeps the relaxed system well posed is the contraction (20). *Euro-area monetary identification*: policy is set for the area as a whole — endogenous in the aggregate, predetermined for each member — so member-level monetary counterfactuals are conditional scenarios on the area instrument, not independent national experiments. *The scenario responses are conditional projections*: a multi-period pin bundles the direct conditioning with the coupled feedback and is not a structural impulse response (Section 12 keeps this distinction explicit).

6 Public finances

Twenty-three economies carry a fiscal block: the six majors (US, euro area, Japan, UK, Canada, China) and the seventeen euro-area member states (Appendix B). The design follows the stock-flow-consistent division of labour shared by the CBO's budget mechanics and the IMF's financial-programming and debt-sustainability frameworks (Escolano, 2010; International Monetary Fund, 2013): the BVAR forecasts a small set of fiscal *flow drivers* jointly with the macroeconomy, while the debt stock and the headline balances are *derived* by the budget identity. Debt is never a free VAR variable — the identity has a near-unit autoregressive root $(1+i)/(1+g)$, and small coefficient errors compound into divergent paths; imposing the identity by construction is the discipline the frameworks demand (`app/src/lib/compute/fiscalAccounting.ts`).

6.1 Estimated drivers

Four drivers enter the VAR as ordinary `lin` variables and inherit the full estimation machinery of Sections 3–4:

- `FREV_GDP`, `FPEXP_GDP`: general-government revenue and primary (non-interest) expenditure, percent of GDP, annual ratios;

- `EFFI`: the *effective* interest rate — net interest over lagged gross debt, percent per annum. The stock reprices only as maturing securities refinance, so this rate lags market yields and is estimated separately from them;

- `SFA`: the stock-flow adjustment, percent of GDP — the deficit–debt discrepancy (intragovernmental holdings, valuation effects, other below-the-line operations), constructed on history as the *exact* residual that closes the recursion below, with the pandemic quarters imputed rather than observed.

The three GDP-ratios carry the stationary `WN` own-lag prior of Section 3.2 — they are mean-reverting shares, and the random-walk center would let the projected ratio drift at the long end; under `WN` the US revenue ratio, for example, reverts to its in-sample mean of about $17\frac{1}{2}$ percent instead of freezing at its end-of-sample value. `EFFI` keeps the `RW` prior: a stock-average rate is genuinely persistent. Each fiscal bloc ships a `fiscal_meta.json` recording the column indices the recursion reads (`idx` = {`gdp`, `deflator`, `revenue`, `primaryExp`, `effRate`, `sfa`}), the seed debt ratio d_0, and the structural constant `sfaDefault` (the long-run SFA mean; e.g. US 1.75, EA 0.33, JP 2.23, UK 3.15, CA 3.19, CN 4.02 percent of GDP).

6.2 The debt recursion

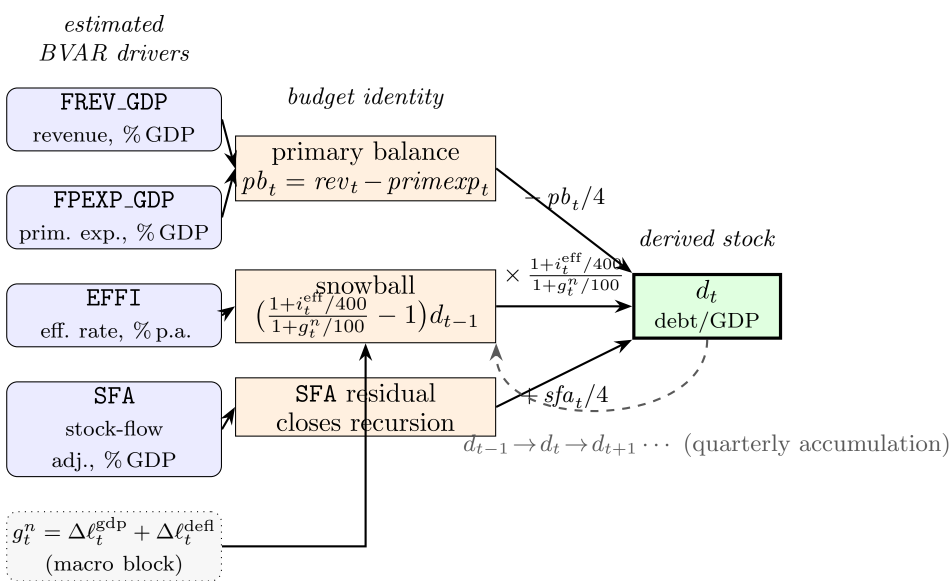

Figure 10: Fiscal stock-flow schematic. The four estimated BVAR drivers (`FREV_GDP`, `FPEXP_GDP`, `EFFI`, `SFA`; Section 6.1) feed the budget identity: revenue and primary expenditure form the primary balance pb_t (§6.1); the effective rate `EFFI` and nominal growth g_t^n (23) form the interest–growth *snowball* (25); and the estimated `SFA` enters as the residual that closes the recursion. These flows accumulate into the derived debt-to-GDP stock d_t by the quarterized roll-forward $d_t = d_{t-1}\,(1+i_t^{\text{eff}}/400)/(1+g_t^n/100) - pb_t/4 + sfa_t/4$ (24), with the lagged stock d_{t-1} fed back each quarter (dashed). Debt is *derived*, never estimated (`computeFiscalBlock`, Section 6.2).

Let d_t denote debt-to-GDP in percent, i_t^{eff} the effective rate in percent per annum, and g_t^n the *quarterly* nominal growth rate in decimal, read off the level-space forecast as

$$g_t^n \;=\; \tfrac{1}{100}\Big[\big(\ell_t^{\text{gdp}} - \ell_{t-1}^{\text{gdp}}\big) + \big(\ell_t^{\text{defl}} - \ell_{t-1}^{\text{defl}}\big)\Big], \tag{23}$$

the sum of the real-GDP and deflator log differences in the stored $100 \ln$ space, divided by 100 (a bloc whose GDP series were nominal would omit the deflator term via the `gdpIsNominal` switch; no shipped bloc currently uses it). With the primary balance $pb_t = rev_t - pexp_t$ (percent of GDP, annual, positive = surplus), the debt stock rolls forward one quarter as

$$d_t \;=\; d_{t-1}\,\frac{1 + i_t^{\text{eff}}/400}{1 + g_t^n} \;-\; \frac{pb_t}{4} \;+\; \frac{\overline{sfa}}{4}, \qquad d_{-1} \equiv d_0, \tag{24}$$

exactly as coded: the annual rate quarterized by 400, the quarterly decimal growth entering directly, and one quarter's worth of the annual flow ratios accruing per step. The quarterly convention is pinned by a back-cast test that reconstructs the observed debt ratio from realized drivers. Two derived flows complete the identity: the interest bill $int_t = i_t^{\text{eff}}\, d_{t-1}/100$ (annual, percent of GDP) and the deficit $def_t = int_t - pb_t$. Subtracting d_{t-1} from (24) isolates the automatic *snowball* from the discretionary balance,

$$\Delta d_t = \underbrace{\Big(\frac{1 + i_t^{\text{eff}}/400}{1 + g_t^n} - 1\Big) d_{t-1}}_{\text{snowball (interest–growth)}} - \frac{pb_t}{4} + \frac{\overline{sfa}}{4} \approx \frac{i_t^{\text{eff}}/400 - g_t^n}{1 + g_t^n}\, d_{t-1} - \frac{pb_t}{4} + \frac{\overline{sfa}}{4}, \tag{25}$$

the quarterized $(i - g)$ form: debt rises mechanically whenever the quarterized effective rate exceeds quarterly nominal growth.

Remark 4 (The SFA in the roll-forward is the structural constant)*.* Equation (24) deliberately uses the constant $\overline{sfa}$ = `sfaDefault`, *not* the forecast `SFA` column, even though the column is estimated and displayed with its own chart, dynamics, and bands. The forecast column responds to everything else in the VAR — including long-run anchors' cross-responses — and can be dragged to spurious extremes that would distort the derived debt path; the long-run mean is the economically correct closure for a projection. The `SFA` *chart* shows the model's forecast of the discrepancy; the debt *arithmetic* uses the stable constant. Conditioning the SFA chart is still meaningful — it moves the rest of the system through the estimated covariances — but it does not enter (24).

The recursion (23)–(25) is `computeFiscalBlock`, the only fiscal recursion wired to the charts: it produces the baseline, scenario, and counterfactual debt paths and, run per posterior draw, the fiscal bands. It is *passive*: lagged debt enters every term, so a single forward sweep is exact, and there is no debt feedback onto the primary balance or the rate — the displayed balance is exactly $rev_t - pexp_t$ and the displayed rate exactly the BVAR forecast. Three derived charts are displayed, in a row labelled **Fiscal block — derived by accounting identity**: the **Primary Balance**, the per-quarter change **Δ Debt-to-GDP** (a pure identity output, not conditionable), and **Debt-to-**

GDP. The remaining identity outputs (interest, deficit, overall balance, snowball) are computed internally but not charted.

Remark 5 (Timing of the debt ratio). The denominator convention follows GFSM 2014 and the Eurostat quarterly Maastricht series: d_t is the *end-of-quarter* gross debt stock divided by *contemporaneous rolling-annual* nominal GDP, $d_t = D_t / \sum_{s=t-3}^{t} Y_s$ — the four-quarter sum *ending in* quarter t, not lagged GDP. All fiscal ratios (`FREV_GDP`, `FPEXP_GDP`, debt/GDP) use the same contemporaneous four-quarter denominator, and d_0 is seeded at the last observed debt stock over the four quarters ending at that observation. The one deliberately *lagged* object is the effective rate, $\texttt{EFFI}_t = 400\, int_t / D_{t-1}$: interest accrues on the pre-existing stock, the financial-programming convention. In the roll-forward (24) the denominator's growth enters through the factor $(1 + i_t^{\text{eff}}/400)/(1 + g_t^n/100)$ with g_t^n the quarterly nominal-GDP growth of the same quarter; in sample, any wedge between quarterly growth and the rolling-sum growth is absorbed into the estimated `SFA` residual by construction, so the identity holds to machine precision on history.

Remark 6 (Coupon-lumpy blocs: even interest distribution). For eight blocs whose recorded interest payments follow an annual or semiannual coupon calendar — `EVEN_INTEREST` = {CL, CO, CR, ID, IL, MX, TR, ZA}, identified by a systematic within-year pattern in the raw interest flows (calendar-$R^2 > 0.5$ with coefficient of variation > 0.25, or near-perfect quarterly alternation) — the lumpy coupon would otherwise inject a mechanical sawtooth into $\texttt{EFFI}_t = 400\, int_t / D_{t-1}$. Before estimation the four quarterly interest payments *within each complete calendar year* are replaced by their common mean, so the annual interest total is preserved *exactly* (a trailing partial year uses the trailing-four-quarter mean); `EFFI` then measures the *accrual* borrowing cost rather than the cash-settlement spike. Primary expenditure keeps the *raw* interest — total minus interest, $te - int$, stays true non-interest spending — and the exact debt-identity residual `SFA` absorbs the genuine cash-versus-accrual timing, which is why these blocs also carry the seasonal dummies of Section 3.3. The debt roll-forward (24) still closes to machine precision ($< 10^{-14}$) (`pipeline/build_fiscal22_columns.py`, `even_distribute_years`). Empirically the evening removes the `EFFI` sawtooth (mean absolute quarterly change falls from ~ 5 to $\lesssim 0.25$ points for CL, MX, ZA) and moves the near-Nyquist companion root that drove it off -1 (e.g. ZA from -0.97 to -0.63).

6.3 Targeting a debt or balance path

The recursion also runs backward. Given a target debt path $\{d_t^{\text{tgt}}\}$, the primary balance required at each quarter is the exact inverse of (24),

$$pb_t \;=\; 4\left[\frac{d_{t-1}^{\text{tgt}} + int_t/4}{1 + g_t^n} \;-\; d_t^{\text{tgt}}\right] \;+\; \overline{sfa}, \tag{26}$$

with the lag taken along the target path and $int_t = i_t^{\text{eff}}\, d_{t-1}^{\text{tgt}}/100$ (`requiredPrimaryBalance`). The tool evaluates the effective rate and growth on the unconditional forecast, computes (26), and pins the combination $rev_t - pexp_t = pb_t$ as a linear-combination constraint (Section 4.2); the smoother

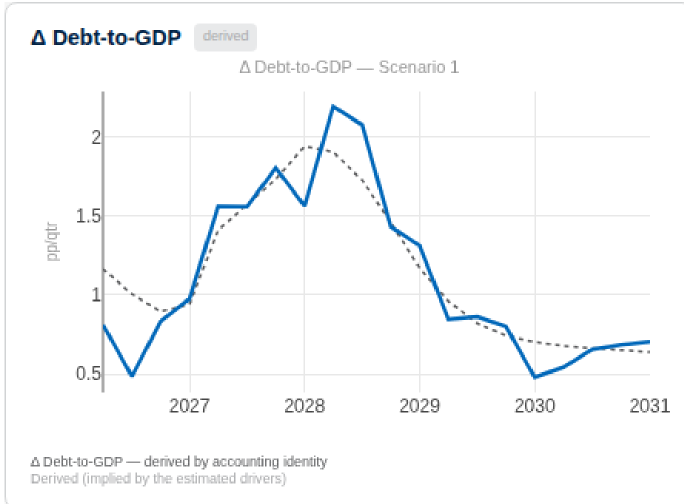

Figure 11: The derived debt block under the tightening scenario: the quarterly change in debt-to-GDP (solid) rises above baseline (dotted) as the snowball term absorbs the higher effective rate, then reverts. Debt panels are conditionable: right-click pins a point, dragging sets a target path, and the smoother back-solves the primary balance that delivers it (§6.3).

finds the most plausible revenue/expenditure mix.

Example 4 (Drawing a debt path)**.** *On a fiscal economy, right-click the* ***Debt-to-GDP*** *chart where the path should pass — on this chart the dragged value is the absolute target,* 100 *meaning* 100 *percent of GDP — and shape the trajectory; right-click a quarter to pin or release it. A* ***Primary Balance*** *target set the same way is hit* exactly *(it is one linear constraint). A* ***Debt-to-GDP*** *target is hit* approximately*: the inversion* (26) *conditions on the model's unconditional rate and growth, which themselves respond a little to the implied consolidation, and the tool does not iterate on that feedback — expect a miss of up to a point or so of GDP over the horizon. For an exact decimal the instrument is the balance target; for a coherent consolidation scenario the debt target is the natural handle. Note that conditioning* any *fiscal variable implicitly conditions the derived row — a note next to the block header says so.*

Remark 7 (Fiscal bands are identity-consistent, and re-centred)*.* Each band draw pushes its own simulated drivers through (24), so the debt band widens exactly as fast as the underlying budget-and-growth uncertainty implies. Because the roll-forward compounds, the per-draw debt distribution is right-skewed: its median sits below the debt path computed at the mean drivers, which would leave the central line hugging the band's upper edge. At render the band is therefore re-centred — each quarter's band midpoint is shifted onto the displayed path, preserving the band's width and asymmetry (the fan-chart convention "central projection ± uncertainty"); at actively pinned quarters the band collapses onto the line. The same treatment applies to the external position below.

7 The external accounts

Three economies — the United States, the euro area, and China — carry an external block that derives the trade balance, the current account, and the net foreign asset (NFA) position from the estimated trade, price, and exchange-rate variables by the balance-of-payments identity (`app/src/lib/compute/externalAccounting.ts`). All three charts are read-only (**derived** badges): to move them, condition the drivers — exports, imports, the exchange rate, growth — and the external row follows.

With b_t the NFA-to-GDP ratio in percent, seeded at the latest published net international investment position ($b_{-1} \equiv nfa_0$: US -66.75, EA $+11.0$, CN $+20.35$ percent of GDP at the current vintage), the per-quarter identity as coded is

$$tb_t = 100\,\frac{X_t - M_t}{Y_t}, \qquad ca_t = tb_t + \frac{i^{\text{NFA}}\,b_{t-1}}{100}, \tag{27}$$

$$val_t = -\,\phi^{\text{FX}}\,b_{t-1}\,\frac{\Delta\ell_t^{\text{FX}}}{100}, \qquad b_t = \frac{b_{t-1}}{1+g_t^n} \;+\; \frac{ca_t}{4} \;+\; val_t, \tag{28}$$

where X_t, M_t, Y_t are the exponentiated export, import, and GDP levels (for the US and the euro area a real-trade-to-real-GDP ratio, validated against the published balance-of-payments ratios; for China a deflated real index with the shipped scale factors, monthly trade aggregated to quarterly), $\Delta\ell_t^{\text{FX}}$ is the quarterly change of the $100\ln$ effective exchange rate (positive = home appreciation, which shrinks the home-currency value of net foreign assets — hence the sign), and g_t^n is quarterly nominal growth as in (23). The stock update divides the inherited ratio by $(1+g_t^n)$ — the automatic denominator effect — adds one quarter of the annual current account, and adds the valuation term; the return on the stock enters through the income term of ca_t, computed on the undiluted lagged ratio (the second-order cross-terms this drops are noted in the code header).

Two inputs are stated modelling assumptions, not estimates, and are flagged as such in the shipped metadata: the return on net foreign assets $i^{\text{NFA}} = 3$ percent per annum, and the share of the position exposed to the exchange rate $\phi^{\text{FX}} = 0.4$. Treat the *level* of the income and valuation terms as indicative and their direction as the reliable part.

8 Long-run anchors

The long-horizon behaviour of the forecasts is inherited from the estimated VAR, whose level dynamics carry roots on or near the unit circle; the far end of a projection can therefore drift away from any value the user regards as a credible steady state — a two-percent inflation objective, a potential growth rate, a stable revenue ratio. The **Long-run targets** panel lets the user pin the terminal forecast of a chosen variable to a number, exactly and instantly. The mechanism is a deterministic counterfactual on the companion drift in the spirit of Villani (2009), implemented in `app/src/lib/compute/driftSetter.ts`.

The shipped consensus set. The tool ships a reviewed default anchor set (`default_anchors.json`; July-2026 transition-default revision: 35 anchors across 28 blocs) tuned under the now-default steady-state transition. Because that path phases the adjustment in rather than shifting the whole path, the set is anchored to *official* long-run values wherever the model shows a genuine terminal gap — central-bank inflation targets throughout (euro-area members at the ECB's two per cent, national targets elsewhere), medium-term potential growth where the raw path is clearly off, and the effective-rate / revenue / expenditure trio on fiscal blocs. This is a real gain of the transition shape: sixteen of these official-target anchors (Czechia, Norway, Portugal, Thailand, Russia, Sweden and more, all to two-to-four per cent) are rejected as harmful under the rigid shift, which would jump their near term, and all are clean under the transition. Anchors are admitted only if the sensitivity clears the guard of §8.1, and the shipped set is verified by the smell test (`anchor_smell.mjs`) and locked by a panel-wide non-explosiveness regression. Two guards still bite — both shape-independent, being properties of the estimated steady state rather than the path: a target that lands within the redundancy threshold of the model's own terminal is dropped, and one that would displace an unanchored headline variable by more than a point at the terminal is dropped or co-anchored. The euro-area primary-expenditure anchor stays removed for the second reason (its cross-drag drives the euro-area policy-rate terminal negative). A handful of high-inflation emerging markets — Colombia, Hungary, Türkiye, Israel, Malaysia — are shipped model-driven and flagged: their estimates link inflation and growth so tightly that pinning one drags the other several points at the steady state, which no univariate anchor and no choice of path can repair; these await re-sourcing or a joint re-estimation. Argentina (hyperinflation decay) and Japan (anchors disabled) also ship unanchored.

8.1 The drift setter

Write the deterministic part of the companion recursion (6) as $Y_{T+h} = T^h Y_T + S_h\, c_\alpha$ with

$$S_h = \sum_{k=0}^{h-1} T^k, \tag{29}$$

the partial sum of companion powers, built once per bloc (H matrix products, cached). A shift δc of the drift moves the forecast by exactly $\Delta Y_{T+h} = S_h\, \delta c$ — an algebraic identity, verified against the iterated recursion to 10^{-11}. We deliberately avoid the closed-form steady state $(I - \sum_\ell B_\ell)^{-1} c$: for near-unit-root level VARs that matrix is near-singular and the "steady state" a meaningless number of order 10^5, whereas the finite-horizon operator (29) is well posed. A target on variable j shifts only j's own equation, $\delta c = e_j\, \delta c_j$, and the coherent system-wide response is the j-th column, $\Delta Y_{T+h} = S_h[:, j]\, \delta c_j$. The scalar is chosen so the *displayed* terminal forecast lands on the target $\psi_j^\star$:

$$\delta c_j = \frac{\psi_j^\star - \psi_j^0}{\mathrm{sens}_j}, \qquad \mathrm{sens}_j = \begin{cases} 4\big(S_H[j,j] - S_{H-1}[j,j]\big), & \texttt{log}\ \text{variable}, \\ S_H[j,j], & \texttt{lin}\ \text{variable}, \end{cases} \tag{30}$$

with ψ_j^0 the current terminal display forecast — the slider's default position, so an unmoved slider is a byte-exact no-op. A variable whose own sensitivity is below the guard $|\text{sens}_j| < 0.3$ has its slider disabled ("near-singular response"): forcing a target there would require an explosive drift.

Two paths to the target: steady-state transition (default) or rigid shift. Equation (30) fixes *how far* the terminal must move; a second choice — the **Path to target** toggle — fixes *how the adjustment is distributed across the horizon.* The constant-drift response $\Delta Y_{T+h} = S_h[:, j]\,\delta c_j$ places the *entire* effect from the first forecast quarter: because $S_1 = I$, a growth variable's displayed deviation at $h = 1$ is $4\,\delta c_j$ — the *full* terminal gap, applied at once and held flat. This **rigid shift** is exact at the terminal and band-coherent, but it *jumps* the near term at the origin and, for a large gap, can manufacture a near-term contraction or deflation. The shipped default is instead a **steady-state transition**: the same terminal δc_j is phased in along the anchored variable's own estimated convergence profile

$$w_j(h) = \frac{S_h[j, j]}{S_H[j, j]}, \qquad w_j(1) \approx \tfrac{1}{S_H[j,j]} \approx 0, \quad w_j(H) = 1, \tag{31}$$

so the near term stays data-consistent (no origin jump) and the forecast *glides* to the target by the end of the horizon at the model's own mean-reversion speed. This is the reactive, no-re-estimation counterpart of a steady-state (mean-adjusted) prior in the spirit of Villani (2009): rather than translating the mean path by a constant, it lets the estimated dynamics carry the forecast from the observed edge to the long-run target. It is finite-horizon by construction — the identity $I - T^h = (I - T)\,S_h$ shows w_j is exactly the normalised $(I - T^h)$ operator — so it never touches the singular level steady state $(I - \sum_\ell B_\ell)^{-1}c$ that motivated (29). The terminal is identical to the rigid shift, so anchored targets are hit under either path; only the near-to-medium term differs. Each horizon is still a rigid translation, so band widths remain invariant (Remark 3). Cross-variable drag phases in at the source anchor's rate applied to the exact terminal cross-response.

Several targets: per-variable or joint. With two or more targets the cross-responses interfere — a drift on equation j moves every other anchored terminal through $S_H[k, j]$. The panel exposes a **Multiple-target solve** toggle:

- **Per-variable** (the default): each δc_k from its own diagonal sensitivity (30). Stable and bounded, but approximate — it ignores the cross-responses, so with several targets each terminal lands slightly off its stated value.
- **Joint**: solve the $K \times K$ display-unit system $M^{\text{lr}}\,\delta c = \psi^\star - \psi^0$, whose diagonal is exactly (30), by Gaussian elimination with partial pivoting. Every anchored terminal then lands exactly — but a near-collinear target set (say two anchored interest rates the dynamics cannot separate in the long run) makes the system ill-conditioned: the exact solution buys terminal precision with huge offsetting drifts (measured up to $|\delta c| \approx 459$ where the diagonal needed ≈ 11) that wreck the path elsewhere. A singular system falls back to per-variable numerically; an

ill-conditioned but solvable one is returned and *flagged* (the joint drift exceeding twice the per-variable maximum trips the flag), and the user is expected to switch back.

Per-variable is the shipped default. The near-term-implausibility that once accompanied a large gap — at the 2026Q1 origin, anchoring the US deflator (model terminal 4.3 against the 2.0 target) once displayed a -2.1 point disinflation shock at the first forecast quarter — was the signature of the *rigid shift*; the steady-state transition of (31) removes it, gliding the same series from 3.5 smoothly down to 2.0 with a first-quarter move of under a tenth of a point. The two toggles are orthogonal: the solve mode (per-variable / joint) sets *how far* each terminal moves; the path (transition / rigid) sets *how the move is phased in*.

Bands, fiscal forwarding, and visibility. The response ΔY is added identically to the central path and to every shipped band quantile — a rigid translation, widths provably invariant (Remark 3) — and, in level space, forwarded into the fiscal recursion (24), so the debt path responds coherently: a real-growth target owns the direction of the debt ratio through the $(i - g)$ snowball, which is why its row carries the tag **sets debt path**. Judgment is flagged: the affected chart title turns amber with a **judgement** badge stating target and estimate, and the grid-level status line counts the anchored variables. Every control starts at the model estimate, and clearing all targets restores the raw model byte-exactly.

Consensus preset. Blocs with publicly defensible long-run values carry an **Apply consensus anchors** preset — deliberately minimal (35 anchors across 28 economies at the July-2026 transition revision): potential growth and the inflation objective per economy, plus the effective-rate/revenue/expenditure trio for fiscal blocs; never policy rates, unemployment, or asset prices, whose collinear anchors would ill-condition the joint system. Each kept anchor was individually verified — under the shipped transition path — to fix a genuine terminal bias without displacing another headline variable. Where a preset ships it is on by default and fully reversible (**Clear**); the model-driven-and-flagged economies of the preceding paragraph ship none and stay entirely model-driven.

On a coupled bloc the target is marginal. The drift setter operates on one bloc's companion. On a GVAR economy it shifts that bloc's own long run; partners are not re-solved unless the second spillover switch, **Long-run target spillovers (GVAR constant shifts)**, is on. That channel is instant — the drift deviation enters the coupled system only through the forcing term d of (19), leaving M and its precomputed inverse untouched, and the receiving country's banner turns amber with **+ spillover from ...** naming the origin — judgment stays visible across borders.

8.2 The in-prior long-run prior

A second, offline mechanism exists: the "prior for the long run" (PLR) of Giannone et al. (2019), which augments the prior of Section 3.2 with dummy observations shrinking a set of long-run linear combinations Hy toward non-cointegration. It is a proper Bayesian prior — it changes the posterior and requires re-estimation — and is therefore part of the estimation pipeline, not the browser. A US head-to-head (run on an earlier, pre-expansion $n = 36$ US specification; the deployed bloc is $n = 40$) found the PLR acts mainly on the *bands*, tightening the GDP-growth 68% band by about 7 percent at unit tightness, while leaving the central long-run path essentially unchanged — the appropriate instrument for principled band tightening, not for pinning a terminal to a chosen number. The two mechanisms are complementary: to pin a long run interactively, use the drift setter; for disciplined bands, the PLR.

Remark 8 (A retired mechanism). An earlier release pinned long runs by entropic tilting (Robertson et al., 2005) — reweighting the posterior draws so the weighted terminal moment hits the target. It degenerated in practice: as the target moved away from the draw cloud the effective sample size $1/\sum_d w_d^2$ collapsed toward one (a joint 13-constraint tilt reached ESS $\to 1$ at $D = 500$), leaving the anchored band resting on a handful of draws precisely where the user most wanted a firm answer. The drift setter replaced it wholesale — deterministic, exact at any target, instant — and the tilt is no longer wired to any interface control (its module remains in the repository, exercised only by its unit test).

9 Policy counterfactuals

A scenario asks what happens if the data come in differently; a policy counterfactual asks what a *deliberate policy change* would cause. The distinction is identification: the smoother's response to a pinned rate path is the estimated historical co-movement, which cannot tell a policy surprise from a response to something else. The counterfactual layer therefore borrows structure from a DSGE model of the user's choice, drawn from the Macroeconomic Model Data Base (MMB; Wieland et al., 2012, 2016), and overlays its identified responses on the scenario.

9.1 Impulse responses and shock inversion

The MMB catalogue in the tool lists 169 models; pre-computed impulse responses ship for 155 of them, each under up to nine interest-rate rules (Taylor, Smets–Wouters, CEE, Gerdesmeier–Roffia, Levin–Wieland–Williams, two Orphanides–Wieland variants, Coenen, Christiano–Motto–Rostagno). A model–rule pair is one JSON file of responses to a monetary (and, where the model has a fiscal sector, a government-spending) shock, with the rule embedded in the model's equilibrium when the responses were generated — the tool never evaluates a policy rule at run time. Rules without computed responses for the selected model are greyed out.

Let $h_r(s)$ denote the interest-rate response at horizon s to a unit monetary shock. Given a user rate adjustment Δr_t on a set of *constrained* quarters, the tool solves for the shock sequence by forward substitution on the lower-triangular Toeplitz system — a new shock only where the user pinned, zero elsewhere:

$$\varepsilon_t = \begin{cases} \dfrac{\Delta r_t - \sum_{j<t} h_r(t-j)\,\varepsilon_j}{h_r(0)}, & t \text{ constrained,} \\ 0, & \text{otherwise,} \end{cases} \tag{32}$$

so at unconstrained quarters the rate follows the decaying response of the earlier shocks — the model's rule, not a snap-back. The implied inflation and output responses are the convolutions $\Delta\pi_t = \sum_{j\le t} h_\pi(t-j)\,\varepsilon_j$ and $\Delta y_t = \sum_{j\le t} h_y(t-j)\,\varepsilon_j$; a parallel inversion runs on the fiscal shock for the spending sliders.

9.2 Re-conditioning and the response modes

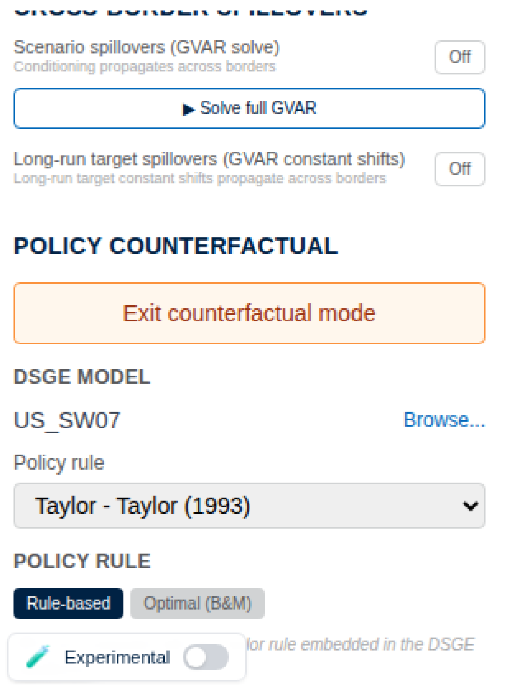

Figure 12: The counterfactual control rail: entering counterfactual mode selects the DSGE model (here US_SW07) and the policy rule — a named rule (Taylor 1993) or the optimal rule of §9.4 — whose impulse responses are layered on the BVAR scenario as described in §9.1.

The counterfactual is not an additive overlay of every variable: the DSGE deltas are imposed on the *key* variables and the BVAR fills in the rest. The **Macro response** selector offers two modes:

- **DSGE/Taylor-rule (IRF-consistent)** (default): the rate is pinned at scenario plus Δr, inflation and output at scenario plus the DSGE deltas; all other variables are re-conditioned through the smoother, so the responses are consistent with both the identified transmission and the estimated covariances. One convention subtlety is handled per variable: the MMB inflation response is a *rate* (cumulated into the 100 ln price level, $\sum_{s\le t} \Delta\pi_s/4$), the output-gap response a percent *level* deviation (applied directly, not cumulated), and a rate response an identity.

- **BVAR-endogenous (let adjust)**: only the rate path is pinned; everything else responds through the estimated co-movements, exactly as in a scenario. The two modes answer different questions — "what does this theory say the move would do" versus "what usually accompanies such a path" — and their disagreement is informative. If the selected pair has no computed responses the tool falls back to this mode and says so in an amber notice.

In both modes the displayed counterfactual is the scenario plus the smoother delta between the counterfactual pass and a reference pass with zero adjustment, converted to display units by the conventions of Section 4.3; a zero adjustment reproduces the scenario exactly.

9.3 Reading the result

The counterfactual draws as a dashed line in a distinct rust colour on every chart; the scenario line stays put. To isolate the policy effect, set **Units** to **% vs base**; in that view the **Level | Rate (IRF)** toggle selects between the cumulative effect on the *level* of each variable (the "% chg." reporting convention) and the quarter-by-quarter response in native units — the latter reproduces the MMB's own IRF plots exactly, since differencing the cumulated log-level deviation and scaling by four recovers the rate response. Uncertainty bands, when computed, belong to the scenario; the counterfactual is a single model-implied path.

Remark 9 (The rule choice is a first-order magnitude decision)*.* The same one-quarter rate move can have an order-of-magnitude different effect across rules. In the shipped responses of the default US model (`US_SW07`, Smets and Wouters, 2007), the peak inflation response to a unit monetary shock under the model's own Smets–Wouters rule is about 32 times the response under the Taylor rule, the output response about 6 times. Treat pass-through magnitudes as model-and-rule dependent, never as a fact; flipping the model while the scenario stays put (each scenario tab carries its own model and rule) is the cheapest robustness check in the tool.

9.4 Optimal policy

Instead of hand-crafting the path, **Optimal (B&M)** computes it from the same sufficient statistics, following Barnichon and Mesters (2023): choose the shock sequence ε minimizing the quadratic loss

$$\mathcal{L} = \sum_{t=1}^{H} \Big[\lambda_\pi \big(\pi_t - \pi^\star\big)^2 + \lambda_y \big(y_t - y^\star\big)^2 \Big] \tag{33}$$

subject to the linear IRF mapping. Stacking the weighted Toeplitz operators, the minimizer is the Tikhonov-regularized least squares

$$\varepsilon^\star = \big(\tilde{\mathcal{H}}'\tilde{\mathcal{H}} + \alpha I\big)^{-1} \tilde{\mathcal{H}}'\,\tilde{b}, \qquad \tilde{\mathcal{H}} = \begin{pmatrix} \sqrt{\lambda_\pi}\,\mathcal{H}_\pi \\ \sqrt{\lambda_y}\,\mathcal{H}_y \end{pmatrix}, \quad \alpha = 10^{-6}, \tag{34}$$

with the implied rate path $\Delta r^\star = \mathcal{H}_r\,\varepsilon^\star$ handed to the IRF-consistent mode. One slider sets the preference weights ($\lambda_\pi = 5w$, $\lambda_y = 5(1-w)$); the $\pi^\star$ and $y^\star$ sliders set the targets. Unlike a hand-set adjustment, the optimal path pins *every* quarter — it is a full plan, not a one-off surprise.

A fiscal lever completes the block: twelve government-spending sliders scale the model's fiscal responses on top of the scenario, for the subset of models (fewer than half) that include a fiscal sector. For projecting the public finances themselves, the fiscal block of Section 6 is the instrument.

10 Scenario drivers: reading a path through a model

The counterfactual layer of Section 9 starts from a deliberate policy move and asks what it would cause. The drivers layer runs the same borrowed structure in the opposite direction. It takes a path the user has *already* built — the displayed scenario, or an imported outside-forecast combination (Section 4.5) — and asks which sequence of a chosen DSGE model's own structural shocks would have produced it over the first twelve forecast quarters. The answer is drawn as a stacked contribution chart in the same units as the main panel, with an explicit grey residual for the part the named shocks cannot reproduce.

The claim is conditional on the model, and on nothing else. The reading is: *through model M's lens, this is the mix of M's shocks that reproduces the path.* It is not a statement about what caused the scenario, and it is not model-free shock labelling; a different model on the same path gives a different mix, and the tool makes no attempt to adjudicate between them. What is validated is same-model recovery: a path is generated from a model's own known shock sequence, decomposed under that same model, and the recovered contribution paths are compared with the truth. Those checks speak to the estimator, not to the model and not to the world. The feature is experimental and says so on screen — the **Drivers** button carries an amber `exp` badge — and the licence conditions of Sections 10.5 and 10.6 are enforced by the interface rather than left to the user's discretion.

10.1 The question and the object

The decomposition answers the question a policy meeting asks of a projection: *which shocks would have produced this path?* It is the forward-looking counterpart of the historical shock decomposition that central-bank models report over the sample, and it is performed the way a structural model is used to interpret an outside forecast — filter the path through the model and read off the shocks — except that the object filtered is a scenario the user drew, not a data vintage.

Formally, let $H_d = 12$ be the decomposition horizon in quarters, q the number of model observables watched, and K the number of shocks (or shock clusters, Section 10.7) the user has named. The input is the scenario's *level deviation* from the baseline, z, expressed in the model's observable space by the conventions of Section 10.4. Because the long-run drift of Section 8 translates baseline and scenario rigidly, it cancels in the deviation by construction, and a zero scenario gives a zero decomposition.

10.2 The stacked system and the estimator

Write $R_{k,o}(s)$ for model M's response of observable o to a unit of structural shock k at horizon s, taken from the same pre-computed MMB response files that feed the counterfactual layer (Section 9.1). A shock arriving in quarter τ moves observable o in quarter $t \geq \tau$ by $R_{k,o}(t-\tau)$, so the path implied by a whole shock sequence is the convolution

$$z_o(t) \;=\; \sum_{k=1}^{K} \sum_{\tau=0}^{t} R_{k,o}(t-\tau)\, \varepsilon_k(\tau) \;+\; u_o(t), \qquad o = 1, \ldots, q, \quad t = 0, \ldots, H_d - 1, \tag{35}$$

with $\varepsilon_k(\tau)$ the size of shock k in quarter τ and $u_o(t)$ the part of the path the named shocks leave unexplained. Stacking observables and quarters gives the linear system

$$z = \Theta\,\varepsilon + u, \qquad \Theta[\, oH_d + t,\; kH_d + \tau\,] = R_{k,o}(t-\tau)\,\mathbf{1}\{t \geq \tau\}, \tag{36}$$

where $z \in \mathbb{R}^{qH_d}$ holds the target path (observable-major, quarter within observable), $\varepsilon \in \mathbb{R}^{KH_d}$ the unknown shock sequence (shock-major, arrival quarter within shock), and Θ is the $qH_d \times KH_d$ stacked response matrix whose every (o,k) block is lower-triangular Toeplitz. Only the first H_d horizons of each response are used; the shipped response arrays carry twenty-one entries, of which index 0 is the pre-impact zero, so $R_{k,o}(s)$ is read at offset $s+1$ — the same convention the validation benches use.

The system is square only by coincidence. With three observables and three shocks it has 36 rows and 36 columns; with the full seven-observable US configuration, 84 and 84; with two shocks and three observables, 36 rows and 24 columns. What is reported is the *minimum-norm least-squares* solution: among all ε minimizing the weighted fit error, the one of smallest Euclidean norm. Let $\nu_j = \|\Theta_{\cdot j}\|_2$ be the response energy of column j, $V = \mathrm{diag}(\nu_1, \ldots, \nu_{KH_d})$, and W the diagonal row weighting of Section 10.5 (unity on the three common observables). The estimator is

$$\hat{\varepsilon} = V^{-1}\tilde{\varepsilon}^{\star}, \qquad \tilde{\varepsilon}^{\star} = \arg\min \|\tilde{\varepsilon}\|_2 \quad \text{over} \quad \arg\min_{\tilde{\varepsilon}} \left\| W\left(\Theta V^{-1}\tilde{\varepsilon} - z\right)\right\|_2^2, \tag{37}$$

computed by forming the Gram matrix of $W\Theta V^{-1}$, adding a ridge $\alpha = 10^{-8}$ times its mean diagonal entry, factorizing once, and then running two sweeps of iterative refinement against that same factorization. Every refinement iterate stays in the row space of Θ, so the sequence converges to the minimum-norm solution and the ridge acts only as a numerical stabilizer, not as a prior on the shocks. Columns whose norm falls below 10^{-12} keep a scale of one, so a shock with no response stays inert instead of being amplified.

The reported contributions are computed from the *raw* responses and the un-normalized shocks,

$$c_{k,o}(t) = \sum_{\tau \leq t} R_{k,o}(t-\tau)\,\hat{\varepsilon}_k(\tau), \qquad u_o(t) = z_o(t) - \sum_{k=1}^{K} c_{k,o}(t), \tag{38}$$

so neither the column scaling nor the row weighting touches what is drawn: $\Theta V^{-1}\tilde{\varepsilon} \equiv \Theta\hat{\varepsilon}$ identically, and the bars plus the residual reproduce the scenario line to eight decimal places in the regression tests.

Whether a residual can exist at all is a property of the configuration, not of the scenario. The drawer computes the rank of Θ by a pivoted Cholesky on the smaller-side Gram matrix with relative pivot tolerance 10^{-12}, and reports the regime in a banner. When the rank reaches the number of rows qH_d the system is *exact-fit*: the named shocks span the observable paths, the residual is identically zero by construction, and it is evidence of nothing. Only when observables are watched beyond what the named shocks can span — the *residual-able* regime, flagged in amber with the rank and the row count — does the grey band carry information. On the shipped menu the two regimes are about twenty orders of magnitude apart in conditioning, so the rank threshold is not a delicate choice: full-rank lenses have Θ condition numbers of order 10^2, while the rank-deficient Canadian lens sits near 3×10^{17}.

10.3 Equal response energy across shocks

The column scaling V^{-1} in (37) is not cosmetic. A minimum-norm criterion applied to the raw columns penalizes each shock by its own size in shock units, which means it quietly prefers whichever shock moves the observables most per unit of ε. Model shocks are not comparable in that metric. Measured on the shipped Smets–Wouters lens (Smets and Wouters, 2007) at three observables and twelve quarters, the stacked column norms run from 1.28 for the monetary shock and 2.12 for the government-spending shock up to 11.58 for the wage markup and 13.84 for the price markup: an order of magnitude between the smallest and the largest. Under the raw criterion the monetary shock is priced out of its own attribution, because explaining a given path with it costs far more in $\|\varepsilon\|$ than explaining the same path with a markup shock.

Normalizing every column to unit response energy replaces that implicit ranking with an explicit neutral one: the criterion is indifferent between two shocks that move the watched observables identically. Where the configuration is identified the scaling is a pure reparameterization and changes nothing — the licence fixtures still recover their generating shock sequence to 10^{-6} — and where it is not identified it removes the bias. On the same three-observable Smets–Wouters configuration, mean off-class leakage falls from 32 percent under the raw criterion to 5 percent under the normalized one. The signature of the choice is easiest to see in the extreme case: given two exactly collinear shocks whose response norms differ two to one, the shipped solver splits the contribution equally, whereas the raw criterion would split it four to one, entirely on the strength of the units the model happens to write its shocks in.

10.4 From the panel to the model's observables

The scenario lives in the BVAR's smoother level space of Section 3.1 — $100 \ln$ for log variables, native units for linear ones — and the model lives in its own observable space. Three observables are common to every MMB model in the menu and are always watched: output, quarterly inflation, and the policy rate. They are built from the active bloc's panel columns by the conversions of the upper block of Table 1. Two factors of four deserve naming, because getting either wrong silently rescales an entire class of shocks: the model's `inflationq` is an annualized quarterly rate, so the price-level deviation is differenced and multiplied by four, while the model's `interest` is already annualized and is taken as an identity, with *no* division by four. These are not assumptions. The shipped Smets–Wouters file carries both the MMB common block and the model's own native observables, and a regression test reads the ratios off it, checking on every shock and every horizon with a non-trivial response that `inflationq/pinfobs` and `interest/robs` both equal 4 to better than 10^{-9}.

Table 1: The observable mapping on the United States bloc under the Smets–Wouters lens. Deviations are scenario minus baseline in the stored level space; all differences use a zero pre-sample value, $\text{dev}_{-1} = 0$. The upper block is the three common observables, watched under every model; the lower block is the four native observables the user may add. Spec column names are the US bloc's.

Observable	**Meaning**	**Built from the panel**	**Drawn as**
`output`	Output	`GDPH` level deviation, identity	4Δ, % q/q ann.
`inflationq`	Inflation	4Δ of the `JGDP` level deviation	identity, % q/q ann.
`interest`	Policy rate	`POLRATE` deviation, identity	identity, %
`dc`	Consumption (Δ)	Δ of the `CH` deviation	$\times 4$, % q/q ann.
`dinve`	Investment (Δ)	Δ of the `FNH` deviation	$\times 4$, % q/q ann.
`dw`	Real wage (Δ)	Δ of the (`LXBC` − `JGDP`) deviation	$\times 4$, % q/q ann.
`labobs`	Hours (level)	`LANAGRA` deviation, level	4Δ, % q/q ann.

Every one of these maps is *linear* — a gain, or a scaled first difference — which is what makes the chart legible: contributions and residual pass through the display operator additively, so the stacked bars and the grey residual sum to the scenario line exactly rather than approximately.

The four native observables of Table 1 require the spec columns `CH`, `FNH`, `LXBC`, `JGDP` and `LANAGRA`, and of the deployed blocs only the United States carries all five. Everywhere else the drawer runs at three observables. The Christiano–Motto–Rostagno lens exposes a parallel set of four, built from the same panel columns; its observation equations were measured from the shipped response file rather than assumed, and the anchor is that its `gdp_obs` equals the first difference of the common `output` column for every non-trend shock, which fixes consumption, investment and the real wage in the same percent quarterly growth units through the trend loading they share with it, and its `hours_obs` in the percent level units of `labobs`. Three of its observables are excluded as duplicates of an already-watched column, for the reason given in Section 10.5. It exposes no financial observables in this pass: credit, net worth and the external-finance premium are not mapped.

10.5 Two rules: the naming budget and the collinearity flag

The decomposition is only as informative as the configuration allows, and two rules keep the interface from offering readings the estimator cannot support. Both come out of the validation bench `pipeline/bench_simple_decomp.m`, which generates paths from a model's own shocks and measures how well the shipped solver recovers them.

The first is the **naming budget**: at q watched observables the user may name at most q shocks. The bench's identified reference point is the Smets–Wouters lens at its native seven observables with all seven usable shocks, where Θ has rank 84 out of 84 rows, noiseless recovery of the generating sequence is accurate to 3.6×10^{-15}, and recovery succeeds in 100 percent of instances at one percent noise. The rule is enforced twice over: the engine refuses a decomposition with more shocks than observables, and the interface disables every un-named shock chip the moment the budget is reached. Adding a native observable raises the budget by one — the chip says so — which is why the four addable observables of Table 1 carry the seven-shock US configuration.

Not every observable a model exposes may be added, and the exclusions are the substance of the rule rather than an exception to it. The Smets–Wouters observables `dy`, `pinfobs` and `robs`, and their Christiano–Motto–Rostagno counterparts, are each an invertible linear transform of an observable already watched: a differenced or rescaled copy of output, inflation or the policy rate. Watching one would raise the naming budget by one while buying no new rank at all, and the identification the budget rule promises would be fictitious. The exclusion is pinned against the data rather than against a curated list: a regression test requires each of the four legitimate additions to buy exactly $H_d = 12$ new rank and each of the three excluded transforms to buy exactly zero, alone and all together.

The second rule is the **collinearity flag**. Two shocks whose stacked responses point in nearly the same direction cannot be told apart by any estimator, whatever the rank test says. For every model the build step computes the cosine between each pair of unit-normalized stacked three-observable, twelve-quarter response paths, and ships every pair with $|\cos| > 0.95$ in the index. Naming both members of a flagged pair is permitted but warned about, with the measured cosine quoted to two decimals; the one-click **Use all** selection instead walks the model's shock list in order and skips any shock that forms a flagged pair with one already chosen. Among the small lenses this bites in four places: the Christiano–Motto–Rostagno lens flags three pairs among its financial and investment shocks (the largest at 0.998), the Smets–Wouters 2003 euro-area lens one pair at 0.963, the Hong Kong lens two (one at 1.000), and the Canadian lens one at 0.975. The Canadian lens is the instructive case: with three usable shocks, two of them near-duplicates, its response columns span only 24 of the 36 stacked rows however many are named, so the greedy selection names two and the decomposition runs in the residual-able regime. The two FRB/US lenses are a different order of problem entirely — 109 and 117 flagged pairs — and are handled by grouping rather than by skipping (Section 10.7). The discipline to take from the flag is not to avoid such shocks but to read them jointly: the sum over a flagged pair is identified, the split within it is not.

Remark 10 (The measurement-error slider is inert at budget-max). Added observables can be held with measurement error, on a strict-to-permissive slider that sets their row weight to $w = 1/\sqrt{1 + me}$ while the three common observables stay exact. The control does nothing when the budget is exhausted, and is disabled there with a tooltip that says so. The reason is arithmetic: at $K = q$ the stacked system is square, and scaling both Θ and z by the same positive diagonal W leaves the solution of (37) exactly unchanged. Row weights matter only when observables outnumber named shocks, which is precisely when the model is being asked to reconcile more information than it has shocks to reconcile it with. Un-name a shock to give the slider a role.

10.6 The class layer and where it is licensed

Individual shock names are not how a projection is discussed. The **Classes** view therefore aggregates contributions into Monetary, Demand and Supply — and, where a model has one, Financial — by a fixed class map shipped with the tool. Aggregation is exact in the sense that class bars are sums of shock bars, and by linearity of the display operator they still reconcile with the scenario line. Whether those sums are *recoverable* is a separate question, and it is the question the second bench, `pipeline/bench_class_recovery.m`, was written to settle. It generates paths from known class-active shock sequences over twenty seeds and applies two pre-committed criteria: the recovered class-total path must correlate at least 0.95 with the true class total in at least 90 percent of active-class instances, and mean off-class leakage must not exceed 25 percent. Both must pass.

Under the shipped solver, exactly one configuration passes: the Smets–Wouters lens with all seven observables watched and all seven of its class-map shocks named, which recovers class totals in 100 percent of instances with 0 percent off-class leakage and a perfectly diagonal attribution matrix. Nothing else does. The same lens at three observables recovers in only 50 percent of instances — it fails the recovery gate even though its leakage, at 5 percent, is excellent. The Christiano–Motto–Rostagno lens at seven observables reaches 78 percent against the 90 percent bar. The FRB/US lens at three observables, the only configuration in which it can be run, reaches 37 percent with 47 percent leakage and fails both gates. The **Classes** toggle is greyed everywhere outside the single licensed configuration, with a tooltip stating that class totals are not reliably separable at that combination and pointing at the results file.

Two consequences of that scope are worth stating plainly. The first is that the licence is a joint property of three coordinates — the model, the set of watched observables, and the set of named shocks — and not of the model alone, nor of the model and the observable set (Remark 11). The second is what the licence does and does not assert: it says the estimator recovers class totals when the truth is generated by that same model at that configuration. It says nothing about whether the Smets–Wouters classes are the right classes, and nothing about the world.

Remark 11 (Un-naming one class member revokes the class view). The licence predicate takes three coordinates — the model, the set of watched observables, and the set of *named shocks* — because the bench held all three fixed: it ran the Smets–Wouters lens at its seven-observable set with all seven class-map shocks active. Drop one member under **Customize**, say the wage markup, and **Classes** greys immediately, even though the model and the observable set are untouched. This is not pedantry. The dropped shock's energy does not vanish; it is redistributed across the classes that remain, which is exactly the quantity the off-class-leakage criterion measures and exactly the quantity the bench never measured for that six-shock split. An earlier build gated on model and observables only, and would have kept the licensed labels and the "100 percent recovery, 0 percent leakage" tooltip on a configuration no bench had seen.

Two of the remaining lenses group their shocks without claiming a class split. The Christiano–Motto–Rostagno lens shows its own class map as four *families* — Monetary, Demand, Supply and Financial — purely as an organising grouping of its nine shocks, and the drawer prints the number the bench refused them on, verbatim, underneath. They are not a licensed class split and should not be quoted as one. Every other model in the menu, including the euro-area, Canadian, Hong Kong, Brazilian and UK lenses, has no class map at all and falls through to one series per named shock under the model's own mnemonics, with the monetary and fiscal shocks given plain-language labels.

10.7 The FRB/US lenses: empirical clusters

The two FRB/US census lenses in the menu each carry 54 shocks: the policy and fiscal shocks, plus 52 equation residuals. They are handled differently from every other model, for two independent reasons. The class view is refused for them by the bench at the only configuration they can be run in, as reported above. And the individual residuals are not separable in the three common observables in the first place: 109 of the 1,431 shock pairs exceed the 0.95 collinearity threshold in one variant, 117 in the other. Naming three of them at random would produce a chart with no identification behind it.

The drawer therefore groups by the structure the response space actually has. The flagged pairs define a graph on the 54 shocks; its connected components are the *clusters*, labelled by the equation name of their largest member and ordered by size. One variant has 23 clusters, the other 21; the largest gathers seventeen or eighteen financial, housing and fiscal residuals, and about three-quarters of the clusters are singletons. Naming a cluster names one column of Θ — the first principal component of its members' unit-normalized response paths, computed by power iteration and signed to agree with the largest member — so the naming budget counts clusters, not members: three clusters at three observables, whatever their membership.

A full-rank Θ is not evidence that a naming is usable, and here the gap is wide. The build step scans every possible cluster triple through the shipped solver over twenty sparse-truth seeds and records which recover their generating sequence to 10^{-6}: 123 of 1,771 triples pass in one variant, 122 of 1,330 in the other. Most of the rest pass the rank test and fail the recovery test. The shipped default is chosen deterministically among the passers — maximize the number of shocks covered, prefer a triple containing the monetary cluster, then take the first in cluster order — and it covers 19 of the 54 shocks with a worst recovery error of 2.5×10^{-9} in one variant, 12 of 54 at 1.3×10^{-7} in the other. Leave the default naming and the drawer says so in an amber notice, because nothing backs the alternative. The obvious naive choice, the three largest clusters by membership, fails the recovery criterion by several orders of magnitude.

Remark 12 (Cluster members are chained, not mutually parallel). Collinearity is not transitive. A cluster is a connected *component* of the $|\cos| > 0.95$ graph, so two of its members are linked through a chain of collinear neighbours and may themselves sit well below the threshold against each other. It is therefore wrong to read a cluster as a bundle of near-identical shocks, and wrong to describe its composite column as a representative member: the composite is a genuine first principal component of a set that is merely connected, and its economic interpretation rests on the equations that happen to be linked, not on a shared mechanism. The drawer states this in the chip tooltip, in the cluster-membership footnote, and in the caveat under the chart. Read a cluster as "whatever in the model moves the watched observables in roughly this direction", and open the membership list before naming it in a briefing.

10.8 Operating the drawer

The **Drivers** button sits in the scenario-action row above the chart grid, and appears only when there is something to decompose — some displayed, isolated or coupled scenario deviation, or an active outside forecast, above a numerical tolerance. It opens a fixed panel on the right of the window, closed with the × or with ESC, and nothing is solved while it is closed. The controls, in the order they appear:

- The **lens**: the model in force, any shocks dropped from it with the reason (a zero response, or a missing observable), and a **Change model** link opening a picker grouped into own-country models, the two US reference models, and the rest. The choice is remembered per bloc.
- **Classes** | **Groups**: aggregate into classes, or show one series per named shock or cluster. **Classes** is disabled unless the configuration is licensed (Section 10.6).
- The **chart view**: **Forecast** draws contribution bars anchored on the baseline path with the scenario line over them — the familiar decomposition picture — while **Deviation from baseline** drops the baseline and draws the contributions around zero. The view follows the main panel's **Units** setting on opening and stays wherever the user puts it thereafter (Remark 13).
- **Customize** opens the rest: **Use all**, which watches every mappable observable and then names shocks up to the budget skipping collinear duplicates; the input source, either the displayed scenario or an imported outside-forecast combination, and for the scenario a **Domestic** | **Global** toggle when both an isolated and a coupled path exist; the observable chips, three padlocked and the addable ones annotated with the budget they buy; the measurement-error slider (Remark 10); the cluster chips on the FRB/US lenses; and an **Experimental: raw shocks** fold, which on the FRB/US lenses replaces cluster columns with individual shocks and on every other lens is where the named shock chips themselves live.
- Below the chart, a **shares** strip reports each series' share of the total, and an expander lists cluster membership where it applies. The share is the Euclidean norm of that series' contribution path divided by the sum of all such norms including the residual's. It is a norm share, not a variance decomposition: it always sums to one hundred and can never show a negative or offsetting contribution as negative.

The chart shows one observable at a time, chosen from a dropdown, but the solve always uses every watched observable jointly — switching the displayed observable does not re-solve anything. The horizon is twelve quarters whatever the display horizon of the main grid. **Reset** returns the drawer to the default Smets–Wouters configuration and clears the remembered model for that bloc.

Example 5 (The zero-click reading on a US scenario)**.** We decompose a scenario we have already built on the United States, without touching a single control, and read the licensed class split. *Build any US scenario — the credit-crunch recipe of Example 2 will do — and press **Drivers**. The drawer opens on the Smets–Wouters lens with all seven observables watched and all seven usable shocks named, which is the one configuration the class bench licenses. The view is **Classes**: three stacked series in blue, amber and teal for Monetary, Demand and Supply, a grey residual, the dotted baseline and the scenario line on top, with output on the vertical axis in percent quarter-on-quarter annualized. The banner reports rank* 84 *of* 84*: the system is exact-fit, so the grey residual is flat at zero and is telling us nothing (Section 10.2). No collinearity or rank warning fires, because the Smets–Wouters lens has no flagged pair. Change the observable dropdown from output to inflation and the same seven-shock solve is re-plotted against a different row block of* (36)*; nothing is recomputed. The strongest sentence this chart supports has the form "through the Smets–Wouters lens, this path is mostly such-and-such a class with such-and-such an offset in the second year". Which classes those are is read off the chart; the qualifier in front of them is not optional.*

Example 6 (Two ways to leave the licence)**.** We deliberately step outside the licensed configuration, twice, to see what the drawer does about it. *Start from the state of Example 5. (1) Open **Customize**, then the **Experimental: raw shocks** fold, which is where a lens without clusters keeps its shock chips, and un-name the wage markup shock. The chart redraws with six shocks — and the **Classes** toggle greys, with a tooltip saying that class totals are not reliably separable at this model, observable and shock-set combination. The view falls back to **Groups**, seven series becoming six, under the model's own shock names. Nothing about the model or the observables changed; what changed is that the wage markup's energy is now redistributed over the classes that remain, and no bench ever measured that split (Remark 11). (2) Re-name the shock, then use **Change model** to select the FRB/US census lens. The observables collapse to the common three, because the native mapping exists only for the two US models that expose one; the naming budget falls to three; and the chips are now* 23 *collinearity clusters rather than* 54 *shocks. The shipped default naming is in force, covering* 19 *of the model's* 54 *shocks, and the caveat under the chart changes: it no longer cites the recovery bench, which never ran an FRB/US lens, and instead states that what backs this lens is a build-time check that the default naming recovers a known shock sequence to* 10^{-6}*, and that other namings are unvalidated. Name a different cluster and an amber notice appears saying exactly that. The same scenario, read through two lenses, gives two different answers with two different warranties; that is the feature working as intended, not a defect.*

Remark 13 (The delta view and the panel need not share units)*.* The drawer's chart view follows the main grid on opening: with **Units** set to **% vs base** it opens in **Deviation from baseline**. The two are not the same object. The drawer's deviation bars are always the *rate* deviation of the panel series, while the main grid's **% vs base** carries its own **Level | Rate (IRF)** sub-toggle and defaults to the cumulative level effect (Section 9.3). Auto-following can therefore put the contribution chart in different units from the panel that prompted the question. The axis title always states the unit in force; read it before comparing a number in the drawer with a number on the grid.

10.9 What the decomposition does not establish

It is worth saying plainly where this stops, in the same voice as what it does. The feature is experimental. Its claim is model-conditional: it reports which mix of a chosen model's own shocks reproduces a path, never what caused the path, and it is not model-free shock labelling. Its validation is same-model recovery, so every number quoted above measures the estimator on paths generated by the model itself, not the model's fidelity to any economy. Different lenses give different answers, and they are not equally validated — the recovery bench ran eight of the models in the menu and none of the FRB/US lenses, and one model in the menu carries neither a bench row nor a build-time recovery scan. Class aggregation is licensed at exactly one point in the configuration space and greyed elsewhere; the families of the Christiano–Motto–Rostagno lens are an organising device, not a licensed split; and the FRB/US clusters are an empirical grouping of response directions, not an economic taxonomy. Within a flagged collinear pair the split is not identified. In the exact-fit regime the residual is zero by construction. The shares strip is a norm share, not a variance decomposition. The seven-observable mode, and with it the only licensed class view, exists on the United States bloc alone.

None of this makes the reading useless; it makes it a reading. Used the way the counterfactual layer is used — flip the lens, keep the scenario, and see what survives — it answers a question the reduced form cannot answer at all, which is what a scenario would have to be made of if a particular theory of the economy were true.

11 Data, vintages, and updates

11.1 Sources

No single provider supplies current quarterly data for the full roster, so the panels are assembled bloc by bloc from a small number of source families, wired by per-bloc build scripts (`pipeline/build_*.py`) and recorded, series by series, in a machine-readable coverage manifest (`coverage_manifest.json` under `app/public/data/`) from which the roster table of Appendix B is generated.

- **FRED** supplies the entire US bloc with exact series identifiers, the China bloc (built from verified real FRED series, with quarterly real activity anchored on a chain-linked volume index), the four global commons, and the BIS credit and residential-property series redistributed through FRED.

Table 2: Repository layout. Estimation in `pipeline/` produces a per-bloc JSON export; `port_gvar_exports.py` ports it into `app/public/data/bvar/`; the front end and its compute kernels consume that data and never re-estimate.

Top-level	Key contents
`pipeline/` (MATLAB + Python)	`export_bvar_gvar24_loop.m` (GVAR–EM estimator); `build_*.py` (FRED / Eurostat / BIS / ECB pulls, trade weights, fiscal drivers); `port_gvar_exports.py`; `_verify/` ground-truth checks.
`app/` (Next.js + compute)	`src/lib/compute/` (Durbin–Koopman, GVAR solve, fiscal accounting); `scripts/` (`precompute_gvar_coupling.mjs`, `verify_gvar24.mjs`, `invert_coupling.{m,py}`); `public/data/bvar/<CC>/` (the deployed JSON the app reads).
`docs/` (manual + runbooks)	`GMAPS_User_Manual.tex`; `REESTIMATING_MODELS.md`, `ADDING_A_COUNTRY.md`, `ARCHITECTURE_FOR_AI.md`.

- **Eurostat** supplies the EU and EFTA economies: chain-linked volumes (`namq_10_gdp`), HICP, unemployment, interest rates, effective exchange rates, house prices; euro members carry the ECB policy and lending rates.
- **OECD quarterly national accounts and national sources** supply the remaining OECD members and emerging markets, with policy rates from national central banks or the BIS.
- **IMF WEO** is a back-fill of last resort: where a bloc's genuine quarterly series end short of the common estimation window, annual WEO projections are temporally disaggregated to fill the macro tail (proportional-Denton within-year profiling, observed quarters held fixed). WEO never overrides observed data, and every bloc whose live tail is WEO-derived is flagged in the manifest and in **Data & coverage**.
- **Derived series** — the stars, the effective interest rate, the stock-flow adjustment, the rest-of-euro-area aggregates — are computed, not pulled (Sections 5.1 and 6).

The **Data & coverage** dialog (top bar) lists, for the economy on screen, every variable's source, coverage span, last observed quarter, and caveats — the on-screen rendering of the manifest. Amber entries end before the panel origin and correspond exactly to the amber chart tails.

11.2 Vintage policy: estimate once, shift quarterly

The shipped panels are the product of two clearly separated steps. *Estimation build*: every bloc harmonized to a common 2025Q4 endpoint, on which all coefficients, draws, weights, and the coupling artifact were estimated and frozen. *Observed refresh*: the live panel then advances — currently to a default last-historical quarter of 2026Q1 — by pulling each newly published quarter from the same sources in the same transforms, with *no* re-estimation: the forecast re-anchors from the unchanged companion (`app/scripts/refresh-quarter.mjs`, fail-closed per bloc). Because the coupled system links economies row by row, the panel origin is *uniform*: at this writing every bloc's history ends at 2026Q1. Blocs whose sources lag reach the origin through the amber, flagged nowcast described next — never through fabricated observations. The one exception is Israel, whose statistics arrive with a long lag; its charts run on Israel's own calendar, about a year behind the panel (a documented row-offset approximation, flagged in **Data & coverage**).

Within a data round the not-yet-published quarters of the remaining series are *nowcast* by the coupled system itself: the panel is truncated to the last quarter at which every cell is observed, every observed cell in the ragged window is pinned at its published value, and one coupled solve (Section 5.3) delivers each missing cell's conditional expectation given everything already observed anywhere in the panel — the star channel carries the leaders' releases into the laggards' estimates. Observed cells are never altered; every imputed cell carries a provenance flag and draws amber (`nowcastCompletion.ts`). The label **NC** in the conditioning list is this nowcast quarter.

The **Monitor** page (top bar) shows what each data round did: per economy, the revision of the current-quarter estimate against what the previous vintage predicted for it, split by channel — **observed** (the economy's own release surprised) versus **cross-block** (the economy published nothing; the move is pure transmission of partners' releases). Source restatements of history are accepted as current best knowledge and logged in the same page. Once a year, after the previous calendar year is fully published for every economy, the whole system is re-estimated and the cycle restarts; only then do the estimated relationships change.

12 Evaluation

The system's forecasting performance is evaluated in a companion repository (`em-bayesian-gvar-eval`) under a protocol designed before the runs; this section summarizes the design and the realized results. Two questions are asked: does the estimation apparatus — Bayesian shrinkage, COVID-as-missing, the EM star fixed point, the endogenous US — earn its keep out of sample, and what does the endogenous US buy that a frozen anchor cannot deliver.

12.1 Design

Models. Nine forecast densities are scored: the proposed EM-Bayesian GVAR in its two variants (`emgvar_endo`, the full endogenous-US loop, and `emgvar_frozen`, the frozen-anchor ablation), six benchmarks estimated on the same truncated panels — a driftless random walk in the scored space, an AR(p) per series (p by AIC, iterated multi-step; Marcellino et al., 2006), a single-country BVAR with the stars dropped (the value-of-coupling ablation), the textbook fixed-weight OLS GVAR of Pesaran et al. (2004) and Dees et al. (2007) with no data augmentation (the headline classical competitor), a single-pass Bayesian GVAR (the value-of-EM ablation), and a Bayesian dynamic factor model — plus a sample-mean climatology sibling of the random walk.

Scheme. Pseudo-real-time recursive expanding window on the final vintage: quarterly origins 2010Q1–2024Q4 (60 raw origins; the four 2020 origins are dropped from the headline tables, leaving 56), horizons $h \in \{1, 2, 4, 8\}$, three targets per bloc (real-GDP growth and inflation in the 4Δ scored space, the short rate in levels), six headline blocs (US, EA, JP, UK, CA, CN). Look-ahead is excluded by construction: every panel is truncated to the origin before any transformation or star is built; trade weights are rolling, as-of-origin three-year averages (the shipped 2021–2023 weights are forbidden in the sweep); and the COVID mask is re-applied in-sample at every origin. The proposed variants require a longer warm-up and score at up to 52 origins. Predictive densities use 200 retained draws per origin with a stationarity backstop that discards explosive posterior draws.

Scoring. Point accuracy by relative RMSE against the random walk. Density accuracy by the continuous ranked probability score (CRPS), the primary metric, computed from the draw ensemble by the standard energy-form estimator

$$\widehat{\text{CRPS}} = \frac{1}{m}\sum_{k=1}^{m}|X_k - y| - \frac{1}{2m^2}\sum_{k=1}^{m}\sum_{l=1}^{m}|X_k - X_l| \tag{39}$$

(Gneiting and Raftery, 2007), which degrades gracefully at modest draw counts; calibration by PIT histograms and interval coverage (Berkowitz, 2001). Because most model pairs are *nested*, predictive-ability inference uses the Giacomini and White (2006) conditional test as the headline (valid under nesting and correct for the expanding scheme), Clark and West (2007) for nested point comparisons, and Diebold and Mariano (1995) with the Harvey et al. (1997) correction only for non-nested pairs; multiplicity is controlled by Romano–Wolf step-down (Romano and Wolf, 2005), and the per-cell *model confidence set* (MCS) of Hansen et al. (2011) reports which models are statistically indistinguishable from the best.

12.2 Results

The numbers below are the nine-model headline run archived in the companion repository as `results/tables_v3.{txt,json}` (30,744 scored cells); it is the canonical run for this section, and any later partial sweep in that repository is a diagnostic, not a restatement.

On *point* accuracy the univariate AR(p), the factor model, and the sample-mean climatology lead — a familiar macro-forecasting result — and the EM-GVAR does not win the horse race outright. Its best point showing is on inflation, where at $h = 1$ the whole Bayesian family lands on top of itself — relative RMSE 0.80 for both proposed variants, 0.80 for the one-pass Bayesian GVAR, 0.80 for the single-country BVAR — against 0.64 for the AR and 0.75 for the factor model, which are ahead of all of them. The evidence that matters is in the density comparison and the model confidence set: the EM-Bayesian GVAR sits *inside* the 90% MCS for **real GDP growth at every horizon**, for **inflation at every horizon**, and for the **short rate at** $h \leq 2$ — statistically indistinguishable from the best model in ten of the twelve (target, horizon) cells. It leaves the set only for the short rate at $h = 4$ and $h = 8$, the two cells where every persistent multivariate model is beaten by the random walk on a near-unit-root series.

Two structural findings follow, and the first runs against the convenient reading. The endogenous-US loop does *not* pay for itself in forecast accuracy: the frozen anchor is weakly better throughout, carrying the lower relative RMSE in eleven of the twelve cells and surviving in the MCS at the short rate at $h = 4$ where the endogenous variant is eliminated, and being eliminated later than it at $h = 8$. Everywhere else the two are in the set together. The endogenous US is therefore justified by what it makes computable — the rest-of-world-to-US feedback of Section 12.3, which is structurally absent under a frozen anchor — and not by out-of-sample accuracy. Second, and central to the design's claim, the apparatus *decisively* beats its classical ancestor: the textbook OLS GVAR is eliminated from five of the twelve MCS cells — growth at $h \in \{1, 4, 8\}$ and inflation at $h \geq 4$ — against two for the endogenous variant and one for the frozen, and it carries badly miscalibrated densities, with one-quarter-ahead 90%-interval coverage of 0.39 on growth and 0.36 on inflation against the 0.90 target.

Two caveats are part of the record. The draw-based densities of the proposed variants are themselves over-confident at short horizons (realized 90% coverage 0.74 at $h = 1$ pooled across targets, rising to 0.82 by $h = 8$): the forecast draws propagate parameter and shock uncertainty but not the smoother uncertainty of the COVID-imputed states. The CRPS and MCS compare score *differentials* and are robust to this; the under-coverage is a calibration caveat, not a ranking artifact. And the log predictive score is numerically unreliable on under-covered clouds (a parametric tail fit explodes on a handful of origins), so ranking rests on CRPS and the MCS only.

12.3 The endogenous-US contrast

What the solver computes is a *conditional-mean scenario response* — pin a path, read the coupled deltas via (19) — not a generalized impulse response or a variance decomposition; the evaluation keeps that distinction explicit. The realized contrast propagates a US +100bp policy-rate path through both topologies: under the frozen anchor the world-to-US direction is structurally absent — not small, absent — while the endogenous configuration returns a non-trivial RoW→US feedback, with sizeable, sign-heterogeneous gaps between the two topologies across partner blocs. These are conditional-projection deltas (the rate is pinned along the whole path, bundling the direct conditioning with the coupled feedback), so their *existence and direction* is the documented result; calibrated spillover *multipliers* await the structural extension (generalized IRFs and connectedness indices on the coupled system, Pesaran and Shin, 1998; Diebold and Yilmaz, 2012), which is constructible from the shipped exports but not part of the tool.

13 Implementation and reproducibility

13.1 Client-side computation

All matrix algebra, filtering, and simulation run in the browser in TypeScript on `Float64Array` storage (IEEE 754 double precision): LU decomposition with partial pivoting for F_t^{-1} (singularity threshold 10^{-15}), Cholesky factorization for shock draws (positive-definiteness threshold 10^{-14}, with a diagonal fallback), and row-major layouts throughout. Table 3 collects the numerical constants a replicator needs; each appears in exactly one place in the code.

Table 3: Numerical constants of the deployed engine.

Constant	Value	Role
σ_ε^2	10^{-12}	hard-pin measurement variance (8); soft-pin threshold
P_0	$10^{-7}I$	initial state covariance
LU pivot	10^{-15}	singularity threshold in F_t^{-1}
Cholesky	10^{-14}	positive-definiteness threshold
D	500	shipped posterior draws per bloc
N_b	50 / 200	band draws, **Quick** / **Full**
band quantiles	q_{16}, q_{84}	scenario bands (baseline adds q_{05}, q_{50}, q_{95})
trust default / clamp	0.7 / $[0.001, 0.999]$	soft-pin dial (15); ≥ 0.999 is hard
GS tolerance / sweeps	10^{-4} / 12	Gauss–Seidel coupled solve
Neumann tolerance / cap	10^{-10} rel. / 2000	inverse-free fixed point
$\rho(M)$ / dim	0.922 / 2060	shipped coupling artifact
artifact precision	6 sig. figures	all exported floats
sens guard	0.3	long-run slider disable (30)
B&M ridge α	10^{-6}	optimal-policy solve (34)
EM TOL / MAXIT	0.03 / 12	estimation star fixed point

Interactivity is engineered so the heavy work never blocks the mouse: no coupled solve fires during a drag (a signal bus outside the rendering state carries the drag flag); exactly one cheap direct-preview solve dispatches on release; the pinned dot is sourced from the click itself through an exact display-to-storage inverse, so it snaps to the cursor instantly while the line and bands reconcile; and each chart panel is memoized so a conditioning edit repaints one panel of the roughly thirty-five, not all of them. The full contract is asserted in the test suite.

13.2 Repository and replication

One repository holds the three concerns: `pipeline/` (MATLAB and Python: estimation and data assembly), `app/` (the front end and its compute kernels), `docs/` (this manual and the operational runbooks). The estimator writes a per-bloc JSON export that a porter script copies into `app/public/data/bvar/`; the deployed application reads only that JSON. Dependencies: a recent MATLAB with the Parallel Computing Toolbox (the bundled `BVAR/MFBVAR` code needs no further packages), Python 3 standard library for the data pulls (a FRED API key via `FRED_API_KEY`), and the pinned Node toolchain for the application.

The end-to-end path from raw data to a deployable build is six steps; the canonical commands are in `REESTIMATING_MODELS.md`:

1. **Build the data**: run the `pipeline/build_*.py` pulls to assemble the harmonized per-bloc

panels.

2. **Estimate**: run `export_bvar_gvar24_loop.m` — the GVAR-EM loop of Algorithm 2 — which exports the eight-file contract per bloc.

3. **Port**: run `port_gvar_exports.py`, which copies the contract into the application, compacting floats to six significant figures.

4. **Regenerate the coupling**: run `scripts/precompute_gvar_coupling.mjs`, which assembles M, inverts $(I - M)$ offline, and writes the gzipped artifact of Section 5.3.

5. **Verify**: run `scripts/verify_gvar24.mjs` (artifact signature fresh; the shipped inverse inverts $(I - M)$; $\rho(M) < 1$ and matches the stored value; Gauss–Seidel agrees with the artifact-seeded closed form), then the type check and the test suite (`npm test`; the multi-second solver canaries run under `npm run test:heavy`).

6. **Deploy**: on a green gate, push `app/**` — a GitHub Action builds the static export and publishes it; the Action only copies files.

Because the model signature (Section 5.3) flips whenever any bloc is re-estimated, it is not possible to ship a coupling artifact that silently disagrees with its blocs; the regenerated per-economy tables of Appendix B enforce the same discipline on this document.

14 Final thoughts

The design bets on a separation of concerns. The baseline is computed once, offline, and shipped — the browser never estimates. The scenario layer imposes user paths without asking where the deviation comes from, in the tradition of Waggoner and Zha (1999); the counterfactual layer adds structural identification exactly where it is needed — the policy transmission — and lets the estimated covariances carry the rest, because no single DSGE model matches the empirical co-movement of thirty variables. The accounting blocks derive what must never be estimated freely. And every judgment device — pins, trust dials, anchors, counterfactuals — is built to be an exact no-op at its neutral setting, so the untouched screen is always the model, to the last digit.

14.1 Limitations

The dynamics are linear and time-invariant: coefficients are fixed at their posterior means for the deterministic engine, with no time-varying parameters, stochastic volatility, or regime switching — restrictive in periods of structural change. Innovations are Gaussian; crisis-period fat tails are not accommodated. Counterfactuals are point-identified under one model–rule pair at a time (the user can flip pairs, but the tool does not average over them), and the Lucas critique applies to the re-conditioned variables whose responses are governed by the estimated, policy-invariant-by-assumption covariances. The coefficients are frozen at the estimation vintage within the year: data revisions are accepted as knowledge and the origin advances, but the parameters see new data only at the annual re-estimation. Scenario responses — including the cross-country ones — are conditional projections, not causal impulse responses (Section 12.3).

14.2 Known issues

Specific, identified defects of the current deployment, documented in the interest of honesty for a live research tool:

1. **Divergence is detected, not prevented.** The coupled solve does not refuse to run if a future roster pushed $\rho(M) \geq 1$: the convergence flag trips, a console warning fires, and the result is not cached — but a (diverged) path would still be displayed. A hard spectral-radius backstop is the durable fix.

2. **The retired tilt module still ships.** The entropic-tilt code (Remark 8) is no longer reachable from the interface but remains in the repository with a latent defect (non-finite long-run moments of unstable draws would propagate through its softmax); it should be removed or finite-masked to prevent accidental reintroduction.

3. **Fiscal metadata is not regenerated by the porter.** The fiscal block reads its drivers by *positional* indices stored in `fiscal_meta.json`; a re-export that changes a bloc's variable order leaves those indices pointing at the wrong columns unless the metadata is rebuilt (a repair script exists; folding it into the porter is the durable fix).

4. **Mid-drag previews use the filtered variance.** The impact-view shading requests the smoothed variance (13) except while a drag is in flight, when the cheaper filtered variance is shown; the run-up to a future pin is momentarily overstated until release.

14.3 Directions

Natural extensions, in rough order of value: a hard contraction guard on the coupled solve; integrating the smoother uncertainty of the imputed states into the predictive draws (the documented source of the short-horizon under-coverage, Section 12.2); structural spillover objects — generalized IRFs and connectedness — on the coupled system; scenario priors for identification inside the BVAR (Farkas and Jakab, 2026); and mixed-frequency estimation to bring monthly indicators into the nowcast.

A Notation

Table 4: Notation. Symbols are local to their sections; the two uses of d (the GVAR forcing vector of Section 5, the debt ratio d_t of Section 6) never co-occur.

Symbol	**Meaning**
Estimation (Sections 3)	
y_t, n, p	bloc observables ($n \times 1$), variable count, lag order ($p = 3$)
c, B_ℓ, Σ	intercept, lag matrices, innovation covariance of (1)
$\hat{\beta}$	stored $(1 + np) \times n$ coefficients; row 0 the intercept
ℓ_t	stored level of a `log` variable, $100 \ln Y_t$
g_t	displayed q/q annualized growth, $4(\ell_t - \ell_{t-1})$
δ_i	own-lag prior mean: 1 (`RW`) or 0 (`WN`)
λ, μ	GLP tightness and sum-of-coefficients hyperparameters
D	shipped posterior draws (500)
State space and smoother (Section 4)	
$\alpha_t, T, c_\alpha, Z, Q, R$	companion state and system matrices (7)
$a_{t\|t-1}, P_{t\|t-1}$	predicted state mean and covariance
v_t, F_t, K_t	innovation, its variance, the Kalman gain
r_t, L_t, N_t, V_t	smoother recursions (12)–(13)
Y^*, H	pin matrix (NaN = free) and forecast horizon (20)
τ, R^{ext}	trust and soft-pin variance (15)
N_b	band draws (50 quick / 200 full)
Global model (Section 5)	
$w_{cj}, \tilde{w}_{cj}$	trade weight of partner j in bloc c; renormalized
x_c^*	bloc c's star columns (DEM, INF, FIN)
$K_c[v][s]$	affine response kernel (17)
x, M, d	deviation state, coupling operator, forcing of $x = Mx + d$
$\rho(M)$	spectral radius (shipped: 0.947)
Accounting blocks (Sections 6–7)	
d_t, d_0	debt-to-GDP (%) and its seed
i_t^{eff}, g_t^n	effective rate (% p.a.) and quarterly nominal growth (decimal)
$pb_t, \overline{sfa}$	primary balance; structural stock-flow constant
b_t, nfa_0	NFA-to-GDP (%) and its seed
$i^{\text{NFA}}, \phi^{\text{FX}}$	NFA return (3% p.a.) and FX-exposed share (0.4)
Anchors and counterfactuals (Sections 8–9)	
$S_h, \delta c_j, \text{sens}_j$	drift operator, drift shift, terminal sensitivity (30)
$\psi_j^\star, \psi_j^0$	target and current terminal display forecast
h_r, h_π, h_y	MMB impulse responses; ε_t the inverted shocks
$\lambda_\pi, \lambda_y, \alpha$	loss weights and ridge of (33)–(34)
Scenario drivers (Section 10)	
H_d	decomposition horizon (12 quarters)
q, K	watched model observables; named shocks or clusters
o, t	observable index and forecast quarter, $t = 0, \ldots, H_d - 1$

Symbol	Meaning
k, τ	shock index and arrival quarter, $\tau = 0, \ldots, H_d - 1$
$R_{k,o}(s)$	response of observable o to shock k at horizon s
Θ	stacked response matrix, $qH_d \times KH_d$ (36)
z, ε, u	observable-space deviation path, shock sequence, residual
ν_j, V	column response norms and their diagonal, $\tilde{\Theta} = \Theta V^{-1}$
W	diagonal row weights, $1/\sqrt{1 + me}$ on added observables
$c_{k,o}(t)$	contribution of shock k to observable o at quarter t (38)

B The deployed economies

The tables of this appendix are *generated* from the deployed data files and regenerated whenever the data change, so they cannot drift from the shipped models. The coverage matrix below (Table 5) is built by `build_coverage_table.py` from the coverage manifest; the roster (Table 6) and the per-economy variable tables that follow are built by `generate_manual_tables.py` — the roster from the same manifest and the per-bloc metadata, the variable tables from each bloc's `spec.json`.

The single grid in Table 5 answers, at a glance, which of the nine variable groups each of the 50 deployed blocs carries: a green check marks a group that is present in that bloc's shipped model, a dash marks a genuine absence. The last two columns record the last quarter shown and the primary source family; an italic tag on the source flags the handful of blocs whose most recent observations are model-extended (WEO-back-filled or under source review) rather than genuinely observed, so a reader can see immediately where the data are firm and where they are provisional.

Table 5: Country × variable-group coverage matrix for all 50 deployed blocs. Columns: **GDP** real activity; **Price** HICP/CPI; **Unemp** unemployment rate; **Pol** policy rate; **Long** 10-year/long government yield; **Cred** credit-to-GDP; **Lend** lending/short rate; **FX** nominal effective exchange rate; **Fisc** the four-variable fiscal block (`FREV`/`FPEXP`/`EFFI`/`SFA`). ✓ = carried; -- = absent in that bloc. *Last obs* is the last quarter shown; an italic tag flags a tail that is model-extended rather than genuinely observed (*WEO* = WEO-back-filled, CN/IN; *review* = under source review, ID/KR/RU/TH; *frozen mirror* = carried from a frozen partner mirror, KR).

Bloc	GDP	Price	Unemp	Pol	Long	Cred	Lend	FX	Fisc	Last obs	Source
AR	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q1	OECD/FRED
AT	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
AU	✓	✓	✓	–	✓	✓	✓	✓	✓	2026Q1	OECD/FRED
BE	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
BR	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	OECD/FRED
CA	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	OECD/FRED
CH	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q1	Eurostat
CL	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	OECD/FRED
CN	✓	✓	–	✓	✓	✓	✓	✓	✓	2026Q1	FRED *WEO*
CO	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	OECD/FRED
CR	✓	✓	✓	–	✓	✓	✓	✓	✓	2026Q1	OECD/FRED
CZ	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	Eurostat
DE	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	Eurostat
DK	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	Eurostat
EA	✓	✓	–	✓	✓	✓	✓	✓	✓	2026Q1	Eurostat
EE	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
ES	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
FI	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
FR	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
GR	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
HU	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	Eurostat
ID	✓	✓	–	✓	✓	✓	✓	✓	✓	2026Q1	OECD/FRED *review*
IE	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
IL	✓	✓	✓	✓	✓	✓	✓	✓	✓	2025Q1	OECD/FRED
IN	✓	✓	–	✓	✓	✓	✓	✓	–	2026Q1	OECD/FRED *WEO*
IS	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q1	OECD/FRED
IT	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat

Bloc	GDP	Price	Unemp	Pol	Long	Cred	Lend	FX	Fisc	Last obs	Source
JP	✓	✓	✓	✓	✓	✓	✓	–	✓	2026Q1	OECD/FRED
KR	✓	✓	✓	✓	✓	✓	✓	–	✓	2026Q1	OECD/FRED *review,frozen mirror*
LT	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
LU	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
LV	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
MX	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	OECD/FRED
MY	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q1	OECD/FRED
NL	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
NO	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	Eurostat
NZ	✓	✓	✓	–	✓	✓	✓	✓	✓	2026Q1	OECD/FRED
PL	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	Eurostat
PT	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
RU	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q1	OECD/FRED *review*
SA	✓	✓	✓	✓	✓	✓	–	✓	–	2026Q1	OECD/FRED
SE	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	Eurostat
SI	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
SK	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
TH	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	OECD/FRED *review*
TR	✓	✓	✓	✓	✓	✓	✓	–	✓	2026Q1	OECD/FRED
TW	✓	✓	✓	–	✓	✓	✓	–	–	2026Q1	OECD/FRED
UK	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	OECD/FRED
US	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	FRED
ZA	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	OECD/FRED

B.1 The deployed roster at a glance

Generated from the shipped `spec.json`, `fiscal_meta.json`, `external_meta.json`, the trade-weights file, and the coverage manifest (`coverage_manifest.json`); the table therefore reflects the deployed data exactly. n counts the variables of the estimated VAR (domestic, global, fiscal, and star columns included). *Coupled* marks membership of the cross-country fixed point (Section 5); the remaining blocs are standalone models. *Fiscal* and *External* mark the accounting blocks of Section 6. *Last obs.* is the bloc's last historical quarter.

Table 6: Deployed economies: dimension, coupling, accounting blocks, last observation.

Code	Economy	n	Coupled	Fiscal	External	Last obs.
`US`	United States	40	✓	✓	✓	2026Q1
`EA`	Euro area (aggregate)	25	✓	✓	✓	2026Q1
`JP`	Japan	25	✓	✓	—	2026Q1
`UK`	United Kingdom	27	✓	✓	—	2026Q1
`CA`	Canada	27	✓	✓	—	2026Q1
`CN`	China	22	✓	✓	✓	2026Q1
`AR`	Argentina	20	✓	—	—	2026Q1
`AT`	Austria	23	—	✓	—	2026Q1
`AU`	Australia	25	✓	✓	—	2026Q1
`BE`	Belgium	23	—	✓	—	2026Q1
`BR`	Brazil	24	✓	✓	—	2026Q1
`CH`	Switzerland	20	✓	—	—	2026Q1
`CL`	Chile	25	✓	✓	—	2026Q1
`CO`	Colombia	26	✓	✓	—	2026Q1
`CR`	Costa Rica	20	✓	✓	—	2026Q1
`CZ`	Czechia	25	✓	✓	—	2026Q1
`DE`	Germany	22	—	✓	—	2026Q1
`DK`	Denmark	26	✓	✓	—	2026Q1
`EE`	Estonia	23	—	✓	—	2026Q1
`ES`	Spain	23	—	✓	—	2026Q1
`FI`	Finland	23	—	✓	—	2026Q1
`FR`	France	23	—	✓	—	2026Q1
`GR`	Greece	22	—	✓	—	2026Q1
`HU`	Hungary	25	✓	✓	—	2026Q1
`ID`	Indonesia	24	✓	✓	—	2026Q1
`IE`	Ireland	23	—	✓	—	2026Q1
`IL`	Israel	25	✓	✓	—	2025Q1
`IN`	India	21	✓	—	—	2026Q1
`IS`	Iceland	21	✓	—	—	2026Q1
`IT`	Italy	23	—	✓	—	2026Q1
`KR`	Korea	25	✓	✓	—	2026Q1
`LT`	Lithuania	23	—	✓	—	2026Q1

Code	Economy	n	Coupled	Fiscal	External	Last obs.
`LU`	Luxembourg	23	—	✓	—	2026Q1
`LV`	Latvia	22	—	✓	—	2026Q1
`MX`	Mexico	26	✓	✓	—	2026Q1
`MY`	Malaysia	21	✓	—	—	2026Q1
`NL`	Netherlands	23	—	✓	—	2026Q1
`NO`	Norway	24	✓	✓	—	2026Q1
`NZ`	New Zealand	25	✓	✓	—	2026Q1
`PL`	Poland	25	✓	✓	—	2026Q1
`PT`	Portugal	23	—	✓	—	2026Q1
`RU`	Russia	17	✓	—	—	2026Q1
`SA`	Saudi Arabia	18	✓	—	—	2026Q1
`SE`	Sweden	26	✓	✓	—	2026Q1
`SI`	Slovenia	23	—	✓	—	2026Q1
`SK`	Slovakia	23	—	✓	—	2026Q1
`TH`	Thailand	26	✓	✓	—	2026Q1
`TR`	Türkiye	23	✓	✓	—	2026Q1
`TW`	Taiwan Province of China	20	✓	—	—	2026Q1
`ZA`	South Africa	25	✓	✓	—	2026Q1

B.2 Variables, transformations, and priors by economy

One block per economy, generated from the shipped `spec.json`. *Trans.*: `log` = the variable enters the VAR as $100\ln(\cdot)$ (growth displayed); `lin` = the variable enters in levels. *Prior*: RW = random-walk own-lag prior mean ($\delta_i = 1$), WN = stationary white-noise own-lag prior mean ($\delta_i = 0$; Section 3.2). *Fin.*: the `isFinancial` tag — financial variables stay *observed* through the COVID window (Section 3.4). *Block*: dom. = domestic, glob. = US-determined global common, fisc. = fiscal driver, row = trade-weighted star, ext. = area-wide instrument carried as weakly exogenous (euro members). The dates give the *shipped* history window the application loads (the recent span the charts display and the state initializes from); the estimation sample is longer and ends at the 2025Q4 estimation vintage (Section 11.2).

Table 7: United States (`US`): $n = 40$, shipped history from 1998Q3, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
`GDPH`	Real Gross Domestic Product	`log`	RW	—	dom.
`CH`	Real Personal Consumption Expenditures	`log`	RW	—	dom.
`FRH`	Real Private Residential Fixed Investment (nom/deflator)	`log`	RW	—	dom.
`FNH`	Real Private Nonresidential Fixed Investment (nom/deflator)	`log`	RW	—	dom.
`XH`	Real Exports of Goods & Services	`log`	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
MH	Real Imports of Goods & Services	log	RW	—	dom.
GH	Real Government Consumption Expenditures & Gross Investment	log	RW	—	dom.
JGDP	Gross Domestic Product: Chain Price Index	log	RW	—	dom.
JC	Personal Consumption Expenditures: Chain Price Index	log	RW	—	dom.
JCXFE	PCE less Food & Energy: Chain Price Index	log	RW	—	dom.
UI	CPI-U: All Items	log	RW	—	dom.
UIXFDG	CPI-U: All Items Less Food & Energy	log	RW	—	dom.
LXBC	Business Sector: Compensation Per Hour (nominal)	log	RW	—	dom.
LANAGRA	All Employees: Total Nonfarm	log	RW	—	dom.
LR	Civilian Unemployment Rate: 16 yr +	lin	RW	—	dom.
IP	Industrial Production Index	log	RW	—	dom.
CUMFG	Capacity Utilization: Manufacturing [SIC]	lin	RW	—	dom.
HST	Housing Starts	log	RW	—	dom.
YPDH	Real Disposable Personal Income	log	RW	—	dom.
CSENT	University of Michigan: Consumer Sentiment	lin	RW	—	dom.
POLRATE	Effective Federal Funds Rate (policy rate)	lin	RW	✓	dom.
FCM2	2-Year Treasury Note Yield at Constant Maturity	lin	RW	✓	dom.
FCM5	5-Year Treasury Note Yield at Constant Maturity	lin	RW	✓	dom.
GLB_USLR	10-Year Treasury Note Yield at Constant Maturity	lin	RW	✓	glob.
FAAA	Moodys Seasoned Aaa Corporate Bond Yield	lin	RW	✓	dom.
GLB_USBAA	Moodys Seasoned Baa Corporate Bond Yield	lin	RW	✓	glob.
FXTWM	Nominal Trade-Weighted Exch Value of US$ vs AFE Currencies	log	RW	✓	dom.
GLB_USEQ	Equity Price Index (OECD US share prices / S&P, public)	log	RW	✓	glob.
GSNE	Non-Fuel Commodity Price Index (IMF, non-energy proxy)	log	RW	✓	dom.
SP500VOL	CBOE Volatility Index for S&P 500 Stock Price Index	lin	RW	✓	dom.
HP	House Price	log	RW	✓	dom.
LEV	Real nonfinancial business debt	log	RW	✓	dom.

ID	Description	Trans.	Prior	Fin.	Block
FREV_GDP	Federal Receipts (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	Federal Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Federal Effective Interest Rate (% p.a.)	lin	RW	✓	fisc.
SFA	Stock-Flow Adjustment (% of GDP)	lin	WN	—	fisc.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
ROWDEM_US	US RoW Demand	lin	RW	✓	row
ROWINF_US	US RoW Inflation	lin	RW	✓	row
ROWFIN_US	US RoW Fin	lin	RW	✓	row

Table 8: Euro area (aggregate) (EA): $n = 25$, shipped history from 1999Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_EA	EA Real GDP	log	RW	—	dom.
CONS_EA	EA Consumption	log	RW	—	dom.
INV_EA	EA Investment	log	RW	—	dom.
GOV_EA	EA Gov Spending	log	RW	—	dom.
EXP_EA	EA Exports	log	RW	—	dom.
IMP_EA	EA Imports	log	RW	—	dom.
HICP_EA	EA HICP Index	log	RW	—	dom.
RATE_EA	EA 3M Interbank Rate	lin	RW	✓	dom.
YIELD10_EA	EA 10Y Gov Bond Yield	lin	RW	✓	dom.
NEER_EA	EA NEER	log	RW	✓	dom.
POLRATE_EA	EA ECB Deposit Facility Rate (policy)	lin	RW	✓	dom.
HPI_EA	EA Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	EA Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	EA Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_EA	EA Credit to NFCs	log	RW	✓	dom.
ROWDEM_EA	EA RoW Demand	lin	RW	✓	row
ROWINF_EA	EA RoW Inflation	lin	RW	✓	row
ROWFIN_EA	EA RoW Fin	lin	RW	✓	row

Table 9: Japan (JP): $n = 25$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_JP	JP Real GDP	log	RW	—	dom.
CONS_JP	JP Consumption	log	RW	—	dom.
INV_JP	JP Investment	log	RW	—	dom.
GOV_JP	JP Gov Spending	log	RW	—	dom.
EXP_JP	JP Exports	log	RW	—	dom.
IMP_JP	JP Imports	log	RW	—	dom.
LR_JP	JP Unemployment Rate	lin	RW	—	dom.
CPI_JP	JP CPI Index	log	RW	—	dom.
RATE_JP	JP 3M Interbank Rate	lin	RW	✓	dom.
POLRATE_JP	JP Central Bank Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
YIELD10_JP	JP 10Y JGB Yield	lin	RW	✓	dom.
HP_JP	JP Real Residential House-Price Index (BIS RPP, 2010=100)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	JP Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	JP Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_JP	JP Credit to NFCs	log	RW	✓	dom.
LENDR_JP	JP Avg Contract Interest Rate on New Loans (% p.a.)	lin	RW	✓	dom.
ROWDEM_JP	JP RoW Demand	lin	RW	✓	row
ROWINF_JP	JP RoW Inflation	lin	RW	✓	row
ROWFIN_JP	JP RoW Fin	lin	RW	✓	row

Table 10: United Kingdom (UK): $n = 27$, shipped history from 1997Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_UK	UK Real GDP	log	RW	—	dom.
CONS_UK	UK Consumption	log	RW	—	dom.
INV_UK	UK Investment	log	RW	—	dom.
GOV_UK	UK Gov Spending	log	RW	—	dom.
EXP_UK	UK Exports	log	RW	—	dom.
IMP_UK	UK Imports	log	RW	—	dom.
IP_UK	UK Ind Production Index	log	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
LR_UK	UK Unemployment Rate	lin	RW	—	dom.
CPI_UK	UK CPI Index	log	RW	—	dom.
RATE_UK	UK 3M Interbank Rate	lin	RW	✓	dom.
POLRATE_UK	UK Central Bank Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
YIELD10_UK	UK 10Y Gilt Yield	lin	RW	✓	dom.
HP_UK	UK Real Residential House-Price Index (BIS RPP, 2010=100)	log	RW	✓	dom.
NEER_UK	UK NEER	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	UK Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	UK Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	UK Effective Rate (% p.a.)	lin	RW	✓	fisc.
SFA	UK Stock-Flow Adj (% of GDP)	lin	WN	—	fisc.
CREDIT_GDP_UK	UK Credit to NFCs	log	RW	✓	dom.
LENDR_UK	UK Effective Rate on New Loans to PNFCs (% p.a., BoE CFMBJ82)	lin	RW	✓	dom.
ROWDEM_UK	UK RoW Demand	lin	RW	✓	row
ROWINF_UK	UK RoW Inflation	lin	RW	✓	row
ROWFIN_UK	UK RoW Fin	lin	RW	✓	row

Table 11: Canada (CA): $n = 27$, shipped history from 1998Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_CA	CA Real GDP	log	RW	—	dom.
CONS_CA	CA Consumption	log	RW	—	dom.
INV_CA	CA Investment	log	RW	—	dom.
GOV_CA	CA Gov Spending	log	RW	—	dom.
EXP_CA	CA Exports	log	RW	—	dom.
IMP_CA	CA Imports	log	RW	—	dom.
IP_CA	CA Ind Production (Index 2015=100)	log	RW	—	dom.
LR_CA	CA Unemployment Rate	lin	RW	—	dom.
CPI_CA	CA CPI Index	log	RW	—	dom.
RATE_CA	CA 3M Interbank Rate	lin	RW	✓	dom.
POLRATE_CA	CA Central Bank Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
YIELD10_CA	CA 10Y Gov Bond Yield	lin	RW	✓	dom.
HP_CA	CA Real Residential House-Price Index (BIS RPP, 2010=100)	log	RW	✓	dom.

ID	Description	Trans.	Prior	Fin.	Block
NEER_CA	CA NEER	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	CA Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	CA Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment (% of GDP)	lin	WN	—	fisc.
CREDIT_GDP_CA	CA Credit to NFCs	log	RW	✓	dom.
LENDR_CA	CA Bank Lending Rate (effective business interest rate, % p.a.)	lin	RW	✓	dom.
ROWDEM_CA	CA RoW Demand	lin	RW	✓	row
ROWINF_CA	CA RoW Inflation	lin	RW	✓	row
ROWFIN_CA	CA RoW Fin	lin	RW	✓	row

Table 12: China (CN): $n = 22$, shipped history from 1999Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_CN	CN Real GDP (quarterly, chain-linked SA volume index, base 2011Q1=100)	log	RW	—	dom.
CPI_CN	CN CPI	log	RW	—	dom.
RATE_CN	CN 3M Interbank Rate	lin	RW	✓	dom.
EXP_CN	CN Exports (USD)	log	RW	—	dom.
IMP_CN	CN Imports (USD)	log	RW	—	dom.
REER_CN	CN Real Effective FX	log	RW	✓	dom.
EQ_CN	Shanghai Share Prices	log	RW	✓	dom.
POLRATE_CN	CN Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_CN	CN Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	CN Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	CN Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_CN	CN Credit to NFCs	log	RW	✓	dom.
CORP_CN	CN 10Y Government Bond Yield (sovereign proxy)	lin	RW	✓	dom.
ROWDEM_CN	CN RoW Demand	lin	RW	✓	row

ID	Description	Trans.	Prior	Fin.	Block
ROWINF_CN	CN RoW Inflation	lin	RW	✓	row
ROWFIN_CN	CN RoW Fin	lin	RW	✓	row

Table 13: Argentina (AR): $n = 20$, shipped history from 2004Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_AR	AR Real GDP	log	RW	—	dom.
CONS_AR	AR Real Household Consumption	log	RW	—	dom.
INV_AR	AR Real Gross Fixed Capital Formation	log	RW	—	dom.
GOV_AR	AR Real Government Consumption	log	RW	—	dom.
EXP_AR	AR Real Exports	log	RW	—	dom.
IMP_AR	AR Real Imports	log	RW	—	dom.
CPI_AR	AR CPI Index	log	RW	—	dom.
LR_AR	AR Unemployment Rate	lin	RW	—	dom.
RATE_AR	AR BADLAR Short-Term Rate	lin	RW	✓	dom.
NEER_AR	AR Nominal Effective Exchange Rate	log	RW	✓	dom.
POLRATE_AR	AR Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
CREDIT_GDP_AR	AR Credit to NFCs	log	RW	✓	dom.
LENDR_AR	AR Lending Rate (IMF MFS162, % p.a.; extreme volatility)	lin	RW	✓	dom.
ROWDEM_AR	AR RoW Demand	lin	RW	✓	row
ROWINF_AR	AR RoW Inflation	lin	RW	✓	row
ROWFIN_AR	AR RoW Fin	lin	RW	✓	row

Table 14: Austria (AT): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_AT	AT GDP	log	RW	—	dom.
CONS_AT	AT CONS	log	RW	—	dom.
INV_AT	AT INV	log	RW	—	dom.
GOV_AT	AT GOV	log	RW	—	dom.
EXP_AT	AT EXP	log	RW	—	dom.
IMP_AT	AT IMP	log	RW	—	dom.
HICP_AT	AT HICP Index	log	RW	—	dom.
LR_AT	AT Unemployment Rate	lin	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
HPI_AT	AT Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_AT	AT NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_AT	AT 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_AT	AT ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_AT	AT 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_AT	AT Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_AT	AT Rest-of-EA HICP	log	RW	—	row
FREV_GDP	AT Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	AT Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_AT	AT Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_AT	AT NFC Bank Lending Rate (ECB MIR new business, % p.a.)	lin	RW	✓	dom.

Table 15: Australia (AU): $n = 25$, shipped history from 1999Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_AU	AU Real GDP	log	RW	—	dom.
CONS_AU	AU Real Household Consumption	log	RW	—	dom.
GOV_AU	AU Real Govt Consumption	log	RW	—	dom.
INV_AU	AU Real Investment (GFCF)	log	RW	—	dom.
EXP_AU	AU Real Exports	log	RW	—	dom.
IMP_AU	AU Real Imports	log	RW	—	dom.
CPI_AU	AU CPI All-items Index	log	RW	—	dom.
LR_AU	AU Unemployment Rate	lin	RW	—	dom.
NEER_AU	AU NEER Index	log	RW	✓	dom.
RATE_AU	AU Cash Rate (Interbank Overnight)	lin	RW	✓	dom.
YIELD10_AU	AU 10Y Govt Bond Yield	lin	RW	✓	dom.
HP_AU	AU Real Residential House-Price Index (BIS RPP, 2010=100)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.

ID	Description	Trans.	Prior	Fin.	Block
FREV_GDP	AU Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	AU Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	AU Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	AU Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_AU	AU Credit to NFCs	log	RW	✓	dom.
LENDR_AU	AU Business Lending Rate (RBA F7 outstanding large-business total, % p.a.)	lin	RW	✓	dom.
ROWDEM_AU	AU RoW Demand	lin	RW	✓	row
ROWINF_AU	AU RoW Inflation	lin	RW	✓	row
ROWFIN_AU	AU RoW Fin	lin	RW	✓	row

Table 16: Belgium (BE): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_BE	BE GDP	log	RW	—	dom.
CONS_BE	BE CONS	log	RW	—	dom.
INV_BE	BE INV	log	RW	—	dom.
GOV_BE	BE GOV	log	RW	—	dom.
EXP_BE	BE EXP	log	RW	—	dom.
IMP_BE	BE IMP	log	RW	—	dom.
HICP_BE	BE HICP Index	log	RW	—	dom.
LR_BE	BE Unemployment Rate	lin	RW	—	dom.
HPI_BE	BE Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_BE	BE NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_BE	BE 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_BE	BE ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_BE	BE 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_BE	BE Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_BE	BE Rest-of-EA HICP	log	RW	—	row
FREV_GDP	BE Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	BE Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.

ID	Description	Trans.	Prior	Fin.	Block
CREDIT_GDP_BE	BE Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_BE	BE NFC Bank Lending Rate (ECB MIR new business, % p.a.)	lin	RW	✓	dom.

Table 17: Brazil (BR): $n = 24$, shipped history from 2012Q2, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_BR	BR Real GDP	log	RW	—	dom.
CONS_BR	BR Real Household Consumption	log	RW	—	dom.
INV_BR	BR Real Gross Fixed Capital Formation	log	RW	—	dom.
GOV_BR	BR Real Government Consumption	log	RW	—	dom.
EXP_BR	BR Real Exports	log	RW	—	dom.
IMP_BR	BR Real Imports	log	RW	—	dom.
CPI_BR	BR CPI Index	log	RW	—	dom.
LR_BR	BR Unemployment Rate	lin	RW	—	dom.
NEER_BR	BR Nominal Effective Exchange Rate	log	RW	✓	dom.
POLRATE_BR	BR Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_BR	BR Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	BR Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	BR Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	BR Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	BR Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_BR	BR Credit to NFCs	log	RW	✓	dom.
LENDR_BR	BR Corporate Bank Lending Rate (BCB SGS 20718, % p.a.)	lin	RW	✓	dom.
ROWDEM_BR	BR RoW Demand	lin	RW	✓	row
ROWINF_BR	BR RoW Inflation	lin	RW	✓	row
ROWFIN_BR	BR RoW Fin	lin	RW	✓	row

Table 18: Switzerland (CH): $n = 20$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_CH	CH GDP	log	RW	—	dom.
CONS_CH	CH CONS	log	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
INV_CH	CH INV	log	RW	—	dom.
GOV_CH	CH GOV	log	RW	—	dom.
EXP_CH	CH EXP	log	RW	—	dom.
IMP_CH	CH IMP	log	RW	—	dom.
HICP_CH	CH HICP Index	log	RW	—	dom.
LR_CH	CH Unemployment Rate	lin	RW	—	dom.
NEER_CH	CH NEER	log	RW	✓	dom.
POLRATE_CH	CH Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_CH	CH Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
CREDIT_GDP_CH	CH Credit to NFCs	log	RW	✓	dom.
LENDR_CH	CH Corporate Lending Rate (SNB new investment loans, fixed rate, % p.a.)	lin	RW	✓	dom.
ROWDEM_CH	CH RoW Demand	lin	RW	✓	row
ROWINF_CH	CH RoW Inflation	lin	RW	✓	row
ROWFIN_CH	CH RoW Fin	lin	RW	✓	row

Table 19: Chile (CL): $n = 25$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_CL	CL Real GDP	log	RW	—	dom.
CONS_CL	CL Real Household Consumption	log	RW	—	dom.
INV_CL	CL Real Gross Fixed Capital Formation	log	RW	—	dom.
GOV_CL	CL Real Government Consumption	log	RW	—	dom.
EXP_CL	CL Real Exports	log	RW	—	dom.
IMP_CL	CL Real Imports	log	RW	—	dom.
CPI_CL	CL CPI Index	log	RW	—	dom.
LR_CL	CL Unemployment Rate	lin	RW	—	dom.
RATE_CL	CL 3M Interbank Rate	lin	RW	✓	dom.
POLRATE_CL	CL Central Bank Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
YIELD10_CL	CL 10Y Gov Bond Yield	lin	RW	✓	dom.
HP_CL	CL Real Residential House-Price Index (BIS RPP, 2010=100)	log	RW	✓	dom.
NEER_CL	CL NEER	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.

ID	Description	Trans.	Prior	Fin.	Block
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	CL Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	CL Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	CL Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	CL Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_CL	CL Credit to NFCs	log	RW	✓	dom.
ROWDEM_CL	CL RoW Demand	lin	RW	✓	row
ROWINF_CL	CL RoW Inflation	lin	RW	✓	row
ROWFIN_CL	CL RoW Fin	lin	RW	✓	row

Table 20: Colombia (CO): $n = 26$, shipped history from 2005Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_CO	CO Real GDP	log	RW	—	dom.
CONS_CO	CO Real Household Consumption	log	RW	—	dom.
INV_CO	CO Real Gross Fixed Capital Formation	log	RW	—	dom.
GOV_CO	CO Real Government Consumption	log	RW	—	dom.
EXP_CO	CO Real Exports	log	RW	—	dom.
IMP_CO	CO Real Imports	log	RW	—	dom.
CPI_CO	CO CPI Index	log	RW	—	dom.
LR_CO	CO Unemployment Rate	lin	RW	—	dom.
RATE_CO	CO 3M Interbank Rate	lin	RW	✓	dom.
POLRATE_CO	CO Central Bank Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
YIELD10_CO	CO 10Y Gov Bond Yield	lin	RW	✓	dom.
HP_CO	CO Real Residential House-Price Index (BIS RPP, 2010=100)	log	RW	✓	dom.
NEER_CO	CO NEER	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	CO Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	CO Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	CO Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	CO Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_CO	CO Credit to NFCs	log	RW	✓	dom.

ID	Description	Trans.	Prior	Fin.	Block
LENDR_CO	CO Bank Lending Rate (IMF IFS FILR, % p.a.)	lin	RW	✓	dom.
ROWDEM_CO	CO RoW Demand	lin	RW	✓	row
ROWINF_CO	CO RoW Inflation	lin	RW	✓	row
ROWFIN_CO	CO RoW Fin	lin	RW	✓	row

Table 21: Costa Rica (CR): $n = 20$, shipped history from 2010Q3, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_CR	CR Real GDP (Industrial Production proxy)	log	RW	—	dom.
EXP_CR	CR Goods Exports (USD)	log	RW	—	dom.
IMP_CR	CR Goods Imports (USD)	log	RW	—	dom.
CPI_CR	CR CPI Index	log	RW	—	dom.
LR_CR	CR Unemployment Rate	lin	RW	—	dom.
RATE_CR	CR Monetary Policy Rate	lin	RW	✓	dom.
NEER_CR	CR NEER	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	CR Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	CR Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	CR Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	CR Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_CR	CR Credit to NFCs	log	RW	✓	dom.
LENDR_CR	CR Bank Lending Rate (IMF IFS FILR, % p.a.)	lin	RW	✓	dom.
ROWDEM_CR	CR RoW Demand	lin	RW	✓	row
ROWINF_CR	CR RoW Inflation	lin	RW	✓	row
ROWFIN_CR	CR RoW Fin	lin	RW	✓	row

Table 22: Czechia (CZ): $n = 25$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_CZ	CZ GDP	log	RW	—	dom.
CONS_CZ	CZ CONS	log	RW	—	dom.
INV_CZ	CZ INV	log	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
GOV_CZ	CZ GOV	log	RW	—	dom.
EXP_CZ	CZ EXP	log	RW	—	dom.
IMP_CZ	CZ IMP	log	RW	—	dom.
HICP_CZ	CZ HICP Index	log	RW	—	dom.
LR_CZ	CZ Unemployment Rate	lin	RW	—	dom.
YIELD10_CZ	CZ 10Y Gov Bond Yield	lin	RW	✓	dom.
NEER_CZ	CZ NEER	log	RW	✓	dom.
POLRATE_CZ	CZ Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_CZ	CZ Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	CZ Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	CZ Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	CZ Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	CZ Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_CZ	CZ Credit to NFCs	log	RW	✓	dom.
LENDR_CZ	CZ NFC bank lending rate (% p.a.)	lin	RW	✓	dom.
ROWDEM_CZ	CZ RoW Demand	lin	RW	✓	row
ROWINF_CZ	CZ RoW Inflation	lin	RW	✓	row
ROWFIN_CZ	CZ RoW Fin	lin	RW	✓	row

Table 23: Germany (DE): $n = 22$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_DE	DE GDP	log	RW	—	dom.
CONS_DE	DE CONS	log	RW	—	dom.
INV_DE	DE INV	log	RW	—	dom.
GOV_DE	DE GOV	log	RW	—	dom.
EXP_DE	DE EXP	log	RW	—	dom.
IMP_DE	DE IMP	log	RW	—	dom.
HICP_DE	DE HICP Index	log	RW	—	dom.
LR_DE	DE Unemployment Rate	lin	RW	—	dom.
HPI_DE	DE Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_DE	DE NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund)	lin	RW	✓	dom.

ID	Description	Trans.	Prior	Fin.	Block
POLRATE_DE	DE ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_DE	DE 3M Rate	lin	RW	✓	ext.
ROWEA_GDP_DE	DE Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_DE	DE Rest-of-EA HICP	log	RW	—	row
FREV_GDP	DE Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	DE Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_DE	DE Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_DE	DE NFC Bank Lending Rate (ECB MIR new business, % p.a.)	lin	RW	✓	dom.

Table 24: Denmark (DK): $n = 26$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_DK	DK GDP	log	RW	—	dom.
CONS_DK	DK CONS	log	RW	—	dom.
INV_DK	DK INV	log	RW	—	dom.
GOV_DK	DK GOV	log	RW	—	dom.
EXP_DK	DK EXP	log	RW	—	dom.
IMP_DK	DK IMP	log	RW	—	dom.
HICP_DK	DK HICP Index	log	RW	—	dom.
LR_DK	DK Unemployment Rate	lin	RW	—	dom.
RATE_DK	DK 3M Rate	lin	RW	✓	dom.
YIELD10_DK	DK 10Y Gov Bond Yield	lin	RW	✓	dom.
NEER_DK	DK NEER	log	RW	✓	dom.
POLRATE_DK	DK Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_DK	DK Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	DK Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	DK Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	DK Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	DK Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_DK	DK Credit to NFCs	log	RW	✓	dom.

ID	Description	Trans.	Prior	Fin.	Block
LENDR_DK	DK Corporate Lending Rate (ECB MIR loans to NFCs, new business, % p.a.)	lin	RW	✓	dom.
ROWDEM_DK	DK RoW Demand	lin	RW	✓	row
ROWINF_DK	DK RoW Inflation	lin	RW	✓	row
ROWFIN_DK	DK RoW Fin	lin	RW	✓	row

Table 25: Estonia (EE): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_EE	EE GDP	log	RW	—	dom.
CONS_EE	EE CONS	log	RW	—	dom.
INV_EE	EE INV	log	RW	—	dom.
GOV_EE	EE GOV	log	RW	—	dom.
EXP_EE	EE EXP	log	RW	—	dom.
IMP_EE	EE IMP	log	RW	—	dom.
HICP_EE	EE HICP Index	log	RW	—	dom.
LR_EE	EE Unemployment Rate	lin	RW	—	dom.
HPI_EE	EE Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_EE	EE NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_EE	EE 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_EE	EE ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_EE	EE 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_EE	EE Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_EE	EE Rest-of-EA HICP	log	RW	—	row
FREV_GDP	EE Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	EE Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_EE	EE Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_EE	EE ECB MIR Composite Cost of Borrowing, NFCs (new business, % p.a.)	lin	RW	✓	dom.

Table 26: Spain (ES): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_ES	ES GDP	log	RW	—	dom.
CONS_ES	ES CONS	log	RW	—	dom.
INV_ES	ES INV	log	RW	—	dom.
GOV_ES	ES GOV	log	RW	—	dom.
EXP_ES	ES EXP	log	RW	—	dom.
IMP_ES	ES IMP	log	RW	—	dom.
HICP_ES	ES HICP Index	log	RW	—	dom.
LR_ES	ES Unemployment Rate	lin	RW	—	dom.
HPI_ES	ES Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_ES	ES NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_ES	ES 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_ES	ES ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_ES	ES 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_ES	ES Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_ES	ES Rest-of-EA HICP	log	RW	—	row
FREV_GDP	ES Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	ES Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_ES	ES Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_ES	ES NFC Bank Lending Rate (ECB MIR composite, %)	lin	RW	✓	dom.

Table 27: Finland (FI): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_FI	FI GDP	log	RW	—	dom.
CONS_FI	FI CONS	log	RW	—	dom.
INV_FI	FI INV	log	RW	—	dom.
GOV_FI	FI GOV	log	RW	—	dom.
EXP_FI	FI EXP	log	RW	—	dom.
IMP_FI	FI IMP	log	RW	—	dom.
HICP_FI	FI HICP Index	log	RW	—	dom.
LR_FI	FI Unemployment Rate	lin	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
HPI_FI	FI Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_FI	FI NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_FI	FI 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_FI	FI ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_FI	FI 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_FI	FI Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_FI	FI Rest-of-EA HICP	log	RW	—	row
FREV_GDP	FI Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	FI Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_FI	FI Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_FI	FI ECB MIR Composite Cost of Borrowing, NFCs (new business, % p.a.)	lin	RW	✓	dom.

Table 28: France (FR): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_FR	FR GDP	log	RW	—	dom.
CONS_FR	FR CONS	log	RW	—	dom.
INV_FR	FR INV	log	RW	—	dom.
GOV_FR	FR GOV	log	RW	—	dom.
EXP_FR	FR EXP	log	RW	—	dom.
IMP_FR	FR IMP	log	RW	—	dom.
HICP_FR	FR HICP Index	log	RW	—	dom.
LR_FR	FR Unemployment Rate	lin	RW	—	dom.
HPI_FR	FR Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_FR	FR NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_FR	FR 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_FR	FR ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_FR	FR 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.

ID	Description	Trans.	Prior	Fin.	Block
ROWEA_GDP_FR	FR Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_FR	FR Rest-of-EA HICP	log	RW	—	row
FREV_GDP	FR Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	FR Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_FR	FR Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_FR	FR NFC Bank Lending Rate (ECB MIR new business, % p.a.)	lin	RW	✓	dom.

Table 29: Greece (GR): $n = 22$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_GR	GR GDP	log	RW	—	dom.
CONS_GR	GR CONS	log	RW	—	dom.
INV_GR	GR INV	log	RW	—	dom.
GOV_GR	GR GOV	log	RW	—	dom.
EXP_GR	GR EXP	log	RW	—	dom.
IMP_GR	GR IMP	log	RW	—	dom.
HICP_GR	GR HICP Index	log	RW	—	dom.
LR_GR	GR Unemployment Rate	lin	RW	—	dom.
NEER_GR	GR NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_GR	GR 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_GR	GR ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_GR	GR 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_GR	GR Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_GR	GR Rest-of-EA HICP	log	RW	—	row
FREV_GDP	GR Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	GR Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_GR	GR Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_GR	GR NFC Bank Lending Rate (ECB MIR new-business floating, %)	lin	RW	✓	dom.

Table 30: Hungary (HU): $n = 25$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_HU	HU GDP	log	RW	—	dom.
CONS_HU	HU CONS	log	RW	—	dom.
INV_HU	HU INV	log	RW	—	dom.
GOV_HU	HU GOV	log	RW	—	dom.
EXP_HU	HU EXP	log	RW	—	dom.
IMP_HU	HU IMP	log	RW	—	dom.
HICP_HU	HU HICP Index	log	RW	—	dom.
LR_HU	HU Unemployment Rate	lin	RW	—	dom.
YIELD10_HU	HU 10Y Gov Bond Yield	lin	RW	✓	dom.
NEER_HU	HU NEER	log	RW	✓	dom.
POLRATE_HU	HU Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_HU	HU Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	HU Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	HU Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	HU Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	HU Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_HU	HU Credit to NFCs	log	RW	✓	dom.
LENDR_HU	HU NFC bank lending rate (% p.a.)	lin	RW	✓	dom.
ROWDEM_HU	HU RoW Demand	lin	RW	✓	row
ROWINF_HU	HU RoW Inflation	lin	RW	✓	row
ROWFIN_HU	HU RoW Fin	lin	RW	✓	row

Table 31: Indonesia (ID): $n = 24$, shipped history from 2005Q3, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_ID	ID Real GDP (SA, constant prices)	log	RW	—	dom.
CONS_ID	ID Real Private Consumption (SA)	log	RW	—	dom.
INV_ID	ID Real Gross Fixed Capital Formation (SA)	log	RW	—	dom.
GOV_ID	ID Real Government Consumption (SA)	log	RW	—	dom.
EXP_ID	ID Real Exports of Goods & Services (SA)	log	RW	—	dom.
IMP_ID	ID Real Imports of Goods & Services (SA)	log	RW	—	dom.
CPI_ID	ID CPI (all items, index)	log	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
RATE_ID	ID Monetary Policy-related Rate (BI rate)	lin	RW	✓	dom.
NEER_ID	ID Nominal Effective Exchange Rate (BIS broad)	log	RW	✓	dom.
POLRATE_ID	ID Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_ID	ID Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	ID Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	ID Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	ID Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	ID Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_ID	ID Credit to NFCs	log	RW	✓	dom.
LENDR_ID	ID IFS Lending Rate (% p.a.)	lin	RW	✓	dom.
ROWDEM_ID	ID RoW Demand	lin	RW	✓	row
ROWINF_ID	ID RoW Inflation	lin	RW	✓	row
ROWFIN_ID	ID RoW Fin	lin	RW	✓	row

Table 32: Ireland (IE): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_IE	IE GDP	log	RW	—	dom.
CONS_IE	IE CONS	log	RW	—	dom.
INV_IE	IE INV	log	RW	—	dom.
GOV_IE	IE GOV	log	RW	—	dom.
EXP_IE	IE EXP	log	RW	—	dom.
IMP_IE	IE IMP	log	RW	—	dom.
HICP_IE	IE HICP Index	log	RW	—	dom.
LR_IE	IE Unemployment Rate	lin	RW	—	dom.
HPI_IE	IE Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_IE	IE NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_IE	IE 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_IE	IE ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_IE	IE 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.

ID	Description	Trans.	Prior	Fin.	Block
ROWEA_GDP_IE	IE Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_IE	IE Rest-of-EA HICP	log	RW	—	row
FREV_GDP	IE Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	IE Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_IE	IE Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_IE	IE NFC Bank Lending Rate (ECB MIR composite, %)	lin	RW	✓	dom.

Table 33: Israel (IL): $n = 25$, shipped history from 2000Q1, last observation 2025Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_IL	IL Real GDP	log	RW	—	dom.
CONS_IL	IL Real Household Consumption	log	RW	—	dom.
INV_IL	IL Real Gross Fixed Capital Formation	log	RW	—	dom.
GOV_IL	IL Real Government Consumption	log	RW	—	dom.
EXP_IL	IL Real Exports	log	RW	—	dom.
IMP_IL	IL Real Imports	log	RW	—	dom.
CPI_IL	IL CPI Index (All Items)	log	RW	—	dom.
LR_IL	IL Unemployment Rate	lin	RW	—	dom.
RATE_IL	IL 3M Treasury Bill Yield	lin	RW	✓	dom.
POLRATE_IL	IL Policy Rate (BoI)	lin	RW	✓	dom.
NEER_IL	IL NEER	log	RW	✓	dom.
HP_IL	IL Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	IL Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	IL Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	IL Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	IL Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_IL	IL Credit to NFCs	log	RW	✓	dom.
CORP_IL	IL Tel-Bond Corporate Bond Yield (5y govt + Tel-Bond spread, BoI)	lin	RW	✓	dom.
ROWDEM_IL	IL RoW Demand	lin	RW	✓	row
ROWINF_IL	IL RoW Inflation	lin	RW	✓	row
ROWFIN_IL	IL RoW Fin	lin	RW	✓	row

Table 34: India (IN): $n = 21$, shipped history from 2011Q4, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_IN	IN Real GDP (chain volume, SA)	log	RW	—	dom.
CONS_IN	IN Real Household Consumption (chain vol, SA)	log	RW	—	dom.
INV_IN	IN Real Gross Fixed Capital Formation (chain vol, SA)	log	RW	—	dom.
GOV_IN	IN Real Government Consumption (chain vol, SA)	log	RW	—	dom.
EXP_IN	IN Real Exports of Goods & Services (chain vol, SA)	log	RW	—	dom.
IMP_IN	IN Real Imports of Goods & Services (chain vol, SA)	log	RW	—	dom.
CPI_IN	IN CPI (all items, index)	log	RW	—	dom.
RATE_IN	IN Call Money / Immediate Interbank Rate	lin	RW	✓	dom.
RATE3M_IN	IN Short-term 3-Month Interbank Rate	lin	RW	✓	dom.
YIELD10_IN	IN Long-term 10Y Government Bond Yield	lin	RW	✓	dom.
NEER_IN	IN Nominal Effective Exchange Rate (BIS broad, index)	log	RW	✓	dom.
POLRATE_IN	IN Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_IN	IN Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
CREDIT_GDP_IN	IN Credit to NFCs	log	RW	✓	dom.
ROWDEM_IN	IN RoW Demand	lin	RW	✓	row
ROWINF_IN	IN RoW Inflation	lin	RW	✓	row
ROWFIN_IN	IN RoW Fin	lin	RW	✓	row

Table 35: Iceland (IS): $n = 21$, shipped history from 2000Q2, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_IS	IS Real GDP	log	RW	—	dom.
CONS_IS	IS Real Private Consumption	log	RW	—	dom.
INV_IS	IS Real Investment	log	RW	—	dom.
GOV_IS	IS Real Govt Consumption	log	RW	—	dom.
EXP_IS	IS Real Exports	log	RW	—	dom.
IMP_IS	IS Real Imports	log	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
CPI_IS	IS CPI Index	log	RW	—	dom.
LR_IS	IS Unemployment Rate	lin	RW	—	dom.
RATE_IS	IS 3-Month Interbank Rate	lin	RW	✓	dom.
NEER_IS	IS NEER	log	RW	✓	dom.
POLRATE_IS	IS Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_IS	IS Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
CREDIT_GDP_IS	IS Credit to NFCs	log	RW	✓	dom.
LENDR_IS	IS Bank Lending Rate (IMF IFS FILR, % p.a.)	lin	RW	✓	dom.
ROWDEM_IS	IS RoW Demand	lin	RW	✓	row
ROWINF_IS	IS RoW Inflation	lin	RW	✓	row
ROWFIN_IS	IS RoW Fin	lin	RW	✓	row

Table 36: Italy (IT): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_IT	IT GDP	log	RW	—	dom.
CONS_IT	IT CONS	log	RW	—	dom.
INV_IT	IT INV	log	RW	—	dom.
GOV_IT	IT GOV	log	RW	—	dom.
EXP_IT	IT EXP	log	RW	—	dom.
IMP_IT	IT IMP	log	RW	—	dom.
HICP_IT	IT HICP Index	log	RW	—	dom.
LR_IT	IT Unemployment Rate	lin	RW	—	dom.
HPI_IT	IT Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_IT	IT NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_IT	IT 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_IT	IT ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_IT	IT 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_IT	IT Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_IT	IT Rest-of-EA HICP	log	RW	—	row
FREV_GDP	IT Revenue (% of GDP)	lin	WN	—	fisc.

ID	Description	Trans.	Prior	Fin.	Block
FPEXP_GDP	IT Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_IT	IT Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_IT	IT NFC Bank Lending Rate (ECB MIR composite, %)	lin	RW	✓	dom.

Table 37: Korea (KR): $n = 25$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_KR	KR Real GDP	log	RW	—	dom.
CONS_KR	KR Real Household Consumption	log	RW	—	dom.
INV_KR	KR Real Gross Fixed Capital Formation	log	RW	—	dom.
GOV_KR	KR Real Government Consumption	log	RW	—	dom.
EXP_KR	KR Real Exports	log	RW	—	dom.
IMP_KR	KR Real Imports	log	RW	—	dom.
CPI_KR	KR CPI (all items, index)	log	RW	—	dom.
LR_KR	KR Unemployment Rate	lin	RW	—	dom.
RATE_KR	KR Money Market Rate (3-month, short-term)	lin	RW	✓	dom.
POLRATE_KR	KR Central Bank Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
YIELD10_KR	KR Government Bond Yield (long-term)	lin	RW	✓	dom.
HP_KR	KR Real Residential House-Price Index (BIS RPP, 2010=100)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	KR Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	KR Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	KR Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	KR Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_KR	KR Credit to NFCs	log	RW	✓	dom.
CORP_KR	KR Corporate Bond Yield (3Y, AA-, BoK ECOS)	lin	RW	✓	dom.
ROWDEM_KR	KR RoW Demand	lin	RW	✓	row
ROWINF_KR	KR RoW Inflation	lin	RW	✓	row
ROWFIN_KR	KR RoW Fin	lin	RW	✓	row

Table 38: Lithuania (LT): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_LT	LT GDP	log	RW	—	dom.
CONS_LT	LT CONS	log	RW	—	dom.
INV_LT	LT INV	log	RW	—	dom.
GOV_LT	LT GOV	log	RW	—	dom.
EXP_LT	LT EXP	log	RW	—	dom.
IMP_LT	LT IMP	log	RW	—	dom.
HICP_LT	LT HICP Index	log	RW	—	dom.
LR_LT	LT Unemployment Rate	lin	RW	—	dom.
HPI_LT	LT Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_LT	LT NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_LT	LT 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_LT	LT ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_LT	LT 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_LT	LT Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_LT	LT Rest-of-EA HICP	log	RW	—	row
FREV_GDP	LT Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	LT Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_LT	LT Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_LT	LT ECB MIR Composite Cost of Borrowing, NFCs (new business, % p.a.)	lin	RW	✓	dom.

Table 39: Luxembourg (LU): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_LU	LU GDP	log	RW	—	dom.
CONS_LU	LU CONS	log	RW	—	dom.
INV_LU	LU INV	log	RW	—	dom.
GOV_LU	LU GOV	log	RW	—	dom.
EXP_LU	LU EXP	log	RW	—	dom.
IMP_LU	LU IMP	log	RW	—	dom.
HICP_LU	LU HICP Index	log	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
LR_LU	LU Unemployment Rate	lin	RW	—	dom.
HPI_LU	LU Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_LU	LU NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_LU	LU 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_LU	LU ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_LU	LU 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_LU	LU Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_LU	LU Rest-of-EA HICP	log	RW	—	row
FREV_GDP	LU Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	LU Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_LU	LU Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_LU	LU ECB MIR Bank Lending Rate to NFCs (outstanding amounts, % p.a.)	lin	RW	✓	dom.

Table 40: Latvia (LV): $n = 22$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_LV	LV GDP	log	RW	—	dom.
CONS_LV	LV CONS	log	RW	—	dom.
INV_LV	LV INV	log	RW	—	dom.
GOV_LV	LV GOV	log	RW	—	dom.
EXP_LV	LV EXP	log	RW	—	dom.
IMP_LV	LV IMP	log	RW	—	dom.
HICP_LV	LV HICP Index	log	RW	—	dom.
LR_LV	LV Unemployment Rate	lin	RW	—	dom.
HPI_LV	LV Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_LV	LV NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_LV	LV 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_LV	LV ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.

ID	Description	Trans.	Prior	Fin.	Block
RATE_LV	LV 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_LV	LV Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_LV	LV Rest-of-EA HICP	log	RW	—	row
FREV_GDP	LV Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	LV Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_LV	LV Credit to NFCs (% of GDP)	lin	RW	✓	dom.

Table 41: Mexico (MX): $n = 26$, shipped history from 2001Q4, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_MX	MX Real GDP (chain volume, SA)	log	RW	—	dom.
CONS_MX	MX Real Household Consumption (chain vol, SA)	log	RW	—	dom.
INV_MX	MX Real Gross Fixed Capital Formation (SA)	log	RW	—	dom.
GOV_MX	MX Real Government Consumption (chain vol, SA)	log	RW	—	dom.
EXP_MX	MX Real Exports (chain volume, SA)	log	RW	—	dom.
IMP_MX	MX Real Imports (chain volume, SA)	log	RW	—	dom.
CPI_MX	MX CPI (all items, index)	log	RW	—	dom.
LR_MX	MX Unemployment Rate	lin	RW	—	dom.
RATE_MX	MX Money Market Rate (short-term)	lin	RW	✓	dom.
POLRATE_MX	MX Central Bank Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
YIELD10_MX	MX Government Bond Yield (long-term, S13)	lin	RW	✓	dom.
HP_MX	MX Real Residential House-Price Index (BIS RPP, 2010=100)	log	RW	✓	dom.
NEER_MX	MX Nominal Effective Exchange Rate (index)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	MX Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	MX Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	MX Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.

ID	Description	Trans.	Prior	Fin.	Block
SFA	MX Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_MX	MX Credit to NFCs	log	RW	✓	dom.
LENDR_MX	MX Bank Lending Rate (IMF IFS FILR, % p.a.)	lin	RW	✓	dom.
ROWDEM_MX	MX RoW Demand	lin	RW	✓	row
ROWINF_MX	MX RoW Inflation	lin	RW	✓	row
ROWFIN_MX	MX RoW Fin	lin	RW	✓	row

Table 42: Malaysia (MY): $n = 21$, shipped history from 2015Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_MY	MY Real GDP (chain volume, SA)	log	RW	—	dom.
CONS_MY	MY Real Private Consumption (SA)	log	RW	—	dom.
INV_MY	MY Real Gross Fixed Capital Formation (SA)	log	RW	—	dom.
GOV_MY	MY Real Government Consumption (SA)	log	RW	—	dom.
EXP_MY	MY Real Exports of Goods & Services (SA)	log	RW	—	dom.
IMP_MY	MY Real Imports of Goods & Services (SA)	log	RW	—	dom.
CPI_MY	MY CPI Index (all items)	log	RW	—	dom.
LR_MY	MY Unemployment Rate	lin	RW	—	dom.
RATE_MY	MY Overnight Policy Rate (BNM)	lin	RW	✓	dom.
NEER_MY	MY Nominal Effective Exchange Rate	log	RW	✓	dom.
POLRATE_MY	MY Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_MY	MY Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
CREDIT_GDP_MY	MY Credit to NFCs	log	RW	✓	dom.
LENDR_MY	MY IFS Lending Rate (% p.a.)	lin	RW	✓	dom.
ROWDEM_MY	MY RoW Demand	lin	RW	✓	row
ROWINF_MY	MY RoW Inflation	lin	RW	✓	row
ROWFIN_MY	MY RoW Fin	lin	RW	✓	row

Table 43: Netherlands (NL): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_NL	NL GDP	log	RW	—	dom.
CONS_NL	NL CONS	log	RW	—	dom.
INV_NL	NL INV	log	RW	—	dom.
GOV_NL	NL GOV	log	RW	—	dom.
EXP_NL	NL EXP	log	RW	—	dom.
IMP_NL	NL IMP	log	RW	—	dom.
HICP_NL	NL HICP Index	log	RW	—	dom.
LR_NL	NL Unemployment Rate	lin	RW	—	dom.
HPI_NL	NL Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_NL	NL NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_NL	NL 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_NL	NL ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_NL	NL 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_NL	NL Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_NL	NL Rest-of-EA HICP	log	RW	—	row
FREV_GDP	NL Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	NL Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_NL	NL Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_NL	NL NFC Bank Lending Rate (ECB MIR new business, % p.a.)	lin	RW	✓	dom.

Table 44: Norway (NO): $n = 24$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_NO	NO GDP	log	RW	—	dom.
CONS_NO	NO CONS	log	RW	—	dom.
INV_NO	NO INV	log	RW	—	dom.
GOV_NO	NO GOV	log	RW	—	dom.
EXP_NO	NO EXP	log	RW	—	dom.
IMP_NO	NO IMP	log	RW	—	dom.
HICP_NO	NO HICP Index	log	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
LR_NO	NO Unemployment Rate	lin	RW	—	dom.
NEER_NO	NO NEER	log	RW	✓	dom.
POLRATE_NO	NO Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_NO	NO Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	NO Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	NO Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	NO Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	NO Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_NO	NO Credit to NFCs	log	RW	✓	dom.
LENDR_NO	NO NFC bank lending rate (% p.a.)	lin	RW	✓	dom.
ROWDEM_NO	NO RoW Demand	lin	RW	✓	row
ROWINF_NO	NO RoW Inflation	lin	RW	✓	row
ROWFIN_NO	NO RoW Fin	lin	RW	✓	row

Table 45: New Zealand (NZ): $n = 25$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_NZ	NZ Real GDP	log	RW	—	dom.
CONS_NZ	NZ Real Consumption	log	RW	—	dom.
INV_NZ	NZ Real Investment	log	RW	—	dom.
GOV_NZ	NZ Real Govt Consumption	log	RW	—	dom.
EXP_NZ	NZ Real Exports	log	RW	—	dom.
IMP_NZ	NZ Real Imports	log	RW	—	dom.
CPI_NZ	NZ CPI Index	log	RW	—	dom.
LR_NZ	NZ Unemployment Rate	lin	RW	—	dom.
RATE_NZ	NZ Policy Rate (OCR)	lin	RW	✓	dom.
YIELD10_NZ	NZ 10Y Gov Bond Yield	lin	RW	✓	dom.
HP_NZ	NZ Real Residential House-Price Index (BIS RPP, 2010=100)	log	RW	✓	dom.
NEER_NZ	NZ NEER	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	NZ Government Revenue (% of GDP)	lin	WN	—	fisc.

ID	Description	Trans.	Prior	Fin.	Block
FPEXP_GDP	NZ Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	NZ Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	NZ Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_NZ	NZ Credit to NFCs	log	RW	✓	dom.
LENDR_NZ	NZ business lending rate (% p.a.)	lin	RW	✓	dom.
ROWDEM_NZ	NZ RoW Demand	lin	RW	✓	row
ROWINF_NZ	NZ RoW Inflation	lin	RW	✓	row
ROWFIN_NZ	NZ RoW Fin	lin	RW	✓	row

Table 46: Poland (PL): $n = 25$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_PL	PL GDP	log	RW	—	dom.
CONS_PL	PL CONS	log	RW	—	dom.
INV_PL	PL INV	log	RW	—	dom.
GOV_PL	PL GOV	log	RW	—	dom.
EXP_PL	PL EXP	log	RW	—	dom.
IMP_PL	PL IMP	log	RW	—	dom.
HICP_PL	PL HICP Index	log	RW	—	dom.
LR_PL	PL Unemployment Rate	lin	RW	—	dom.
YIELD10_PL	PL 10Y Gov Bond Yield	lin	RW	✓	dom.
NEER_PL	PL NEER	log	RW	✓	dom.
POLRATE_PL	PL Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_PL	PL Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	PL Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	PL Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	PL Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	PL Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_PL	PL Credit to NFCs	log	RW	✓	dom.
LENDR_PL	PL NFC bank lending rate (% p.a.)	lin	RW	✓	dom.
ROWDEM_PL	PL RoW Demand	lin	RW	✓	row
ROWINF_PL	PL RoW Inflation	lin	RW	✓	row
ROWFIN_PL	PL RoW Fin	lin	RW	✓	row

Table 47: Portugal (PT): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_PT	PT GDP	log	RW	—	dom.
CONS_PT	PT CONS	log	RW	—	dom.
INV_PT	PT INV	log	RW	—	dom.
GOV_PT	PT GOV	log	RW	—	dom.
EXP_PT	PT EXP	log	RW	—	dom.
IMP_PT	PT IMP	log	RW	—	dom.
HICP_PT	PT HICP Index	log	RW	—	dom.
LR_PT	PT Unemployment Rate	lin	RW	—	dom.
HPI_PT	PT Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_PT	PT NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_PT	PT 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_PT	PT ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_PT	PT 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_PT	PT Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_PT	PT Rest-of-EA HICP	log	RW	—	row
FREV_GDP	PT Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	PT Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_PT	PT Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_PT	PT NFC Bank Lending Rate (ECB MIR composite, %)	lin	RW	✓	dom.

Table 48: Russia (RU): $n = 17$, shipped history from 2003Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_RU	RU Real GDP (constant 2010 LCU, SA)	log	RW	—	dom.
CPI_RU	RU CPI Index (all-items)	log	RW	—	dom.
EXP_RU	RU Goods Exports (FOB, USD)	log	RW	—	dom.
LR_RU	RU Unemployment Rate	lin	RW	—	dom.
RATE_RU	RU Interbank Call-Money Rate	lin	RW	✓	dom.
NEER_RU	RU Nominal Effective Exchange Rate	log	RW	✓	dom.
POLRATE_RU	RU Policy Rate (BIS CBPOL)	lin	RW	✓	dom.

ID	Description	Trans.	Prior	Fin.	Block
HP_RU	RU Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
CREDIT_GDP_RU	RU Credit to NFCs	log	RW	✓	dom.
LENDR_RU	RU Lending Rate (IMF MFS162 spliced w/ CBR loans-to-NFC up-to-1y, % p.a.)	lin	RW	✓	dom.
ROWDEM_RU	RU RoW Demand	lin	RW	✓	row
ROWINF_RU	RU RoW Inflation	lin	RW	✓	row
ROWFIN_RU	RU RoW Fin	lin	RW	✓	row

Table 49: Saudi Arabia (SA): $n = 18$, shipped history from 2010Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_SA	SA Real GDP	log	RW	—	dom.
CONS_SA	SA Real Household Consumption	log	RW	—	dom.
INV_SA	SA Real Gross Fixed Capital Formation	log	RW	—	dom.
GOV_SA	SA Real Government Consumption	log	RW	—	dom.
EXP_SA	SA Real Exports	log	RW	—	dom.
IMP_SA	SA Real Imports	log	RW	—	dom.
CPI_SA	SA CPI Index	log	RW	—	dom.
LR_SA	SA Unemployment Rate	lin	RW	—	dom.
NEER_SA	SA NEER	log	RW	✓	dom.
POLRATE_SA	SA Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
CREDIT_GDP_SA	SA Credit to NFCs	log	RW	✓	dom.
ROWDEM_SA	SA RoW Demand	lin	RW	✓	row
ROWINF_SA	SA RoW Inflation	lin	RW	✓	row
ROWFIN_SA	SA RoW Fin	lin	RW	✓	row

Table 50: Sweden (SE): $n = 26$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_SE	SE GDP	log	RW	—	dom.
CONS_SE	SE CONS	log	RW	—	dom.
INV_SE	SE INV	log	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
GOV_SE	SE GOV	log	RW	—	dom.
EXP_SE	SE EXP	log	RW	—	dom.
IMP_SE	SE IMP	log	RW	—	dom.
HICP_SE	SE HICP Index	log	RW	—	dom.
LR_SE	SE Unemployment Rate	lin	RW	—	dom.
RATE_SE	SE 3M Rate	lin	RW	✓	dom.
YIELD10_SE	SE 10Y Gov Bond Yield	lin	RW	✓	dom.
NEER_SE	SE NEER	log	RW	✓	dom.
POLRATE_SE	SE Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_SE	SE Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	SE Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	SE Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	SE Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	SE Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_SE	SE Credit to NFCs	log	RW	✓	dom.
LENDR_SE	SE NFC bank lending rate (% p.a.)	lin	RW	✓	dom.
ROWDEM_SE	SE RoW Demand	lin	RW	✓	row
ROWINF_SE	SE RoW Inflation	lin	RW	✓	row
ROWFIN_SE	SE RoW Fin	lin	RW	✓	row

Table 51: Slovenia (SI): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_SI	SI GDP	log	RW	—	dom.
CONS_SI	SI CONS	log	RW	—	dom.
INV_SI	SI INV	log	RW	—	dom.
GOV_SI	SI GOV	log	RW	—	dom.
EXP_SI	SI EXP	log	RW	—	dom.
IMP_SI	SI IMP	log	RW	—	dom.
HICP_SI	SI HICP Index	log	RW	—	dom.
LR_SI	SI Unemployment Rate	lin	RW	—	dom.
HPI_SI	SI Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_SI	SI NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_SI	SI 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.

ID	Description	Trans.	Prior	Fin.	Block
POLRATE_SI	SI ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_SI	SI 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_SI	SI Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_SI	SI Rest-of-EA HICP	log	RW	—	row
FREV_GDP	SI Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	SI Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_SI	SI Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_SI	SI ECB MIR Composite Cost of Borrowing, NFCs (new business, % p.a.)	lin	RW	✓	dom.

Table 52: Slovakia (SK): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_SK	SK GDP	log	RW	—	dom.
CONS_SK	SK CONS	log	RW	—	dom.
INV_SK	SK INV	log	RW	—	dom.
GOV_SK	SK GOV	log	RW	—	dom.
EXP_SK	SK EXP	log	RW	—	dom.
IMP_SK	SK IMP	log	RW	—	dom.
HICP_SK	SK HICP Index	log	RW	—	dom.
LR_SK	SK Unemployment Rate	lin	RW	—	dom.
HPI_SK	SK Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_SK	SK NEER (trade-weighted, IC42 partners)	log	RW	✓	dom.
SPREAD_SK	SK 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_SK	SK ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA IG Corporate Bond Yield (ECB all-bonds 10Y, %)	lin	RW	✓	ext.
RATE_SK	SK 3M Rate	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_SK	SK Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_SK	SK Rest-of-EA HICP	log	RW	—	row
FREV_GDP	SK Revenue (% of GDP)	lin	WN	—	fisc.

ID	Description	Trans.	Prior	Fin.	Block
FPEXP_GDP	SK Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_SK	SK Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_SK	SK ECB MIR Composite Cost of Borrowing, NFCs (new business, % p.a.)	lin	RW	✓	dom.

Table 53: Thailand (TH): $n = 26$, shipped history from 2003Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_TH	TH Real GDP (SA, constant prices)	log	RW	—	dom.
CONS_TH	TH Real Private Consumption (SA)	log	RW	—	dom.
INV_TH	TH Real Gross Fixed Capital Formation (SA)	log	RW	—	dom.
GOV_TH	TH Real Government Consumption (SA)	log	RW	—	dom.
EXP_TH	TH Real Exports of Goods & Services (SA)	log	RW	—	dom.
IMP_TH	TH Real Imports of Goods & Services (SA)	log	RW	—	dom.
CPI_TH	TH CPI (all items, index)	log	RW	—	dom.
LR_TH	TH Unemployment Rate	lin	RW	—	dom.
RATE_TH	TH Policy Rate (BOT)	lin	RW	✓	dom.
YIELD10_TH	TH 10-Year Govt Bond Yield	lin	RW	✓	dom.
NEER_TH	TH Nominal Effective Exchange Rate (BIS broad)	log	RW	✓	dom.
POLRATE_TH	TH Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_TH	TH Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	TH Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	TH Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	TH Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	TH Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_TH	TH Credit to NFCs	log	RW	✓	dom.
LENDR_TH	TH IFS Lending Rate (% p.a.)	lin	RW	✓	dom.
ROWDEM_TH	TH RoW Demand	lin	RW	✓	row
ROWINF_TH	TH RoW Inflation	lin	RW	✓	row

ID	Description	Trans.	Prior	Fin.	Block
ROWFIN_TH	TH RoW Fin	lin	RW	✓	row

Table 54: Türkiye (TR): $n = 23$, shipped history from 2000Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_TR	TR Real GDP	log	RW	—	dom.
CONS_TR	TR Real Household Consumption	log	RW	—	dom.
INV_TR	TR Real Gross Fixed Capital Formation	log	RW	—	dom.
GOV_TR	TR Real Government Consumption	log	RW	—	dom.
EXP_TR	TR Real Exports	log	RW	—	dom.
IMP_TR	TR Real Imports	log	RW	—	dom.
CPI_TR	TR CPI (all items, index)	log	RW	—	dom.
LR_TR	TR Unemployment Rate	lin	RW	—	dom.
RATE_TR	TR Monetary Policy-related Rate	lin	RW	✓	dom.
POLRATE_TR	TR Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_TR	TR Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	TR Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	TR Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	TR Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	TR Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_TR	TR Credit to NFCs	log	RW	✓	dom.
ROWDEM_TR	TR RoW Demand	lin	RW	✓	row
ROWINF_TR	TR RoW Inflation	lin	RW	✓	row
ROWFIN_TR	TR RoW Fin	lin	RW	✓	row

Table 55: Taiwan Province of China (TW): $n = 20$, shipped history from 1997Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_TW	TW Real GDP	log	RW	—	dom.
CONS_TW	TW Real Household Consumption	log	RW	—	dom.
GOV_TW	TW Real Government Consumption	log	RW	—	dom.
INV_TW	TW Real Gross Fixed Capital Formation	log	RW	—	dom.
EXP_TW	TW Real Exports of Goods and Services	log	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
IMP_TW	TW Real Imports of Goods and Services	log	RW	—	dom.
CPI_TW	TW CPI (all items, index, 2021=100)	log	RW	—	dom.
LR_TW	TW Unemployment Rate	lin	RW	—	dom.
RATE_TW	TW CBC Discount Rate (policy)	lin	RW	✓	dom.
MMRATE_TW	TW Interbank Call-Loan Rate (money market)	lin	RW	✓	dom.
YIELD10_TW	TW 10-Year Government Bond Yield	lin	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
CREDIT_GDP_TW	TW Credit to NFCs	log	RW	✓	dom.
LENDR_TW	TW 5-Major-Banks New Loan Rate (CBC EH45 total, % p.a.)	lin	RW	✓	dom.
ROWDEM_TW	TW RoW Demand	lin	RW	✓	row
ROWINF_TW	TW RoW Inflation	lin	RW	✓	row
ROWFIN_TW	TW RoW Fin	lin	RW	✓	row

Table 56: South Africa (ZA): $n = 25$, shipped history from 2002Q1, last observation 2026Q1.

ID	Description	Trans.	Prior	Fin.	Block
GDP_ZA	ZA Real GDP	log	RW	—	dom.
CONS_ZA	ZA Real Household Consumption	log	RW	—	dom.
INV_ZA	ZA Real Gross Fixed Capital Formation	log	RW	—	dom.
GOV_ZA	ZA Real Government Consumption	log	RW	—	dom.
EXP_ZA	ZA Real Exports	log	RW	—	dom.
IMP_ZA	ZA Real Imports	log	RW	—	dom.
CPI_ZA	ZA CPI Index	log	RW	—	dom.
LR_ZA	ZA Unemployment Rate	lin	RW	—	dom.
RATE_ZA	ZA SARB Policy Rate (Repo)	lin	RW	✓	dom.
YIELD10_ZA	ZA 10-Year Govt Bond Yield	lin	RW	✓	dom.
NEER_ZA	ZA Nominal Effective Exchange Rate	log	RW	✓	dom.
POLRATE_ZA	ZA Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_ZA	ZA Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US equity (S&P 500)	log	RW	✓	glob.
FREV_GDP	ZA Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	ZA Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	ZA Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.

ID	Description	Trans.	Prior	Fin.	Block
SFA	ZA Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_ZA	ZA Credit to NFCs	log	RW	✓	dom.
ROWDEM_ZA	ZA RoW Demand	lin	RW	✓	row
ROWINF_ZA	ZA RoW Inflation	lin	RW	✓	row
ROWFIN_ZA	ZA RoW Fin	lin	RW	✓	row

C Glossary

BVAR

Bayesian vector autoregression; here estimated under the hierarchical Minnesota prior of Giannone et al. (2015) (Section 3).

GVAR, bloc, star

The coupled cross-country system; one economy's model; a bloc's trade-weighted foreign aggregate (Section 5).

Endogenous dominant unit / frozen anchor

The US as a full member of the fixed point, with its own stars and weights row, versus the legacy one-shot exogenous US.

Global commons

The four US-determined series (oil, US 10y, US Baa, US equity) carried by every bloc (Section 5.4).

COVID-as-missing

The 2020Q2–Q4 macro cells treated as missing and imputed inside estimation; financial columns stay observed (Section 3.4).

Nowcast (NC)

The model's estimate of a quarter that has happened but is not yet published; always drawn amber (Section 11.2).

Hard / soft pin

An exact conditioning value versus an outside forecast held with trust τ, i.e. measurement variance (15) (Sections 4.2, 4.5).

Drift setter (δc)

The long-run anchor mechanism: an exact constant shift of a variable's companion drift (Section 8).

SFA, EFFI

The stock-flow adjustment (deficit–debt discrepancy) and the effective interest rate on the debt stock (Section 6).

Snowball
The automatic interest–growth term of the debt identity (25).

Coupling artifact, model signature
The precomputed $(I - M)^{-1}$ and the fingerprint that refuses it when stale (Section 5.3).

CRPS, PIT, MCS
The continuous ranked probability score (39); the probability integral transform; the model confidence set (Section 12).

Drivers
The experimental, model-conditional shock decomposition of a scenario path: the mix of a chosen DSGE model's own structural shocks that reproduces the path over twelve quarters (Section 10).

Naming budget
The rule that at most q shocks may be named when q observables are watched; observables that are invertible transforms of one already watched do not raise it (Section 10.5).

Collinearity flag
The build-time warning on a shock pair whose stacked responses satisfy $|\cos| > 0.95$; the sum over such a pair is identified, the split within it is not (Section 10.5).

Class view and its licence
Aggregation of contributions into Monetary, Demand and Supply, permitted only at the one (model, observables, named shocks) configuration the recovery bench validated (Section 10.6).

Collinearity cluster
A connected component of the $|\cos| > 0.95$ graph, used in place of individual shocks on the FRB/US lenses; members are chained, not mutually parallel (Section 10.7).

Exact-fit / residual-able
The two decomposition regimes: named shocks spanning the watched observable paths, so the residual is zero by construction, versus a genuine unexplained dimension (Section 10.2).

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